MTH645

Fuzzy Logic and Applications

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Lecture # 1

Set: A set is collection of well defined objects.

Classical or Crisp Set

$X, U \rightarrow$ Universe of discourse or Universe set

$X = \mathbb{R}$ whole set, $N = \{1, 2, 3, \dots\}$

$A = \{1, 2, 3\}$ N is subset of A.

$\phi = \{\}$ Null set

$P(A) = \{$ all subsets of $A\}$ power set

Number of elements of $P(A) = 2^{(\text{No. of elements in } A)} = 2^{\#A} = 2^{|A|}$.

Let $X = \{1, 2, 3, 4\}$ Cordinal Number $= R_x = 4$

$P(X) = \{\phi, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}$

$\{3,4\}, \{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}, \{1,2,3,4\}\}$ X

No. of elements of $P(X) = 2^4 = 16$

Let A and B are sets taken from universel set.

$A \cup B =$ A Union B

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$$A \cup B = \{ x / x \in A \text{ or } x \in B \}$$

$X = N$

$A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$

$A \cap B =$ A intersection B

$A \cap B = \{ x / x \in A \text{ and } x \in B \}$

$X = N$, $A = \{1, 2, 3\}$, $B = \{3, 4, 5\}$

$A \cap B = \{3\}$

$\bar{A}$ = Complement of A

$\bar{A} = \{ x / x \notin A, x \in X \}$

if $X = N$, $A = \{1, 2, 3\}$

$\bar{A} = \{4, 5, 6, \dots\}$

if $X = \{1, 2, 3, 4\}$, $A = \{1, 2, 3\}$

$\bar{A} = \{4\}$

$A \setminus B = A | B = A - B =$ A difference B

$A \setminus B = A | B = A - B = \{ x / x \in A \text{ and } x \notin B \}$

$A = \{1, 2, 3\}$, $B = \{1, 3, 5, 7\}$

$A \setminus B = A | B = A - B = \{2\}$

properties:

$A \cup B = B \cup A$ } commutative property.

$A \cap B = B \cap A$ }

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$A \cup (B \cup C) = (A \cup B) \cup C$
$A \cap (B \cap C) = (A \cap B) \cap C$ } Associative property

$A \cup A = A$
$A \cap A = A$ } Idempotency

$A \cup \phi = A$
$A \cap X = A$
$A \cap \phi = \phi$
$A \cup X = X$ } Identity

$A \subseteq B \subseteq C \Rightarrow A \subseteq C$ } Transitivity

$\bar{\bar{A}} = A$ } Involution

$A \cup \bar{A} = X$

$A \cap \bar{A} = \phi$

$\overline{A \cap B} = \bar{A} \cup \bar{B}$
$\overline{A \cup B} = \bar{A} \cap \bar{B}$ } De Morgan's Law

Functions:

Let X and Y are sets.

$f : X \to Y$

$f(x) = y$ $\quad x \in X$, $y \in Y$

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Characteristic Function ($\chi_A(x)$):

Let X be a set and A be a subset of X. Then the function

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

is called the characteristic function of the set A in X.

Example: $A = \{1, 2, 3\}$, $U = \{1, 2, 3, 4, 5\}$

$1 \in A$	$2 \in A$	$3 \in A$	$4 \notin A$
$\chi_A(1) = 1$	$\chi_A(2) = 1$	$\chi_A(3) = 1$	$\chi_A(4) = 0$ ....

Example:

$$U = \{x_1, x_2, x_3, x_4, x_5, x_6\}$$

$$A = \{x_2, x_3, x_5\}$$

$$\mu_A(x) = \begin{cases} 1 ; & x \in A \\ 0 ; & x \notin A \end{cases}$$

$\mu_A(x_1) = 0 ; x_1 \notin A$

$\mu_A(x_2) = 1 ; x_2 \in A$

$\mu_A(x_3) = 1 ; x_3 \in A$

$\mu_A(x_4) = 0 ; x_4 \notin A$

$\mu_A(x_5) = 1 ; x_5 \in A$

$\mu_A(x_6) = 0 ; x_6 \notin A$

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$$\mu_A(x) = \begin{cases} 1 & ; \quad x = x_2, x_3, x_5 \\ 0 & ; \quad x = x_1, x_4, x_6 \end{cases}$$

$$A = \{ (x_1, 0), (x_2, 1), (x_3, 1), (x_4, 0), (x_5, 1), (x_6, 0) \}$$

or $A = \left\{ \frac{x_1}{0}, \frac{x_2}{1}, \frac{x_3}{1}, \frac{x_4}{0}, \frac{x_5}{1}, \frac{x_6}{0} \right\}$

or $A = \left\{ \frac{0}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \frac{0}{x_4}, \frac{1}{x_5}, \frac{0}{x_6} \right\}$

or $A = \{ (x, \mu_A(x)) \mid x \in A \}$

Lecture # 9

$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$

$\chi : A \rightarrow \{0, 1\}$

$\chi : A \cup B \rightarrow \{0, 1\}$

Maximum Operator

$A \cup B \rightarrow \chi_{A \cup B}(x) = \chi_A(x) \vee \chi_B(x)$

$= \max \{ \chi_A(x), \chi_B(x) \}$

Example:

$A = \{1, 2\}, \quad B = \{2, 3\}$

$A \cup B = \{1, 2, 3\}$

$\chi_{A \cup B}(1) = \max (\chi_A(1), \chi_B(1))$

$= \max (1, 0) = 1$

$\chi_{A \cup B}(2) = \max (\chi_A(2), \chi_B(2))$

$= \max (1, 1) = 1$

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$$\chi_{A\cup B}(3) = \max\left(\chi_A(3), \chi_B(3)\right)$$
$$= \max(0, 1) = 1$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$\chi : A \to \{0,1\}$$
$$\chi : A \cap B \to \{0,1\}$$

Minimum operator

$$A \cap B \longrightarrow \chi_{A\cap B}(x) = \chi_A(x) \wedge \chi_B(x)$$
$$= \max\left(\chi_A(x), \chi_B(x)\right)$$

Example:

$$A = \{1,2\},\ B = \{2,3\}$$
$$A \cap B = \{2\}$$
$$\chi_{A\cap B}(1) = \min\left(\chi_A(1), \chi_B(1)\right)$$
$$= \min(1, 0) = 0.$$
$$\chi_{A\cap B}(2) = \min\left(\chi_A(2), \chi_B(2)\right)$$
$$= \min(1, 1) = 1$$
$$\chi_{A\cap B}(3) = \min\left(\chi_A(3), \chi_B(3)\right)$$
$$= \min(0, 1) = 0$$

$$\bar{A} \longrightarrow \chi_{\bar{A}}(x) = 1 - \chi_A(x)$$

Example: $X = \{a, b, c, d\}$

$$A = \{a, b, c\}$$
$$\bar{A} = \{d\}$$

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$\chi_A(x)$	$\chi_{\bar{A}}(x)$
$\chi_A(a) = 1$	$1 - \chi_A(a) = 1 - 1 = 0$
$\chi_A(b) = 1$	$1 - \chi_A(b) = 1 - 1 = 0$
$\chi_A(c) = 1$	$1 - \chi_A(c) = 1 - 1 = 0$
$\chi_A(d) = 0$	$1 - \chi_A(d) = 1 - 0 = 1$
	$\bar{A} = \{d\}$

$A \rightarrow \chi_A(x)$

$$\chi_A(x) = \begin{cases} 1 & ; x \in A \quad \text{member} \\ 0 & ; x \notin A \quad \text{not member} \end{cases}$$

$A \rightarrow \{0, 1\}$

If $A \rightarrow [0, 1]$ $x \in \%$

$(x, 0.7)$ $70\% \in A$

Example:

$X = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 4, 6\}$

$$\chi_A(x) = \begin{cases} 1 & ; x \in A \\ 0 & ; x \notin A \end{cases}$$

$\chi_A(1) = 0$

$\chi_A(2) = 1$

$\chi_A(3) = 0$

$\chi_A(4) = 1$

$\chi_A(5) = 0$

$\chi_A(6) = 1$

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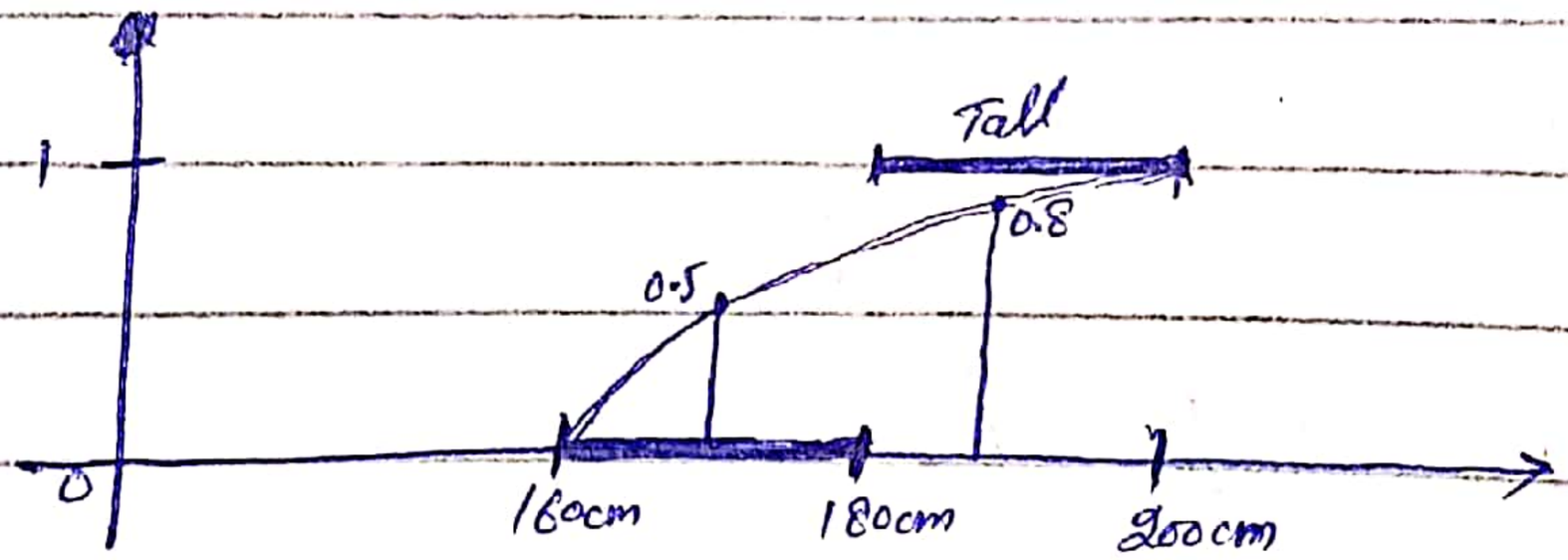
$$A = \{ (x, \chi_A(x)) \mid x \in A, \chi_A(x) \in \{0,1\} \}$$

$$= \{ (2,1), (4,1), (6,1) \}$$

$$= \{ (1,0), (2,1), (3,0), (4,1), (5,0), (6,1) \}$$

$A = \{ \text{Tall men} \}$

$$\mu_A(x) = \begin{cases} 1 & ; 180 \leq x \\ 0 & ; 160 \leq x < 180 \end{cases} \quad cm$$

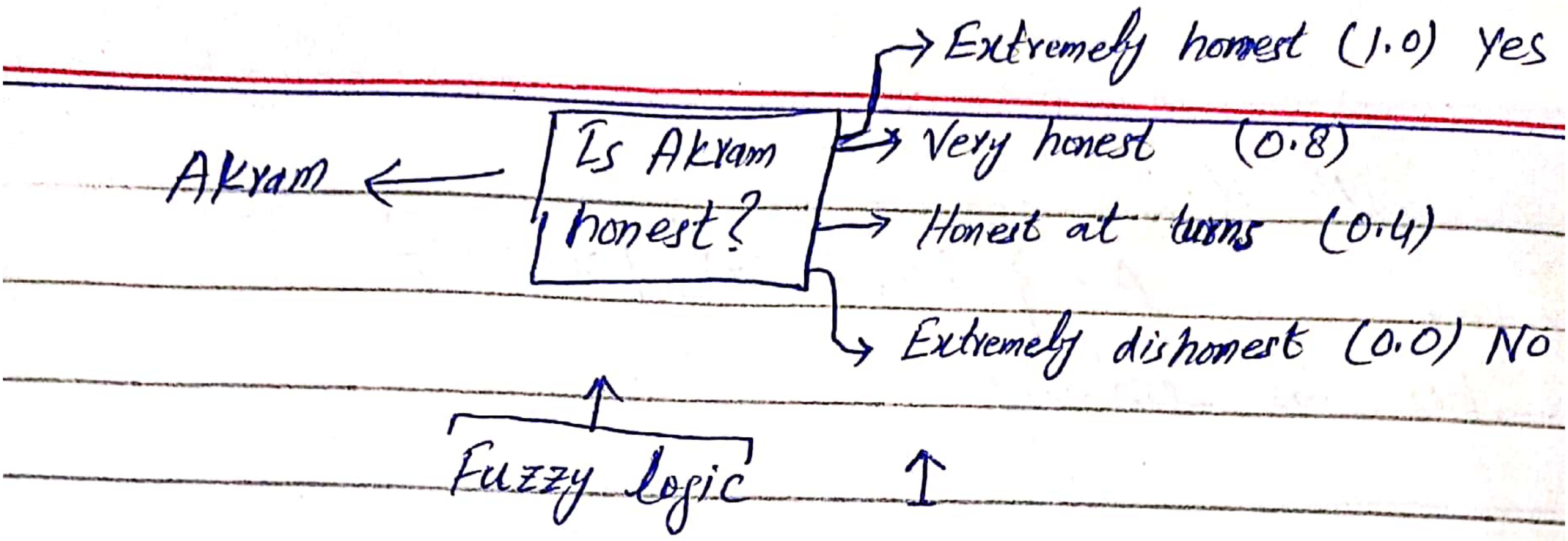


Fuzzy Logic vs Boolean Logic:

Fuzzy logic is an approach to computing based on degree of truth rather than the usual truth or false (0 or 1) Boolean logic on which the modern computer is based.

Akram ← Is Akram honest? → yes (1) True
→ No (0) False

Boolean logic or Traditional binary logic ↑



Lecture #3

Lotfi Aliasker Zadeh ; Introduction :-

February 4, 1921, Baku, Azerbaijan - September 6, 2017, Berkeley, California, United states.

Nationality: Azerbaijan, American, Iranian.

Children: Norm Zada, Stella Zadeh.

Mathematician, Computer scientist, Electrical engineer, Artificial intelligence researcher and professor emeritus of computer science at the University of California, Berkely.

Zadeh for was best know for proposing fuzzy mathematics consisting of these fuzzy-related concepts: fuzzy sets, fuzzy logic, fuzzy algorithms, fuzzy Sementics, fuzzy languages, fuzzy control, fuzzy systems, fuzzy probabilities, fuzzy events and fuzzy information

Fuzzy Set :

A fuzzy set consists of two components: a regular set and a membership function associated

or
crisp or Classical

with it:

$$A = \{ (x, \mu_A(x)) \mid x \in U \}$$

For an ordinary set, A:

$$\mu_A(x) = \begin{cases} 1 & \text{iff } x \in A \\ 0 & \text{iff } x \notin A \end{cases}$$

When U is a finite set $\{x_1, x_2, x_3, \cdots, x_n\}$, the fuzzy set on U may be represented as

$$A = \sum_{i=1}^{n} \frac{x_i}{\mu_A(x_i)}$$

When U is an infinite set, the fuzzy set may be represented as

$$A = \int_x \frac{x}{\mu_A(x)}$$

Example:

Let $U = \{Akram, Aslam, Amir, Asif\}$.

Evaluated by a girl, the fuzzy set "handsome boys" may be characterized as

$$A = \{ (Akram, 0.3), (Aslam, 0.5), (Amir, 0.7), (Asif, 0.9) \}$$

or $$A = \frac{Akram}{0.3} + \frac{Aslam}{0.5} + \frac{Amir}{0.7} + \frac{Asif}{0.9}$$

Example:

$$A = \{ (x_1, 0.1), (x_2, 0.5), (x_3, 0.3), (x_4, 0.8), (x_5, 1), (x_6, 0.2) \}$$

$$A = \{ (1, 0.1), (2, 0.5), (3, 0.3), (4, 0.8), (5, 1), (6, 0.2) \}$$

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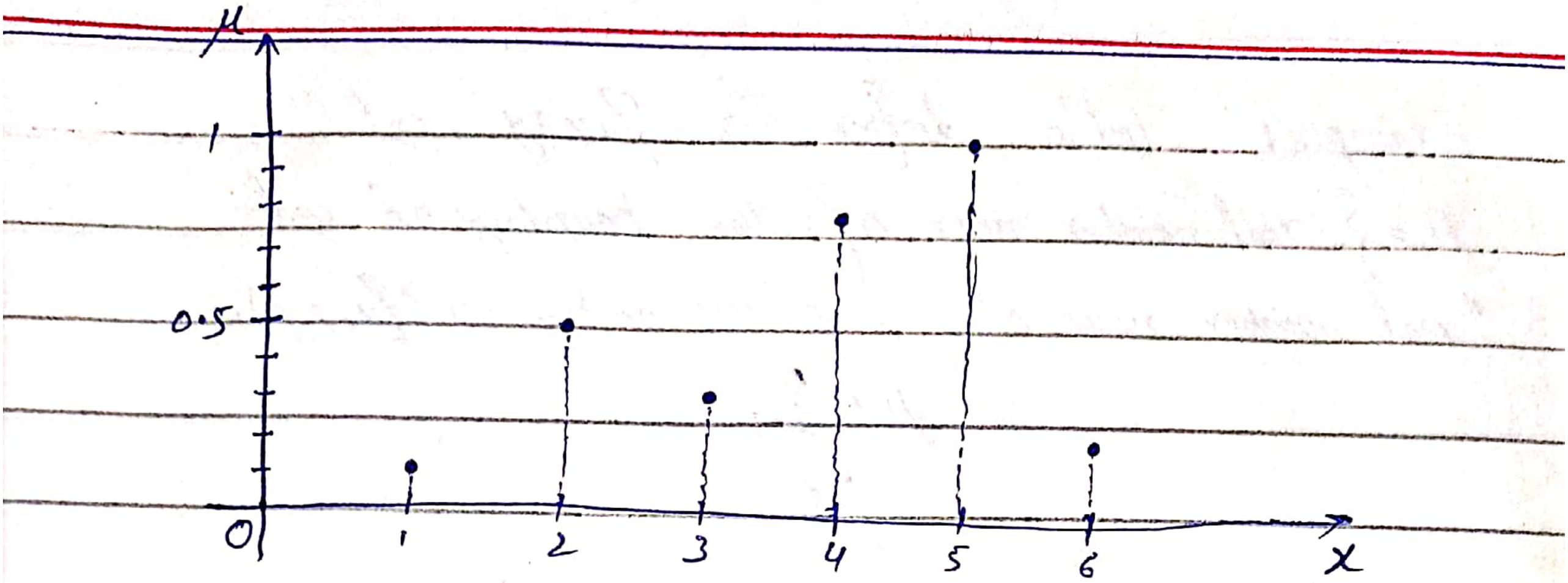


Fuzzy set

Example:

Fuzzy set of numbers close to 10 is

$$A = \left\{ (x, \mu_A(x)) \mid x \in [5, 15], \mu_A(x) = \frac{1}{1 + (x-10)^2} \right\}$$

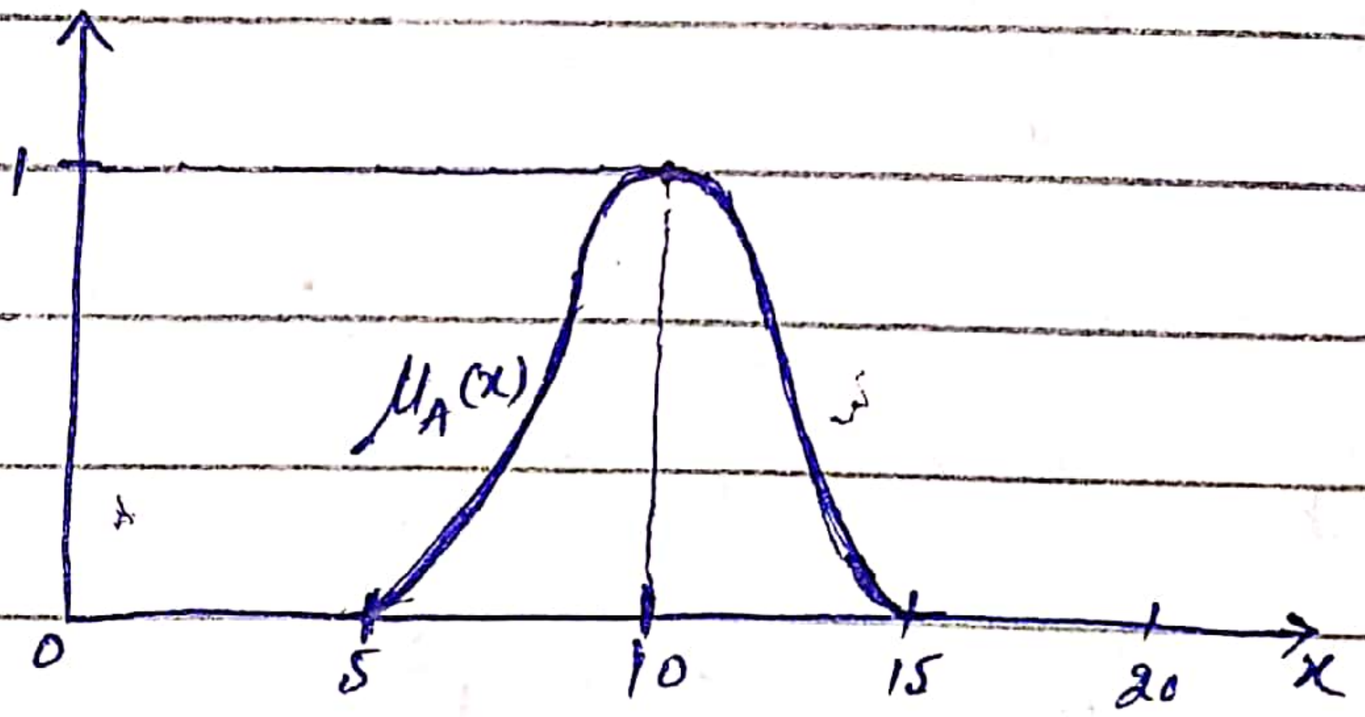
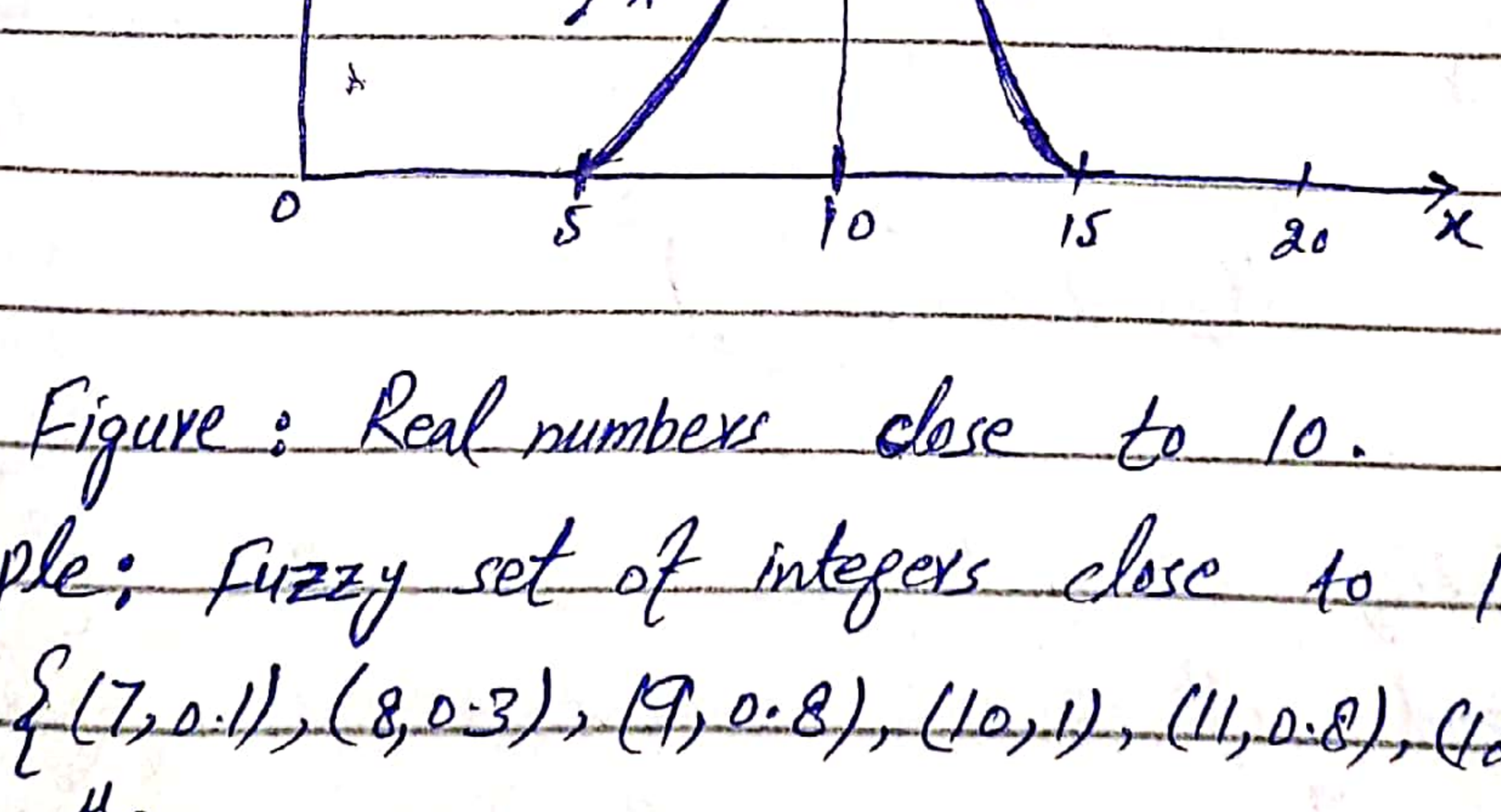


Figure: Real numbers close to 10.

Example: Fuzzy set of integers close to 10 is

$$A = \{(7, 0.1), (8, 0.3), (9, 0.8), (10, 1), (11, 0.8), (12, 0.3), (13, 0.1)\}$$



(Discrete)

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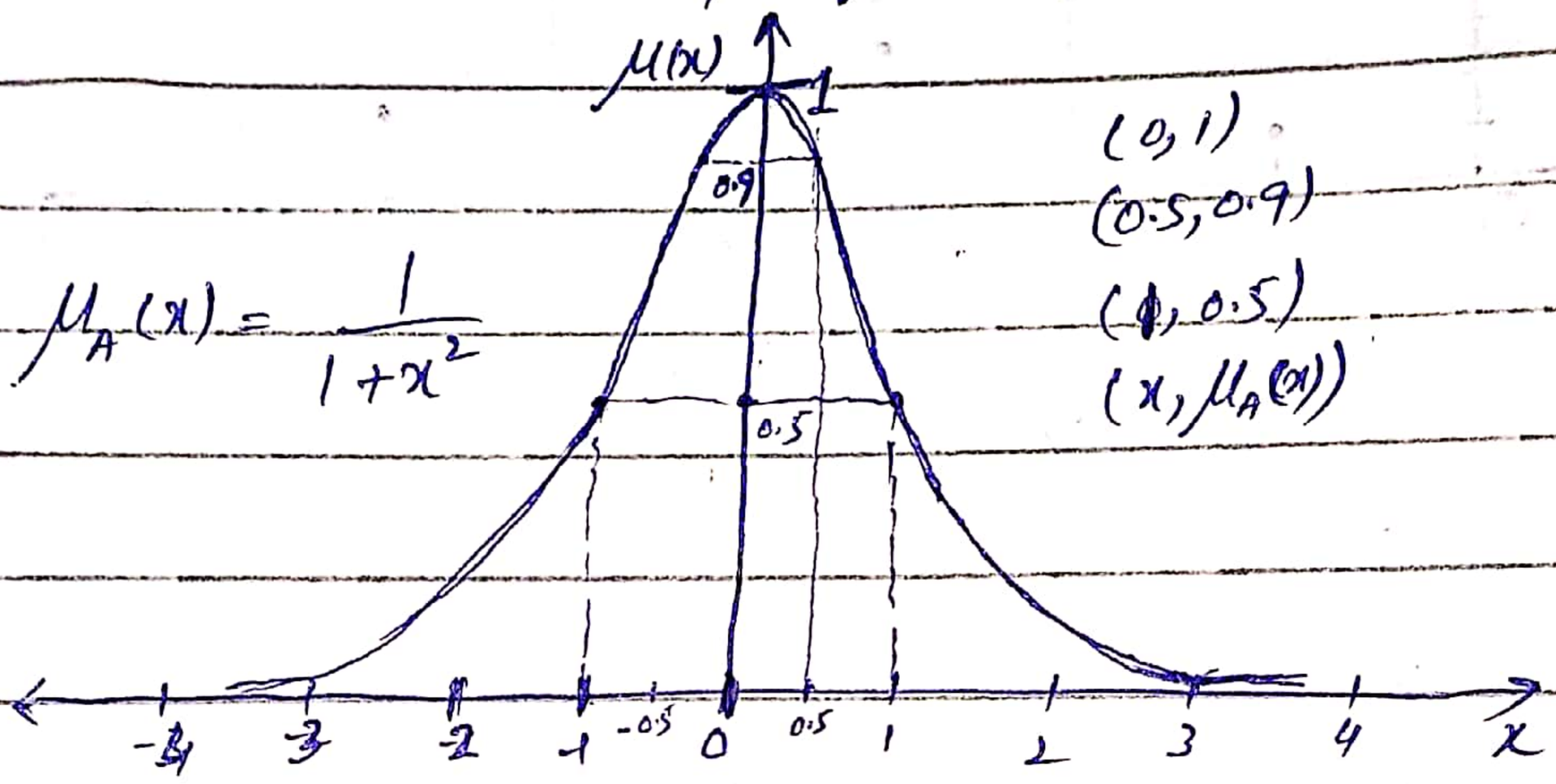
Example: let's define a fuzzy set $A = \{$ real number near $0\}$. The boundary of set "real number near 0" is pretty ambiguous (fuzzy).



$$A = \int \frac{\mu_A(x)}{x} \quad \text{where} \quad \mu_A(x) = \frac{1}{1+x^2}$$

Example: Another fuzzy set $B = \{$ real number very near to $0\}$ can be defined and its membership function is:

$$\mu_B(x) = \left(\frac{1}{1+x^2}\right)^2$$

the possibility of 1 is 0.25 that of 2 is 0.04 and of 3 is 0.01.

By modifying the above function, it is able to denote membership function of fuzzy set A.

$A = \{$ real number nea $a\}$ as

$$\mu_A(x) = \frac{1}{1+(x-a)^2}$$

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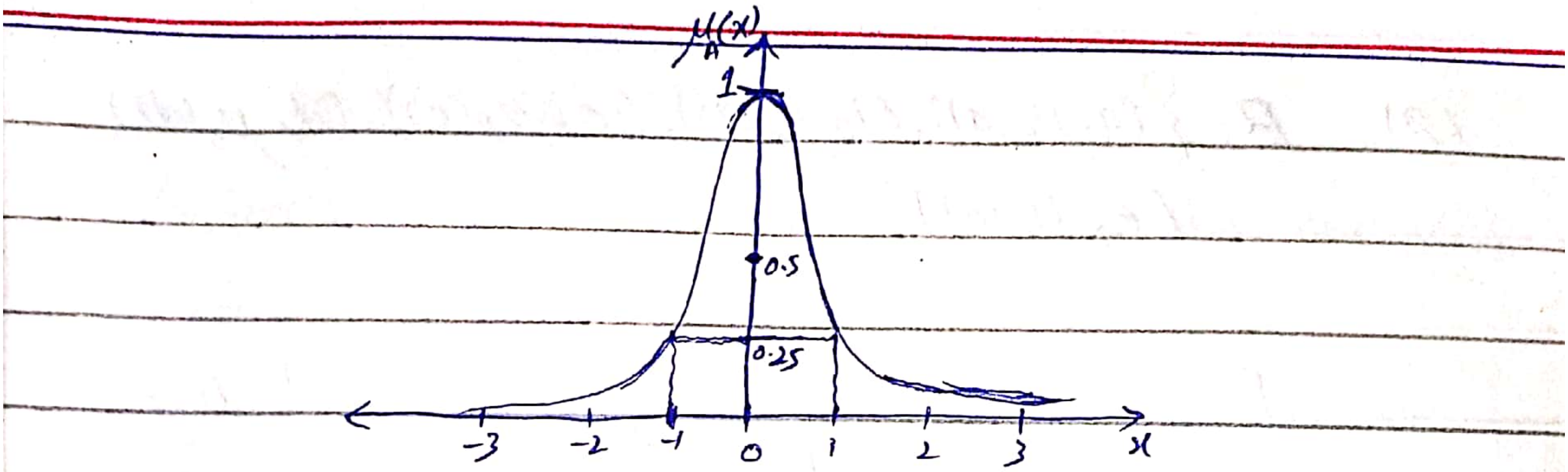

$$A = \int \frac{\mu_A(x)}{x} = \int \frac{\frac{1}{1+(x-a)^2}}{x}$$

Expressions of Fuzzy Set:

★ $A = \{(x, \mu_A(x))\}$

★ $A = \frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \frac{\mu_A(x_3)}{x_3} + \cdots + \frac{\mu_A(x_n)}{x_n} + \cdots$

★ $A = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$ (For discrete case)

★ $A = \int \frac{\mu_A(x)}{x}$ (For continuous case)

Example:

$A = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$ is another form of representation of fuzzy set. Represent the following fuzzy sets by this form.

(1) $A = \{(3, 0.9), (2, 0.3), (4, 1.0), (7, 0.7)\}$

$$A = \frac{0.9}{3} + \frac{0.3}{2} + \frac{1.0}{4} + \frac{0.7}{7}$$

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(2) $B = \{(a, \mu_B(a)), (b, \mu_B(b)), (c, \mu_B(c)), (d, \mu_B(d)), (e, \mu_B(e))\}$

$$B = \frac{\mu_B(a)}{a} + \frac{\mu_B(b)}{b} + \frac{\mu_B(c)}{c} + \frac{\mu_B(d)}{d} + \frac{\mu_B(e)}{e}$$

Prove that $(\tilde{A} \cup \tilde{A}')(x) \geq 0.5$ $\tilde{A}$ = (A tilde)

Membership value of $\tilde{A} \cup \tilde{A}'$ is always greater than or equal to 0.5 ....

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x is $\tilde{A}$. So by definition of compliment, we have

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \longrightarrow (1)$$

$$(\tilde{A} \cup \tilde{A}')(x) = \max(\tilde{A}(x), \tilde{A}'(x)) \quad \forall x \in X$$

$$= \max(\tilde{A}(x), 1 - \tilde{A}(x)) \text{ by eq. (1)}$$

Case I: If $\tilde{A}(x) \geq 0.5$, then obviously

$$1 - \tilde{A}(x) \leq 0.5$$

$$\Rightarrow \tilde{A}'(x) \leq 0.5$$

So $\max(\tilde{A}(x), \tilde{A}'(x)) \geq 0.5$

Case II: If $\tilde{A}(x) \leq 0.5$, then obviously

$$1 - \tilde{A}(x) \geq 0.5$$

$$\Rightarrow \tilde{A}'(x) \geq 0.5$$

So $\max(\tilde{A}(x), \tilde{A}'(x)) \geq 0.5$

Hence $(\tilde{A} \cup \tilde{A}')(x) \geq 0.5$

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Prove that $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Membership value of $\tilde{A} \wedge \tilde{A}'(x)$ is always less then or equal to 0.5.

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x. So by definition of complement, we have

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \longrightarrow (1)$$

$$(\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x))$$

$$= \min(\tilde{A}(x), 1 - \tilde{A}(x)) \quad \text{by eq. (1)}$$

Case I: If $\tilde{A}(x) \geq 0.5$, then obviously

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \leq 0.5$$

So $\min(\tilde{A}(x), \tilde{A}'(x)) \leq 0.5$

Case II: If $\tilde{A}(x) \leq 0.5$, then obviously

$$\tilde{A}'(x) = (1 - \tilde{A}(x)) \geq 0.5$$

So $\min(\tilde{A}(x), \tilde{A}'(x)) \leq 0.5$

Hence $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Lecture #4:

Support of a fuzzy set $\tilde{A}$:

A set is said to be the support of a fuzzy set $\tilde{A}$ if the membership values of elements are greater then zero.

$$\text{Support}(\tilde{A}) = \{x : x \in X \text{ and } \tilde{A}(x) > 0\}$$

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Example: $X = \{x_1, x_2, x_3, x_4, x_5\}$

$\tilde{A} = \{(x_1, 0), (x_2, 0.2), (x_3, 0.9), (x_4, 0.1), (x_5, 0)\}$

$Support(\tilde{A}) = \{x_2, x_3, x_4\}$

Normal Fuzzy set:

A fuzzy set is said to be normal fuzzy set if there exist atleast one element whose membership is one.

Example: $X = \{x_1, x_2, x_3, x_4, x_5\}$

$\tilde{A} = \{(x_1, 0), (x_2, 0.2), \boxed{(x_3, 1)}, (x_4, 0.5), (x_5, 0.8)\}$

Normal fuzzy set ∴

Example: Is $\tilde{A}: \mathbb{R} \to [0,1]$, defined by $\tilde{A}(x) = e^{-x^2}$ a normal fuzzy?

Ans: Yes! because $\tilde{A}(x) = e^{-x^2}$

at $x = 0$

$\tilde{A}(0) = e^{-(0)^2} = e^0 = 1$

Empty Fuzzy set:

A fuzzy set is said to be empty set if all elements of universal set maps on zero. i.e. the membership value of all elements is zero.

Example: $X = \{x_1, x_2, x_3, x_4\}$

$\tilde{A} = \{(x_1, 0), (x_2, 0), (x_3, 0), (x_4, 0)\}$

$\tilde{A}$ is a empty set.

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Cardinality of fuzzy set:

Cardinality of a fuzzy set is sum of all membership values of of the elements of the set.

Examples: $X = \{x_1, x_2, x_3, x_4, x_5\}$

$A = \{x_1, x_2, x_3\}$, $|A| = 3$

$A = \{(x_1, 1), (x_2, 1), (x_3, 1), (x_4, 0), (x_5, 0)\}$

$|A| = 1+1+1+0+0 = 3$

Example: $\tilde{E} = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.7)\}$

$|\tilde{E}| = 0.1+0.5+0.7 = 1.3$

Degree of subsethood of $\tilde{A}$ in $\tilde{B}$

Degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is defined as $S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$

Example: $\tilde{A} = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.5)\}$

$\tilde{B} = \{(x_1, 0.5), (x_2, 0.4), (x_3, 0.2)\}$

$\tilde{A} \cap \tilde{B} = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.2)\}$ $\because \tilde{A} \cap \tilde{B} = \min(\tilde{A}, \tilde{B})$

$|\tilde{A} \cap \tilde{B}| = 0.1+0.3+0.2 = 0.6$

$|\tilde{A}| = 0.1+0.3+0.5 = 0.9$

So degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is

$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|} = \frac{0.6}{0.9} = 0.67$

Note: $S(\tilde{A}, \tilde{B}) \in [0, 1]$ (MCQ's)

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Union of Fuzzy sets:

Let $\tilde{F}(X)$ denote the collection of Fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

$$= \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X$$

We denote $\tilde{C} = \tilde{A} \cup \tilde{B}$

Example: $\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$

$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$

$\tilde{C} = \tilde{A} \cup \tilde{B} = \{(x_1, 0.4), (x_2, 0.6), (x_3, 1), (x_4, 0.8), (x_5, 0.5)\}$

Intersection of Fuzzy sets:

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The interesection of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\}$$

$$= \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x), \quad \forall x \in X$$

We denote $\tilde{C} = \tilde{A} \cap \tilde{B}$

Example: $\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$

$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$

$\tilde{C} = \tilde{A} \cap \tilde{B} = \{(x_1, 0.3), (x_2, 0.2), (x_3, 0.7), (x_4, 0.6), (x_5, 0.3)\}$

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Lecture #05

Complement of fuzzy set:

Let $F(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A} \in \tilde{F}(X)$. The complement of $\tilde{A}$ is the fuzzy set $\tilde{B}$ with

$$\mu_{\tilde{B}}(x) = 1 - \mu_{\tilde{A}}(x), \quad \forall x \in X$$

We denote $\tilde{B} = \bar{\tilde{A}} \overset{or}{=} \tilde{A}'$

Example: $\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$

$\bar{\tilde{A}} = \tilde{A}' = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0), (x_4, 0.4), (x_5, 0.5)\}$

Example: $\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$

$\bar{\tilde{B}} = \tilde{B}' = \{(x_1, 0.6), (x_2, 0.8), (x_3, 0.3), (x_4, 0.2), (x_5, 0.7)\}$

Examples:

Let $X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, Let two fuzzy sets on X are $\tilde{A} = \{(0, 0), (1, 0), (2, 0.1), (3, 0.2), (4, 0.3), (5, 0.8)$
$(6, 0.9), (7, 1), (8, 1), (9, 1)\}$

$\tilde{B} = \{(0, 1), (1, 1), (2, 0.9), (3, 0.8), (4, 0.7), (5, 0.5), (6, 0.4), (7, 0.2), (8, 0.2), (9, 0)\}$

Determine $\tilde{A} \cup \tilde{B}$, $\tilde{A} \cap \tilde{B}$, $\tilde{A}'$

$(\tilde{A} \cup \tilde{B})(x) = \max\{\tilde{A}(x), \tilde{B}(x)\}$

$\tilde{A} \cup \tilde{B} = \{(0, 1), (1, 1), (2, 0.9), (3, 0.8), (4, 0.7), (5, 0.8), (6, 0.9), (7, 1), (8, 1), (9, 1)\}$

$(\tilde{A} \cap \tilde{B})(x) = \min\{\tilde{A}(x), \tilde{B}(x)\}$

$\tilde{A} \cap \tilde{B} = \{(0, 0), (1, 0), (2, 0.1), (3, 0.2), (4, 0.3), (5, 0.5), (6, 0.4), (7, 0.2), (8, 0.2), (9, 0)\}$

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$\tilde{A}'(x) = 1 - \tilde{A}(x)$

$\bar{\tilde{A}} = \tilde{A}' = \{(0,1), (1,1), (2,0.9), (3,0.8), (4,0.7), (5,0.2), (6,0.1), (7,0), (8,0), (9,0)\}$

Also determine De Morgan's Laws

(i) $(\tilde{A} \cup \tilde{B})' = \tilde{A}' \cap \tilde{B}'$

(ii) $(\tilde{A} \cap \tilde{B})' = \tilde{A}' \cup \tilde{B}'$

$\tilde{B}' = \{(0,0), (1,0), (2,0.1), (3,0.2), (4,0.3), (5,0.5), (6,0.6), (7,0.8), (8,0.8), (9,1)\}$

(i) $(\tilde{A} \cup \tilde{B})' = \tilde{A}' \cap \tilde{B}'$

$(\tilde{A} \cup \tilde{B})' = \{(0,0), (1,0), (2,0.1), (3,0.2), (4,0.3), (5,0.2), (6,0.1), (7,0), (8,0), (9,0)\}$

$\tilde{A}' \cap \tilde{B}' = \{(0,0), (1,0), (2,0.1), (3,0.2), (4,0.3), (5,0.2), (6,0.1), (7,0), (8,0), (9,0)\}$

Hence prove that $(\tilde{A} \cup \tilde{B})' = \tilde{A}' \cap \tilde{B}'$

(ii) $(\tilde{A} \cap \tilde{B})' = \tilde{A}' \cup \tilde{B}'$

$(\tilde{A} \cap \tilde{B})' = \{(0,1), (1,1), (2,0.9), (3,0.8), (4,0.7), (5,0.5), (6,0.6), (7,0.8), (8,0.8), (9,1)\}$

$\tilde{A}' \cup \tilde{B}' = \{(0,1), (1,1), (2,0.2), (3,0.8), (4,0.7), (5,0.5), (6,0.6), (7,0.8), (8,0.8), (9,1)\}$

Hence prove that $(\tilde{A} \cap \tilde{B})' = \tilde{A}' \cup \tilde{B}'$

Fuzzy De Morgan's Laws

$\tilde{A} = \{(1,0.2), (3,0.5), (5,1), (7,0.6), (10,0.4)\}$

$\tilde{B} = \{(1,0.5), (3,0.2), (5,0.8), (7,0.7), (10,0.3)\}$

$\tilde{A} \cup \tilde{B} = \{(1,0.5), (3,0.5), (5,1), (7,0.7, (10,0.4)\}$

$(\tilde{A} \cup \tilde{B})' = \{(1,0.5), (3,0.5), (5,0), (7,0.3), (10,0.6)\}$

$\tilde{A}' = \{(1,0.8), (3,0.5), (5,0), (7,0.4), (10,0.6)\}$

$\tilde{B}' = \{(1,0.5), (3,0.8), (5,0.2), (7,0.3), (10,0.7)\}$

$\tilde{A}' \cap \tilde{B}' = \{(1,0.5), (3,0.5), (5,0), (7,0.3), (10,0.6)\}$

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Hence prove that $(\tilde{A} \cup \tilde{B})' = \tilde{A}' \cap \tilde{B}'$

$\tilde{A} \cap \tilde{B} = \{(1, 0.2), (3, 0.2), (5, 0.8), (7, 0.6), (10, 0.3)\}$

$(\tilde{A} \cap \tilde{B})' = \{(1, 0.8), (3, 0.8), (5, 0.2), (7, 0.4), (10, 0.7)\}$

$\tilde{A}' \cup \tilde{B}' = \{(1, 0.8), (3, 0.8), (5, 0.2), (7, 0.4), (10, 0.7)\}$

Hence proved that $(\tilde{A} \cap \tilde{B})' = \tilde{A}' \cup \tilde{B}'$

Union of Continous Fuzzy sets:

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x \le 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x \le 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x \le 50 \\ \frac{x-50}{10} & \text{if } 50 \le x \le 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x \le 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

$A_1 \cup A_2 = A_1(x) \underset{\text{maximum operater}}{\vee} A_2(x) =$

$$\begin{cases} 1 & \text{if } 40 \le x \le 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x \le 55 \\ \frac{x-50}{10} & \text{if } 55 \le x \le 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x \le 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

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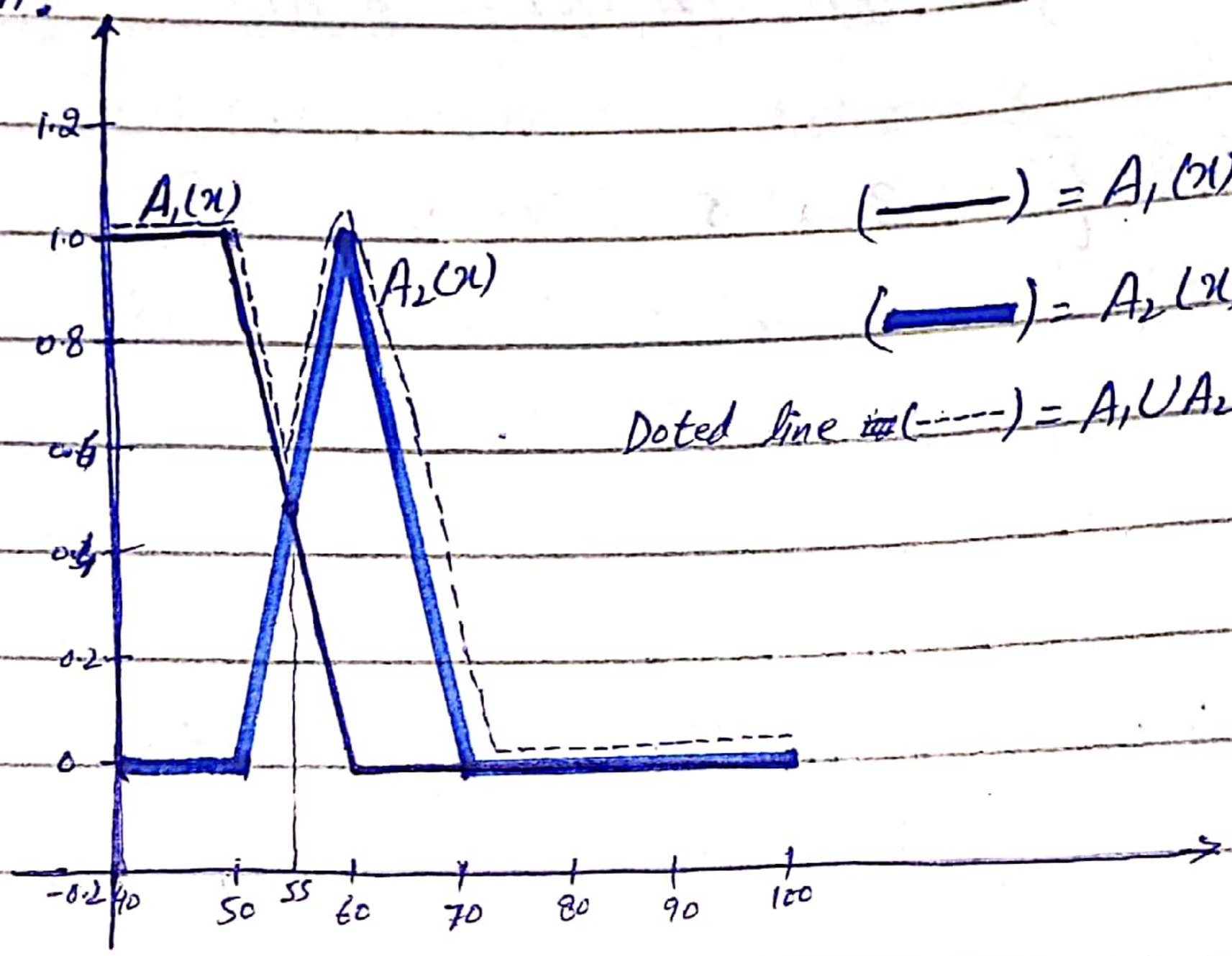
Graph:



Lecture # 6

Intersection of Continuous fuzzy sets:

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

$$A_1 \cap A_2 = A_1(x) \wedge A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 55 \\ 1 - \frac{x-50}{10} & \text{if } 55 \le x 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

↙ minimum operator

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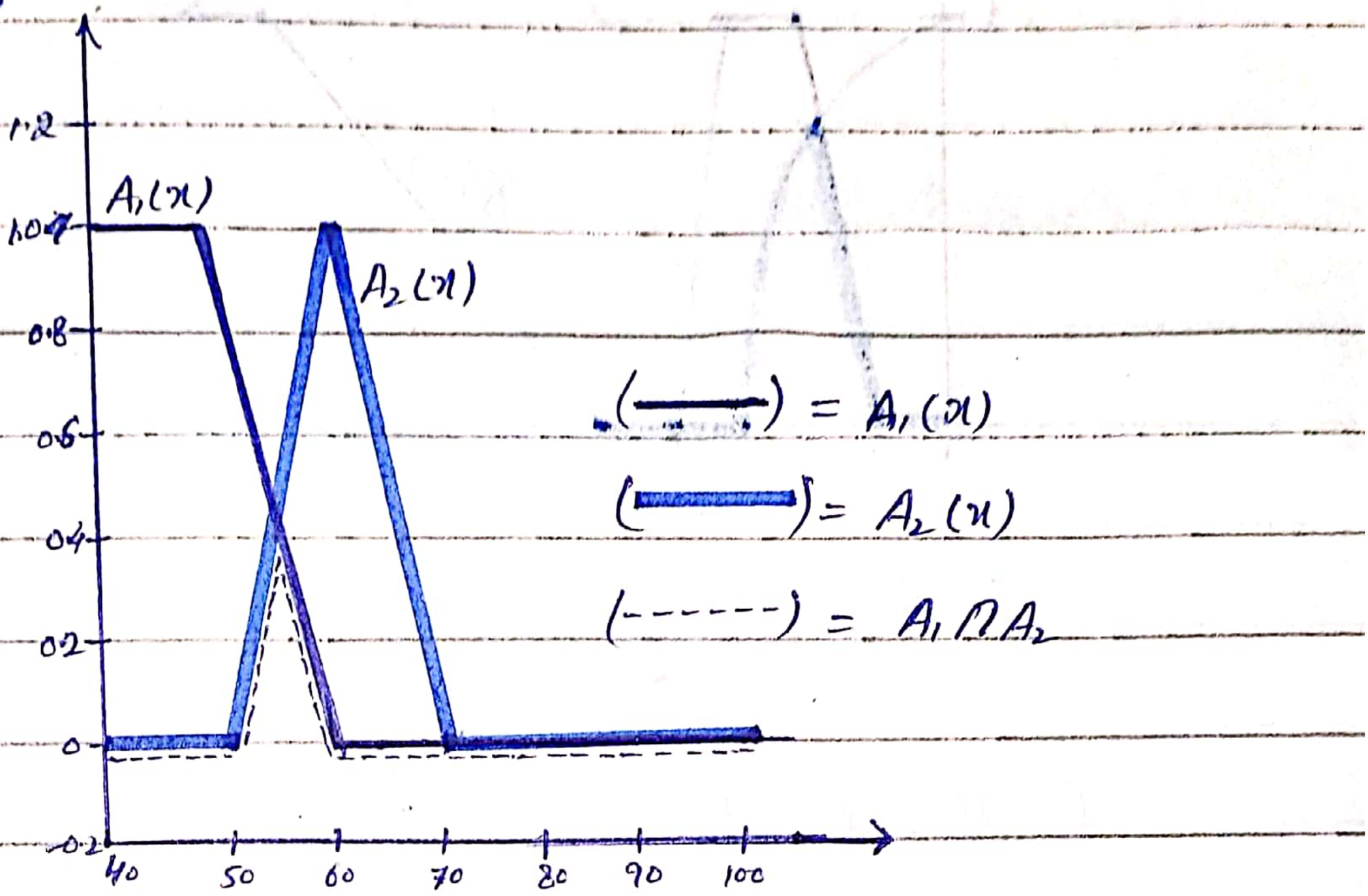
Graph:



Examples:

Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where

X = non negative real numbers

Draw the graph of $\tilde{C} \cap \tilde{D}'$

$$\tilde{C}(x) = \begin{cases} 0 & ; 0 \le x < 1 \\ x-1 & ; 1 \le x < 2 \\ 1 & ; 2 \le x < 3 \\ 4-x & ; 3 \le x < 4 \\ 0 & ; 4 \le x \end{cases}$$

$$\tilde{D}(x) = \begin{cases} e^{x-3} & ; 0 \le x \le 3 \\ 1 & ; 3 \le x < 5 \\ 1 - \frac{x-5}{2} & ; 5 \le x < 10 \\ 0 & ; 10 \le x \end{cases}$$

$$\tilde{D}'(x) = \begin{cases} 1 - e^{x-3} & ; 0 \le x < 3 \\ 0 & ; 3 \le x < 5 \\ + \frac{x-5}{2} & ; 5 \le x < 10 \\ 1 & ; 10 \le x \end{cases}$$

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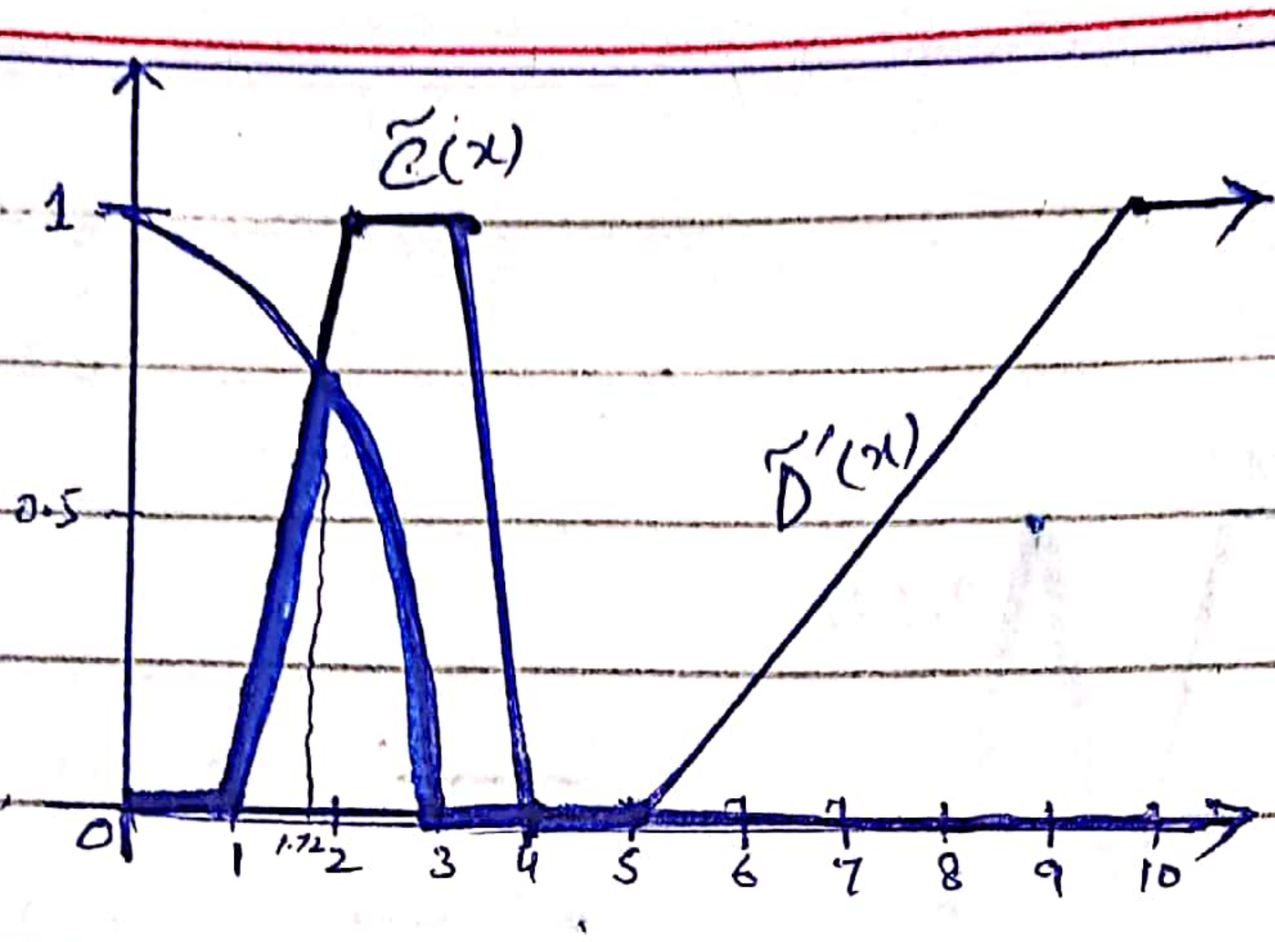

$$(\tilde{C} \cap \tilde{D}')(x) = \min(\tilde{C}(x), \tilde{D}'(x))$$

$$(\tilde{C} \cap \tilde{D}')(x) = \begin{cases} 0 & ; \quad 0 \le x < 1 \\ x-1 & ; \quad 1 \le x < 1.72 \\ 1-e^{x-3} & ; \quad 1.72 \le x < 3 \\ 0 & ; \quad 3 \le x \end{cases}$$

Interval as Fuzzy set:

When interval is defined on real number R, this interval is said to be a subset of R. For instance, if interval is denoted as $A = [a_1, a_3]$, $a_1, a_3 \in R$, $a_1 < a_3$, we may regard this as one kind of sets. Expressing the interval as membership function is shown in the following:

$$\mu_A(x) = \begin{cases} 0 & ; \quad x < a_1 \\ 1 & ; \quad a_1 \le x \le a_3 \\ 0 & ; \quad a_3 < x \end{cases}$$

If $a_1 = a_3$, this interval indicates a point.
That is, $[a_1, a_1] = a_1$

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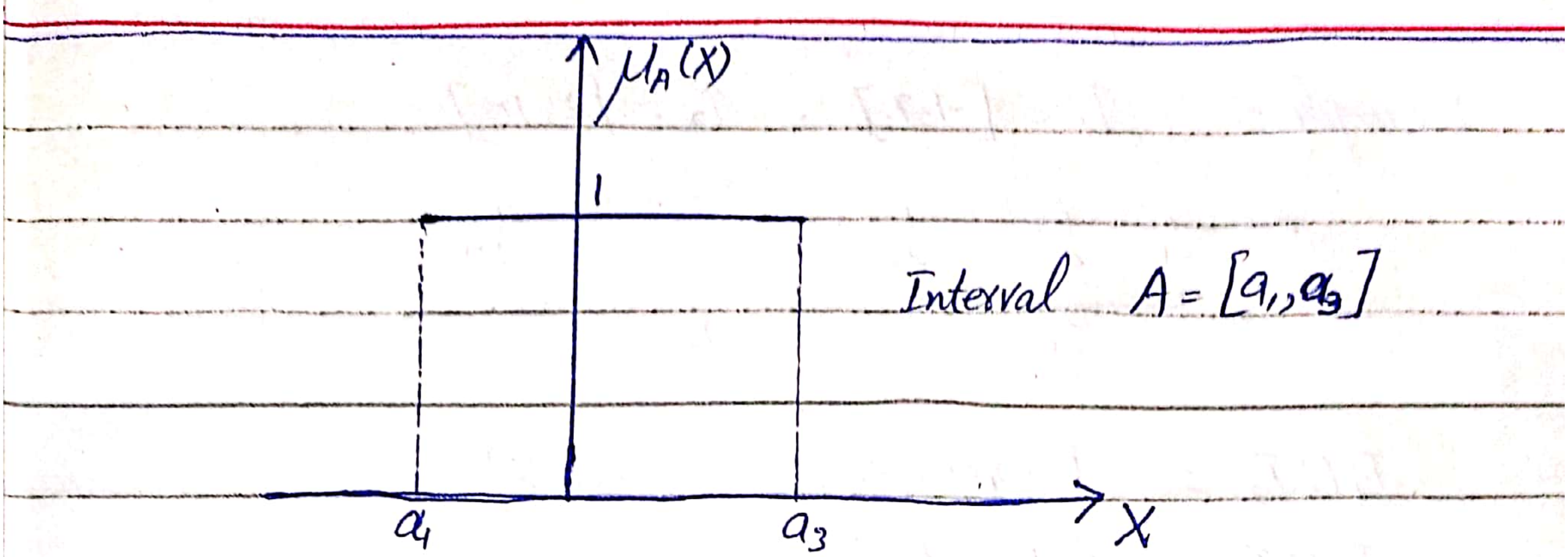


Interval Union:

The union of two intervals $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ is defined to be

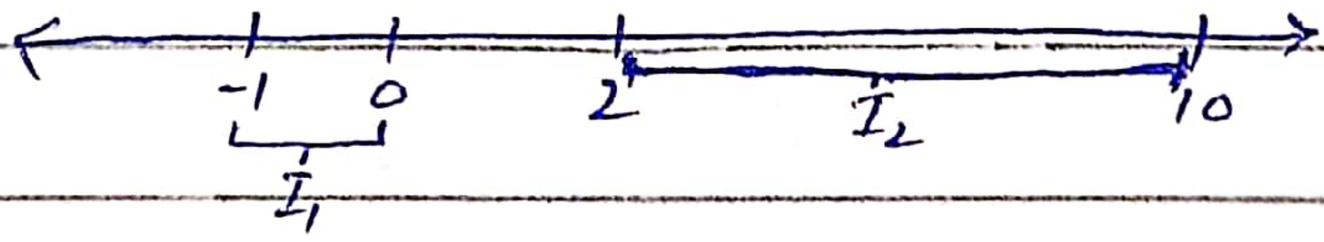
$$I_1 \cup I_2 = [a_1, b_1] \cup [a_2, b_2]$$

$$= [\min\{a_1, a_2\}, \max\{b_1, b_2\}]$$

provided that $[a_1, b_1] \cap [a_2, b_2] \neq \phi$.

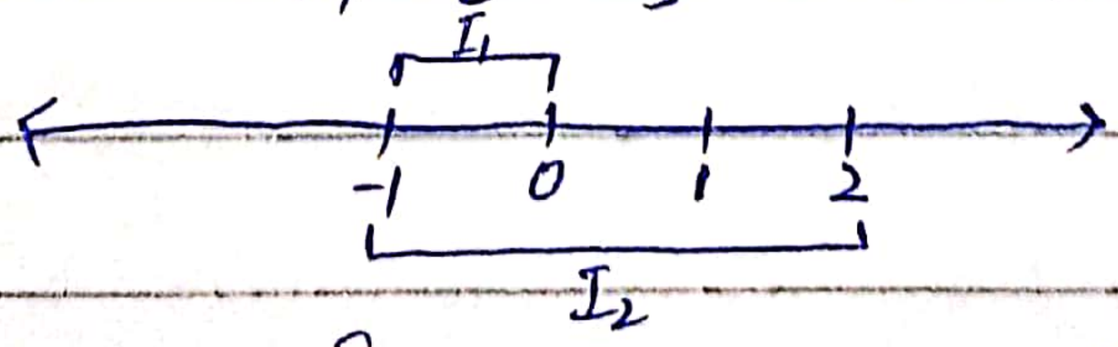
Otherwise, it is undefined (since the result is not an interval).

Example: $I_1 = [-1, 0]$, $I_2 = [2, 10]$



$I_1 \cup I_2$ = undefined.

Example: $I_1 = [-1, 0]$, $I_2 = [-1, 2]$



$$I_1 \cup I_2 = [\min\{-1, -1\}, \max\{0, 2\}]$$

$$I_1 \cup I_2 = [-1, 2]$$

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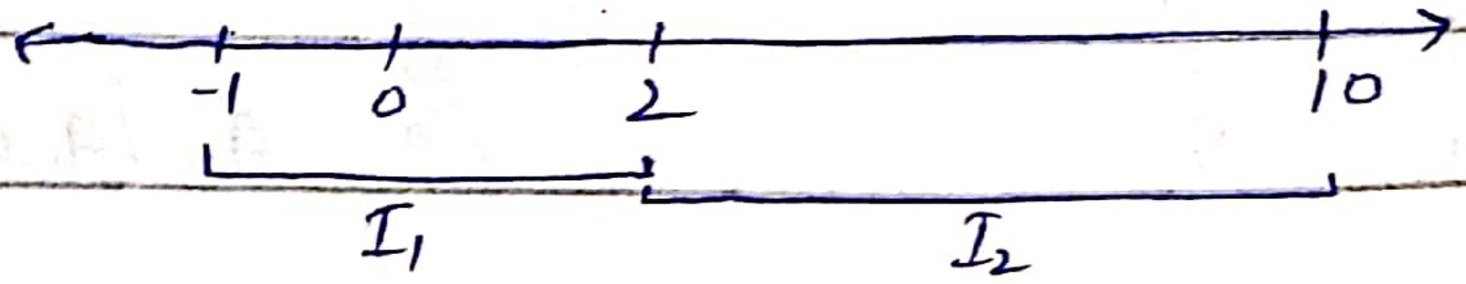
Example: $I_1 = [-1,2]$, $I_2 = [2,10]$



$I_1 \cup I_2 = [\min\{-1,2\}, \max\{2,10\}]$

$I_1 \cup I_2 = [-1,10]$

Lecture #7

Interval Intersection:

The intersection of two intervals $I_1 = [a_1,b_1]$ and $I_2 = [a_2,b_2]$ is difined to be

$$I_1 \cap I_2 = [a_1,b_1] \cap [a_2,b_2]$$

$$= [\max\{a_1,a_2\}, \min\{b_1,b_2\}]$$

and $[a_1,b_1] \cap [a_2,b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

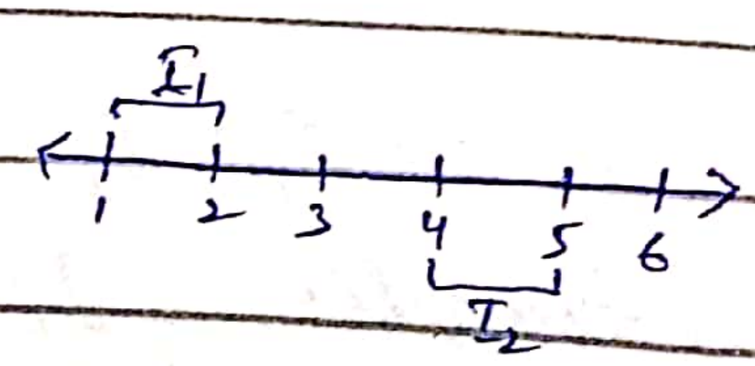
Example: $I_1 = [1,2]$, $I_2 = [4,5]$

$$I_1 \cap I_2 = [1,2] \cap [4,5]$$

$$= [\max\{1,4\}, \min\{2,5\}]$$

$$= [4,2] \neq \text{interval}$$

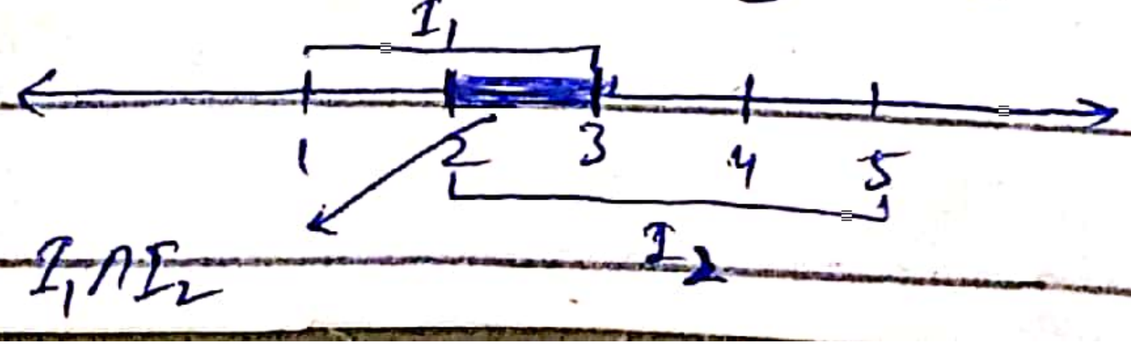
$$= \phi \quad \because a_2 > b_1,\ 4 > 2$$



Example: $I_1 = [1,3]$, $I_2 = [2,5]$

$$I_1 \cap I_2 = [1,3] \cap [2,5]$$

$$= [\max\{1,2\}, \min\{3,5\}] = [2,3]$$



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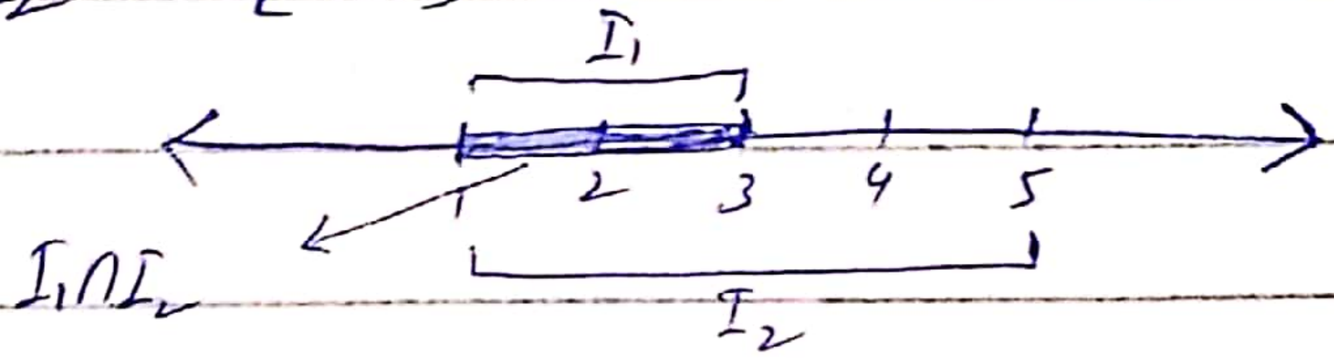
Example: $I_1 = [1,3]$; $I_2 = [1,5]$

$I_1 \cap I_2 = [1,3] \cap [1,5]$

$= [\max\{1,1\}, \min\{3,5\}]$

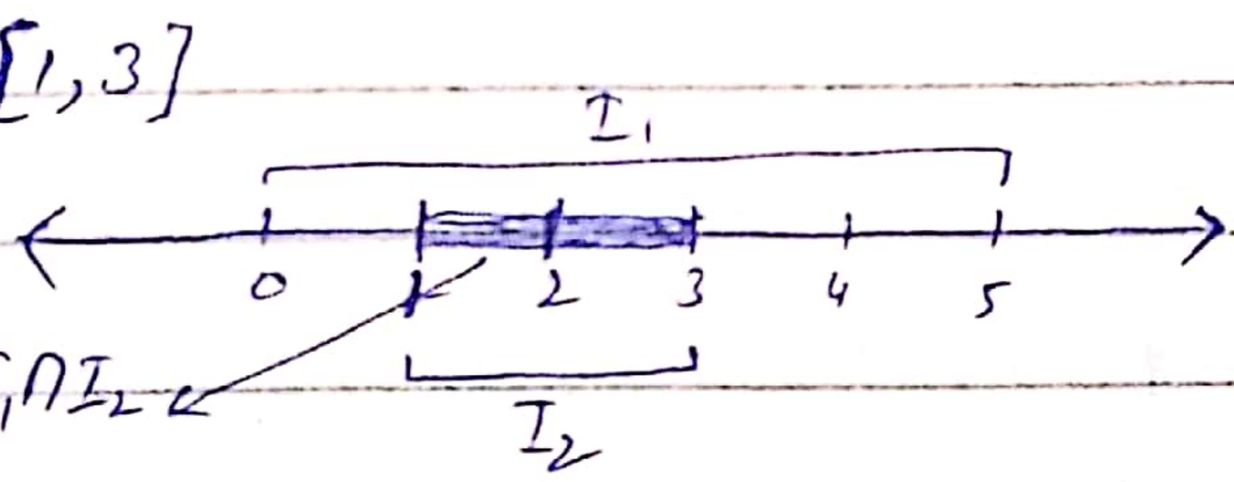
$I_1 \cap I_2 = [1,3]$



Example: $I_1 = [0,5]$, $I_2 = [1,3]$

$I_1 \cap I_2 = [0,5] \cap [1,3] = [\max\{0,1\}, \min\{5,3\}]$

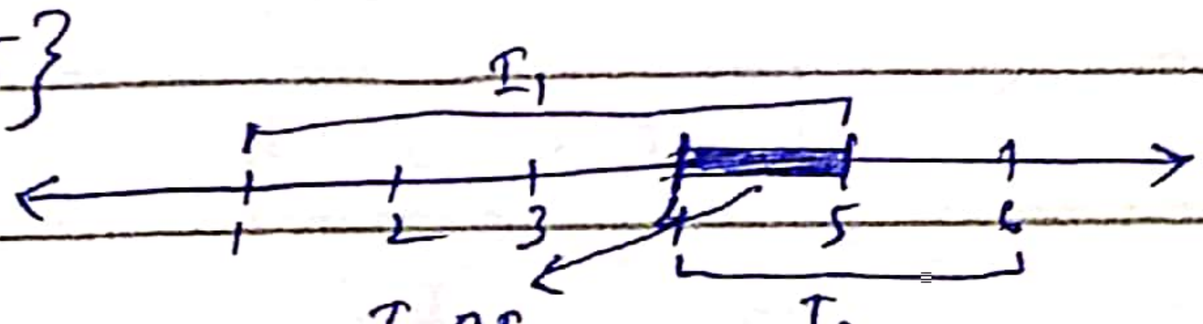
$I_1 \cap I_2 = [1,3]$



Example: $I_1 = [1,5]$, $I_2 = [4,6]$

$I_1 \cap I_2 = [1,5] \cap [4,6] = [\max\{1,4\}, \min\{5,6\}]$

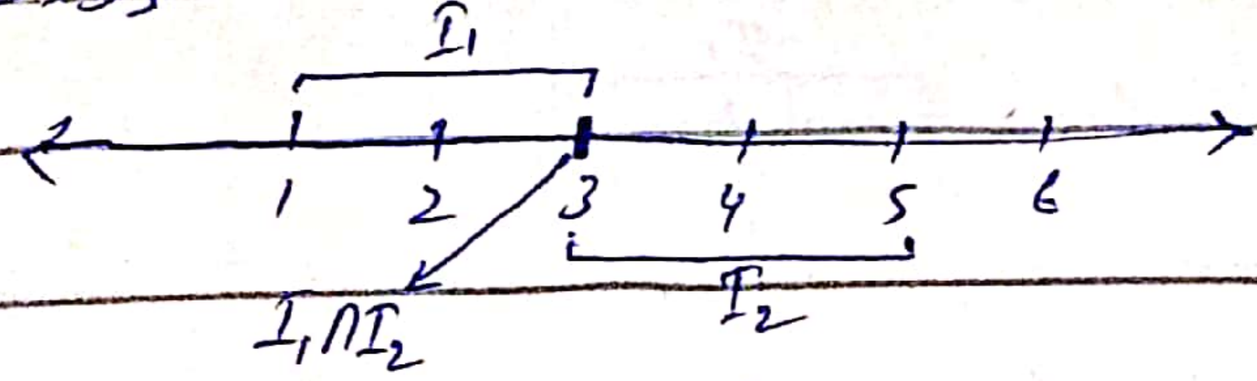
$I_1 \cap I_2 = [4,5]$



Example: $I_1 = [1,3]$, $I_2 = [3,5]$

$I_1 \cap I_2 = [1,3] \cap [3,5] = [\max\{1,3\}, \min\{3,5\}]$

$I_1 \cap I_2 = [3,3] = 3$



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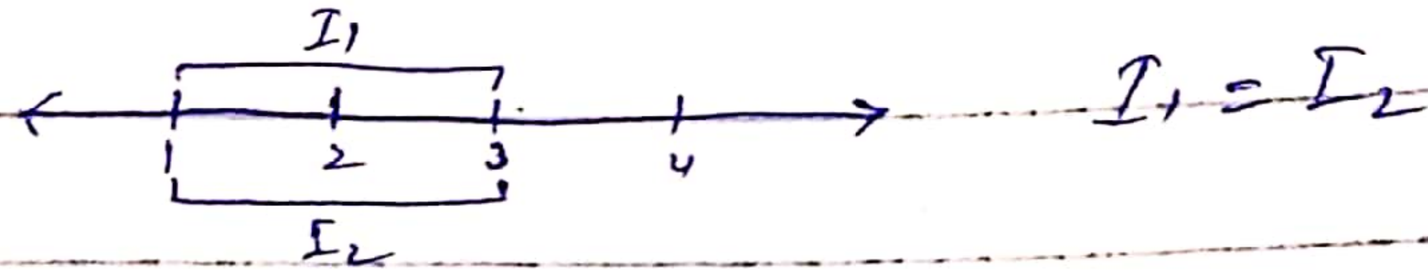
Interval Equality:

Two intervals $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ are said to be equal, denoted by $[a_1, b_1] = [a_2, b_2]$ if and only if $a_1 = a_2$ and $b_1 = b_2$.

Example: $I_1 = [1, 3]$, $I_2 = [1, 2+1]$

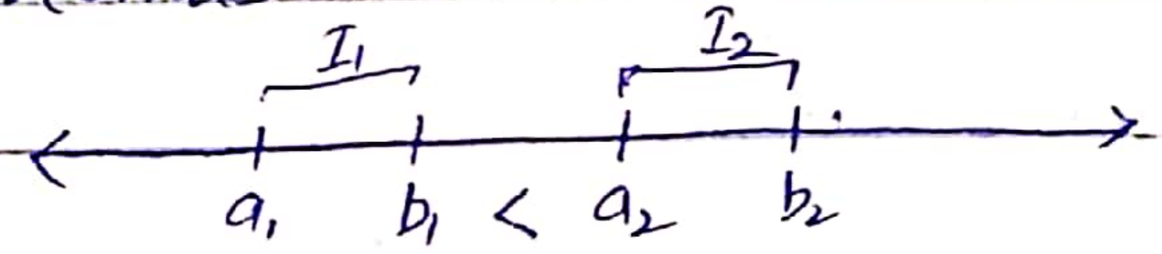
$1 = 1$, $3 = 2+1$



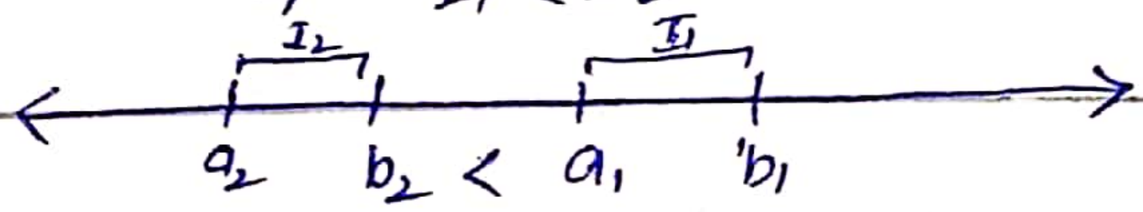
$I_1 = I_2$

Interval Inequality:

Interval $I_1 = [a_1, b_1]$ is said to be less than (respectively, greater than) interval $I_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (respectively, $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (respectively, $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

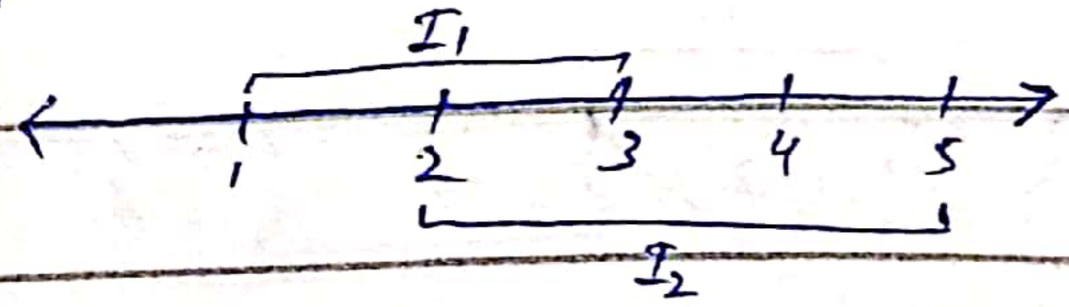


$\Rightarrow I_1 < I_2$



$\Rightarrow I_2 < I_1$ or $I_1 > I_2$

Example: $I_1 = [1, 3]$, $I_2 = [2, 5]$



Cannot be compared.

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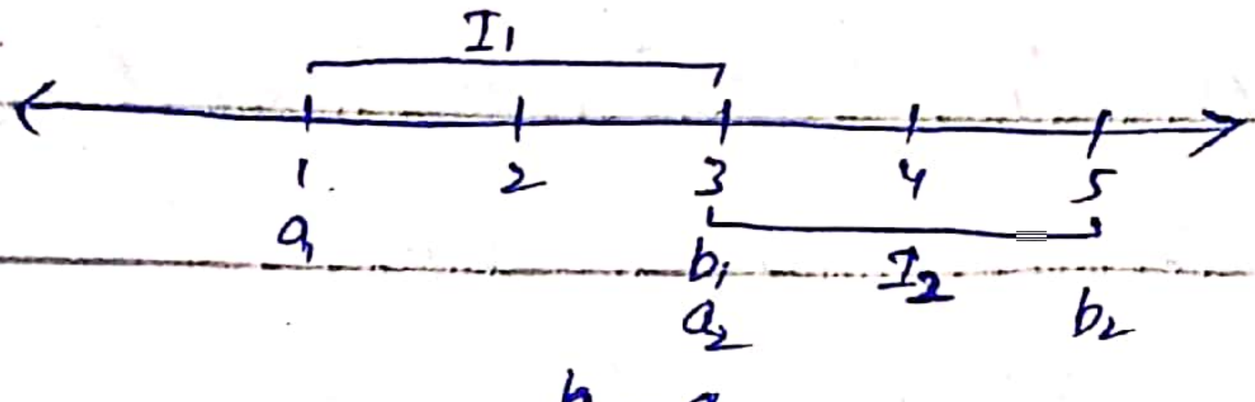
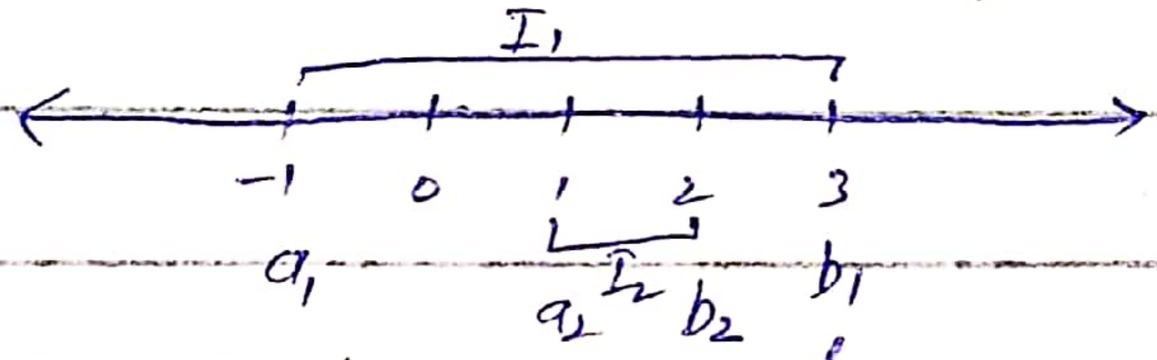
Example: $I_1 = [1,3]$, $I_2 = [3,5]$



$b_1 = a_2$

So cannot be compared.

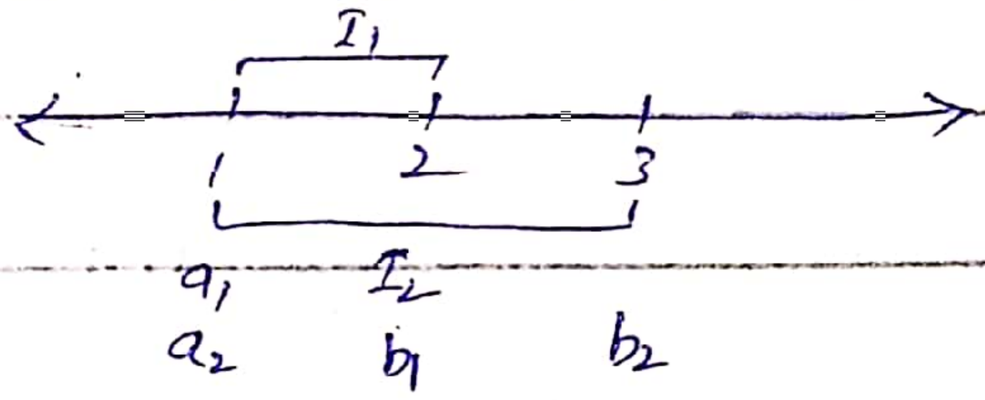
Example: $I_1 = [-1,3]$, $I_2 = [1,2]$



$b_1 \neq a_2$ and $a_1 \neq b_2$

So cannot be compared.

Example: $I_1 = [1,2]$, $I_2 = [1,3]$



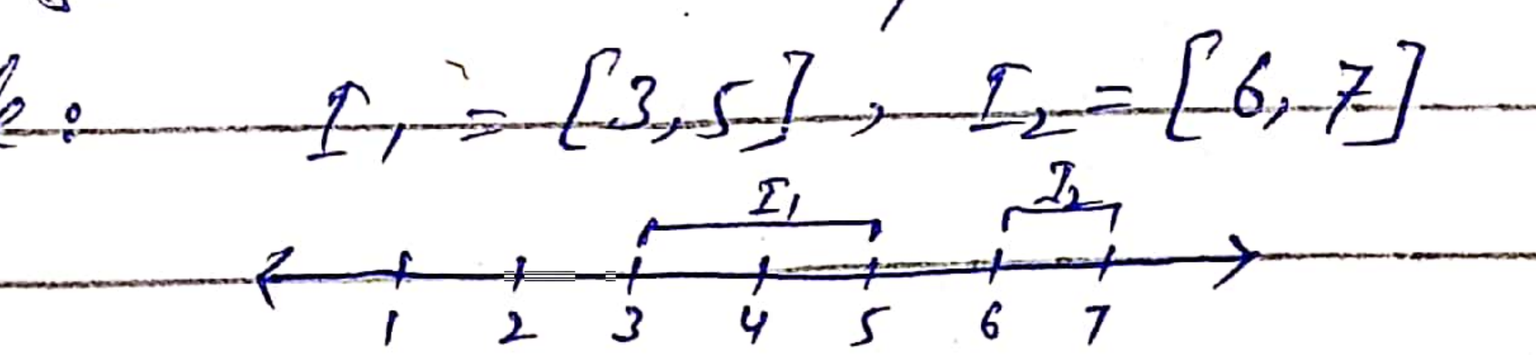
$b_1 \neq a_2$ and $a_1 \neq b_2$

So cannot be compared.

Example: $I_1 = [3,5]$, $I_2 = [6,7]$

So $I_1 < I_2$

Example: $I_3 = [5,6]$, $I_4 = [9,12]$



So $I_3 < I_4$

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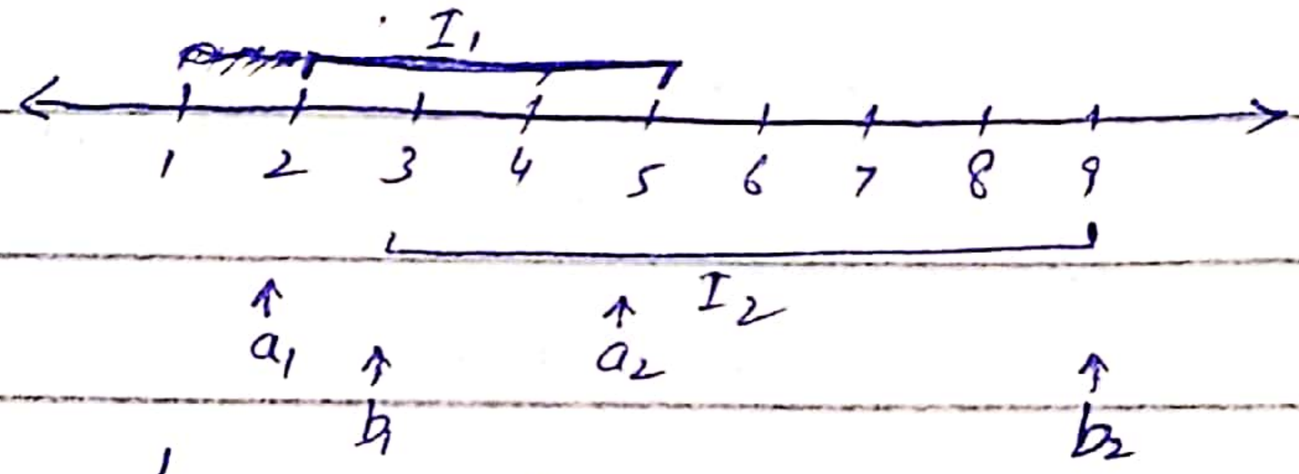
Interval Inclusion:

The interval $I_1 = [a_1, b_1]$ is said to be included in $I_2 = [a_2, b_2]$, denoted by $[a_1, b_1] \subseteq [a_2, b_2]$ if and only if $a_2 \leq a_1$ and $b_1 \leq b_2$. This is equivalent to saying that the interval $I_1 = [a_1, b_1]$ is a subset or subinterval of $I_2 = [a_2, b_2]$.

Example: $I_1 = [2, 5]$, $I_2 = [3, 9]$

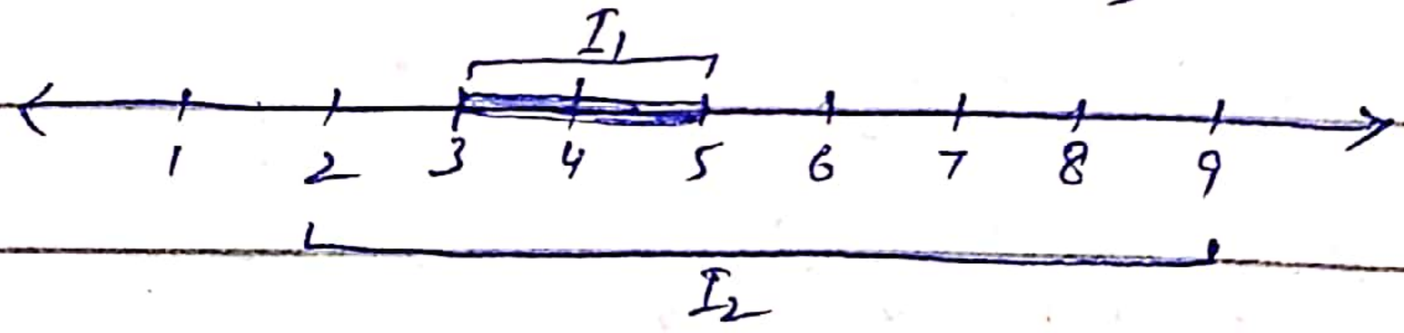


$a_2 \nleq a_1$ and $b_1 < b_2$

So I_1 is not included in I_2

i.e. $I_1 \nsubseteq I_2$

Example: $I_1 = [3, 5]$, $I_2 = [2, 9]$



a_2 a_1 b_1 b_2

$a_2 < a_1$ and $b_1 < b_2$

So $I_1 \subseteq I_2$

Interval Theorem:

The addition and multiplication operations of intervals are commutative and associative but not distributive. More pricisely:

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(1) $X+Y=Y+X$ } commutative

(2) $X \times Y = Y \times X$ }

(3) $X+(Y+Z)=(X+Y)+Z$ } associative

(4) $X \times (Y \times Z) = (X \times Y) \times Z$ }

(5) $X+0=0+X=X$; where $0=[0,0]$

(6) $X \times 1 = 1 \times X = X$; where $1=[1,1]$

(7) $Z(X+Y) \neq ZX+ZY$, except when

(a) $Z=[z,z]$ is a point ; or

(b) $X=Y=0$; or

(c) $xy \geq 0$ for all $x \in X$ and $y \in Y$.

In general, we only have the subdistributive law : $Z(X+Y) \subseteq ZX+ZY$

proof (1)

$$X+Y=\{x+y \mid x \in X;\ y \in Y\}$$
$$=\{y+x \mid y \in Y;\ x \in X\}$$
$$=Y+X$$

proof (2)

$$X \times Y=\{x \times y \mid x \in X;\ y \in Y\}$$
$$=\{y \times x \mid y \in Y;\ x \in X\}$$
$$=Y \times X$$

proof (3)

$$X+(Y+Z)=\{x+a \mid x \in X;\ a \in (Y+Z)\}$$
$$=\{x+(y+z) \mid x \in X;\ y \in Y;\ z \in Z\}$$

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$$X+(Y+Z)=\{(x+y)+z \mid x\in X, y\in Y, z\in Z\}$$
$$=\{b+z \mid b\in(X+Y),\ z\in Z\}$$
$$=(X+Y)+Z$$

proof (4)

$$X\times(Y\times Z)=\{x\times a \mid x\in X,\ a\in(Y\times Z)\}$$
$$=\{x\times(y\times z) \mid x\in X, y\in Y, z\in Z\}$$
$$=\{(x\times y)\times z \mid x\in X, y\in Y, z\in Z\}$$
$$=\{b\times z \mid b\in(X\times Y),\ z\in Z\}$$
$$=(X\times Y)\times Z$$

proof (5)

$$X+0=\{x+o \mid x\in X,\ o\in 0\}$$
$$=\{o+x \mid o\in 0,\ x\in X\}=\{x \mid x\in X\}$$
$$=0+X \qquad =X$$

and $$X\times 0=\{x\times o \mid x\in X, o\in 0\}$$
$$=\{o\times x \mid o\in 0,\ x\in X\}=\{o \mid o\in 0\}$$
$$=0\times X \qquad =0$$

proof (6)

$$X\times 1=\{x\times 1 \mid x\in X,\ 1\in 1\}$$
$$=\{1\times x \mid 1\in 1,\ x\in X\}=\{x \mid x\in X\}$$
$$=1\times X \qquad =X$$

proof (7)(a)

$$z(X+Y)=\{z\cdot a \mid a\in(X+Y)\}$$
$$=\{z\cdot(x+y) \mid x\in X,\ y\in Y\}$$
$$=\{(z\cdot x+z\cdot y) \mid x\in X,\ y\in Y\}$$
$$=zX+zY$$

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proof (7) (b)

$$z(0+0) = z0 = 0 = z0 + z0$$

Lecture # 8

Convex set:

The term convex is applicable to a set A in R^n (non dimensional Euclidian vector space) if the followings are satisfied:

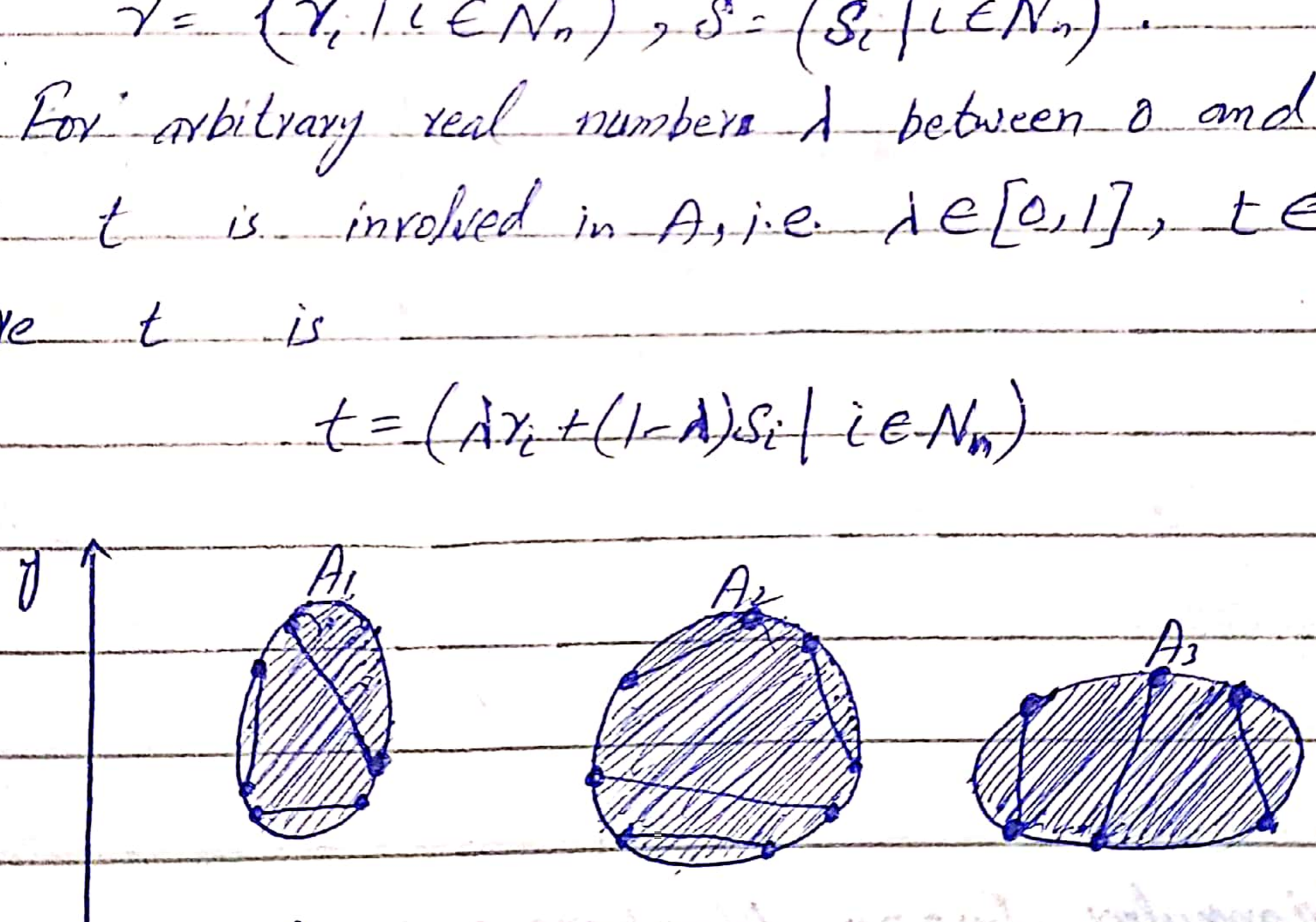
i) Two arbitrary points s and r are defined in A i.e. $s, r \in A$.

$$r = (r_i \mid i \in N_n), \quad s = (s_i \mid i \in N_n).$$

ii) For arbitrary real numbers λ between 0 and 1, point t is involved in A, i.e. $\lambda \in [0,1]$, $t \in A$.

where t is

$$t = (\lambda r_i + (1-\lambda) s_i \mid i \in N_n)$$



A_1, A_2 and A_3 are convex set. A_4, A_5 and A_6 are not convex set.

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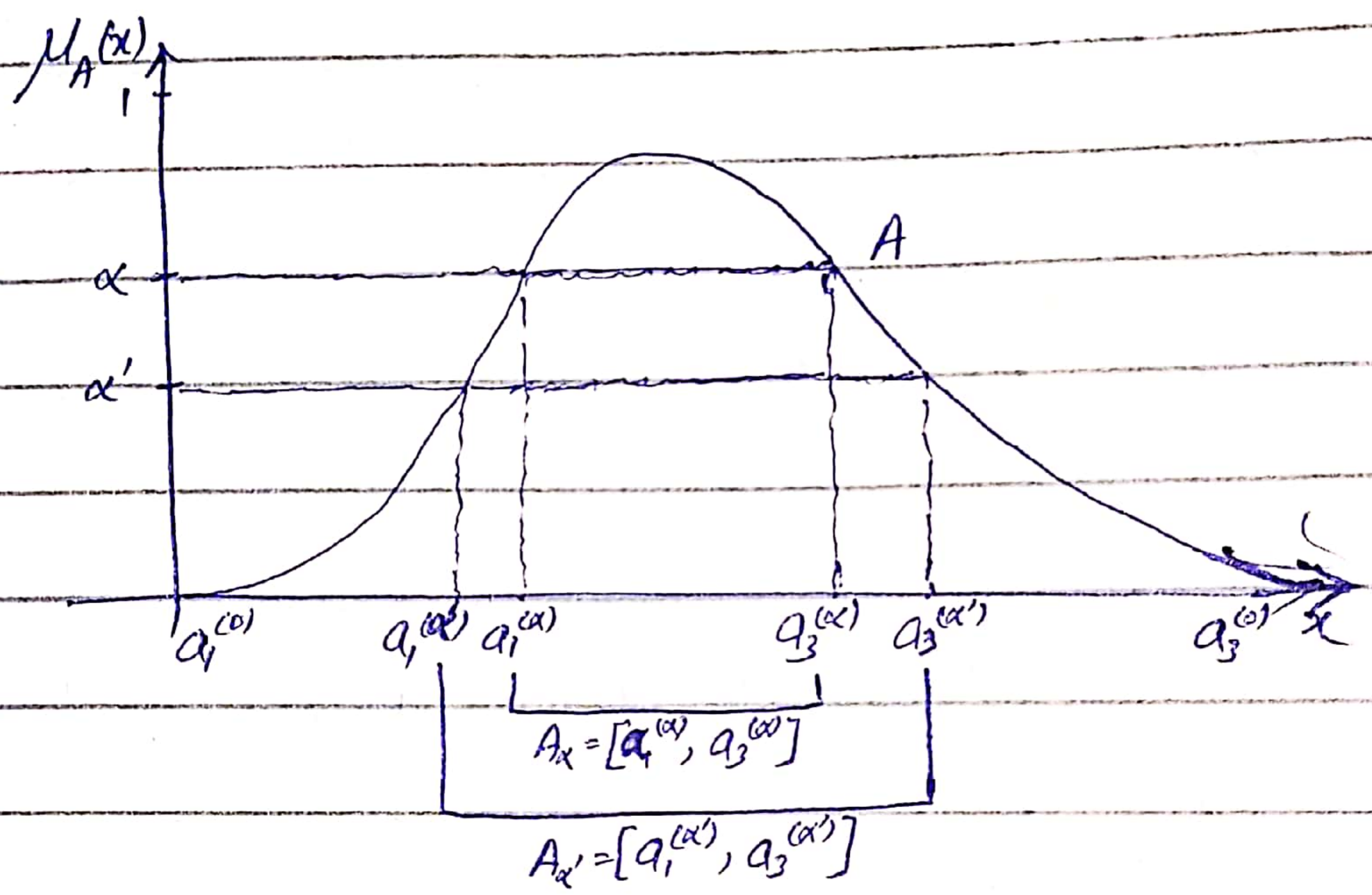
Convex Fuzzy Set:

The convex ~~cod~~ condition is that the line by α-cut is continuous and α-cut interval satisfies the following relation.

$$A_\alpha = [a_1^{(\alpha)}, a_3^{(\alpha)}]$$

$$(\alpha' < \alpha) \Rightarrow (a_1^{(\alpha')} \leq a_1^{(\alpha)}, a_3^{(\alpha')} \geq a_3^{(\alpha)})$$



α-cut of fuzzy number $(\alpha' < \alpha) \Rightarrow (A_\alpha \subset A_{\alpha'})$

The convex condition may also be written as,

$$(\alpha' < \alpha) \Rightarrow (A_\alpha \subset A_{\alpha'})$$

Triangular Fuzzy Numbers:

A fuzzy number represented with three points is called fuzzy number. It defined as follows:

$$\tilde{A} = (a_1, a_2, a_3).$$

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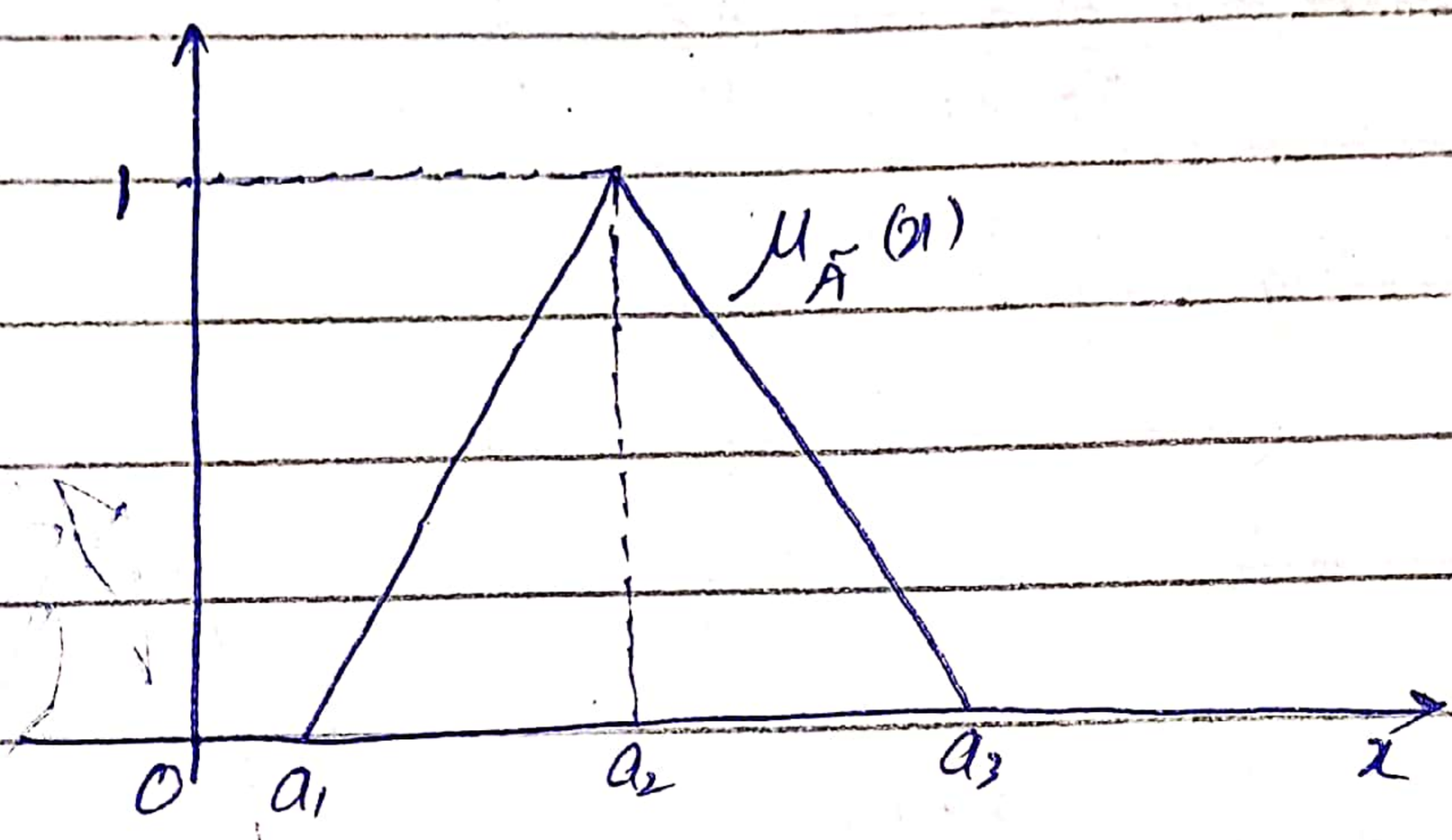
$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; \ x < a_1 \\ \frac{x-a_1}{a_2-a_1} & ; \ a_1 \le x \le a_2 \\ \frac{a_3-x}{a_3-a_2} & ; \ a_2 \le x \le a_3 \\ 0 & ; \ x > a_3 \end{cases}$$



Example:

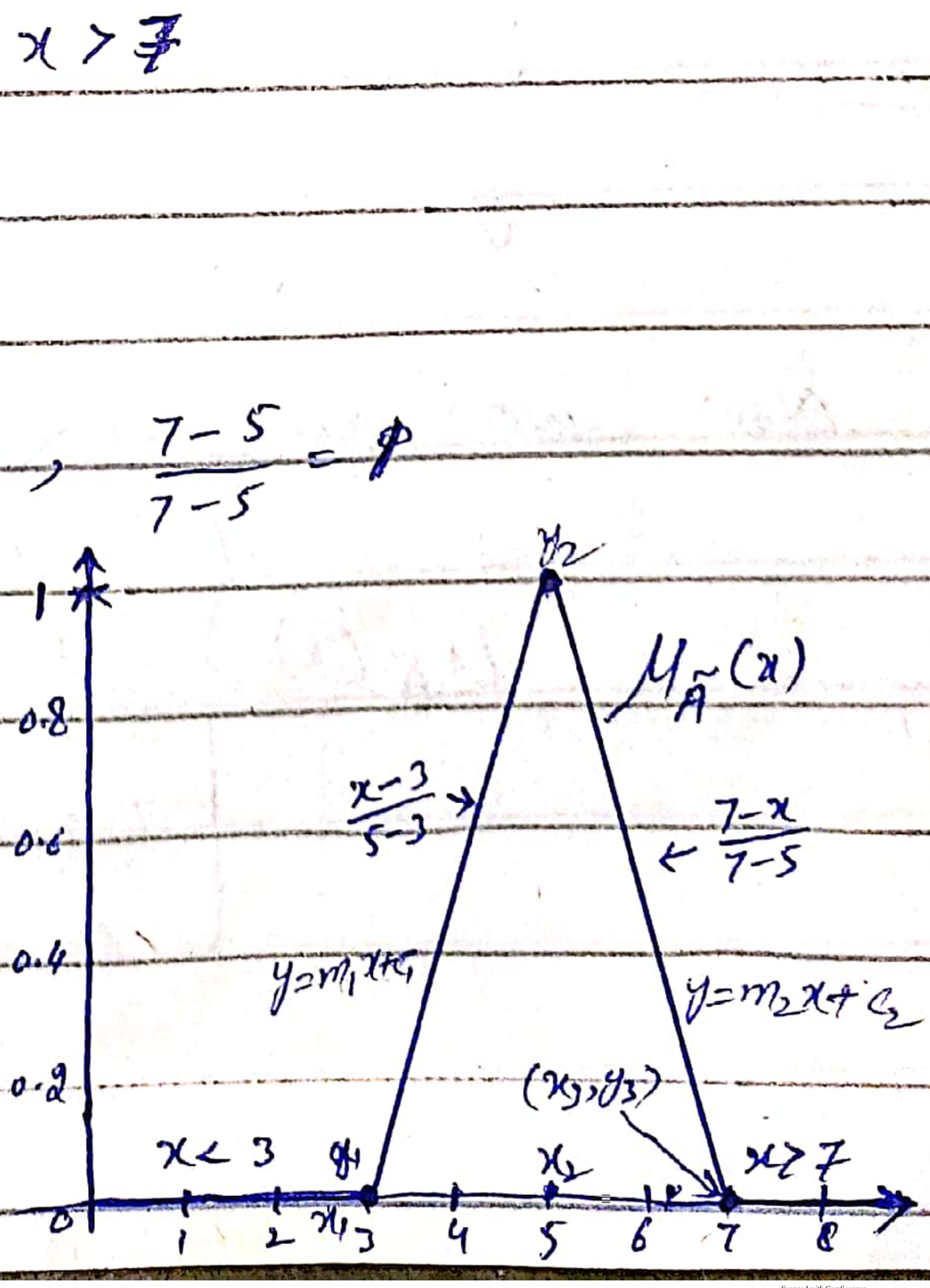
$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; \ x < 3 \\ \frac{x-3}{5-3} & ; \ 3 \le x \le 5 \\ \frac{7-x}{7-5} & ; \ 5 \le x \le 7 \\ 0 & ; \ x > 7 \end{cases}$$

$$\tilde{A} = (3, 5, 7)$$

put $x = 3$, $\frac{3-3}{5-3} = 0$

put $x = 5$, $\frac{5-3}{5-3} = 1$, $\frac{7-5}{7-5} = 1$

put $x = 7$, $\frac{7-7}{7-5} = 0$



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If graph is given then, How to find function?

$$y = mx + c$$

↙ slope

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\text{Rise}}{\text{Run}}$$

$$m_1 = \frac{1-0}{5-3} = \frac{1}{2}$$

$$m_2 = \frac{0-1}{7-5} = -\frac{1}{2}$$

$$y = \frac{1}{2}x + c_1$$

put $(x_1, y_1) = (3, 0)$

$$0 = \frac{1}{2}(3) + c_1$$

$$\boxed{c_1 = \frac{-3}{2}}$$

$$y = \frac{1}{2}x - \frac{3}{2}$$

$$y = \frac{x-3}{2} = \frac{x-3}{5-3}$$

Now

$$y = -\frac{1}{2}x + c_2$$

put $(x_3, y_3) = (7, 0)$

$$0 = -\frac{1}{2}(7) + c_2$$

$$\boxed{c_2 = \frac{7}{2}}$$

$$y = -\frac{1}{2}x + \frac{7}{2}$$

$$y = \frac{7-x}{2} = \frac{7-x}{7-5}$$

Now we can write function:

$$\mu_A(x) \begin{cases} 0 & ; \quad x < 3 \\ \frac{x-3}{5-3} & ; \quad 3 \le x \le 5 \\ \frac{7-x}{7-5} & ; \quad 5 \le x \le 7 \\ 0 & \quad 7 < x \end{cases}$$

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Lecture # 9

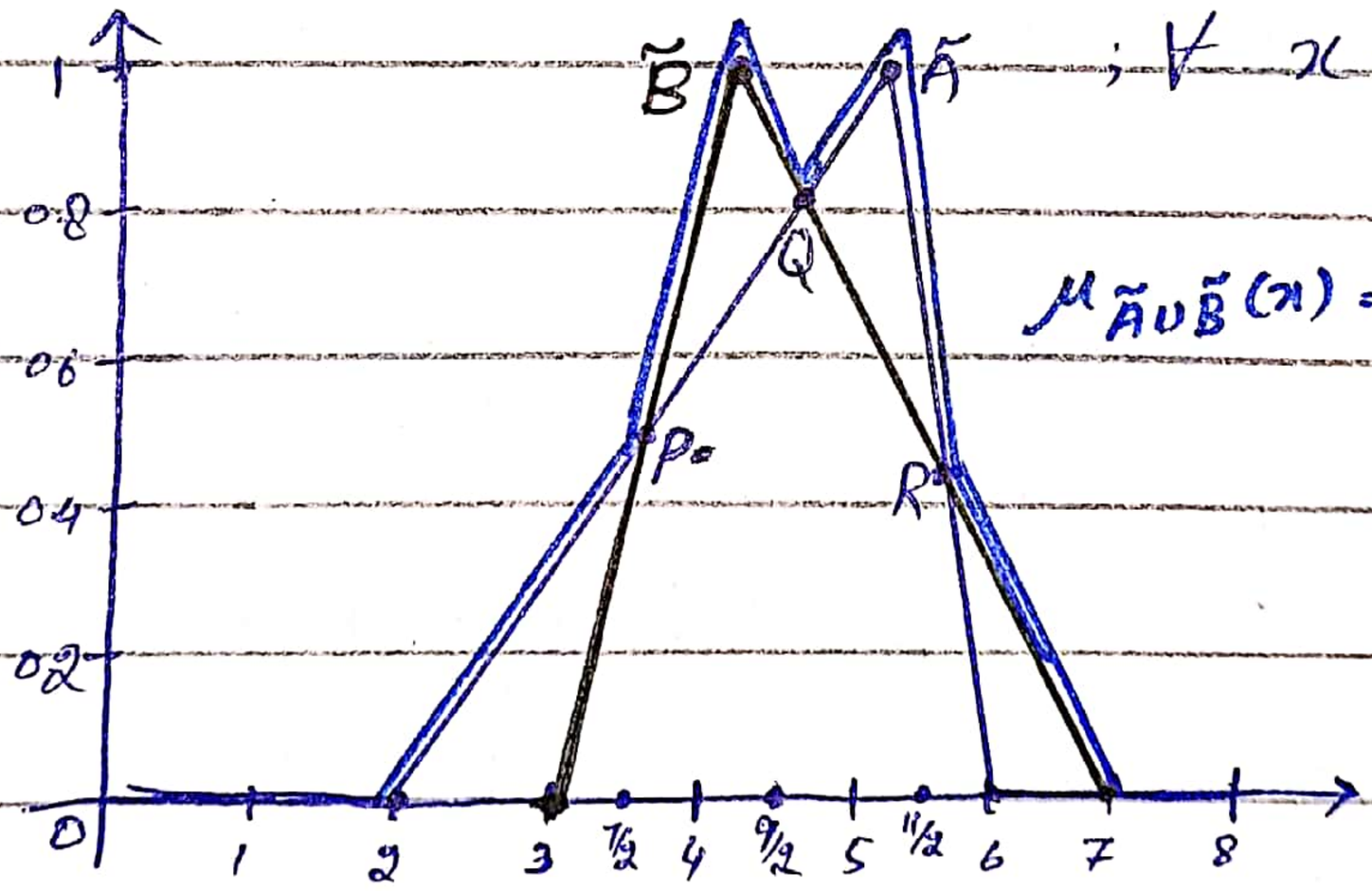
Union and Intersection of Triangular Fuzzy Numbers:

$\tilde{A} = (2, 5, 6)$, $\tilde{B} = (3, 4, 7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; \ x < 2 \\ \frac{x-2}{5-2} & ; \ 2 \le x \le 5 \\ \frac{6-x}{6-5} & ; \ 5 \le x \le 6 \\ 0 & ; \ x > 6 \end{cases} \qquad \mu_{\tilde{B}}(x) = \begin{cases} 0 & ; \ x < 3 \\ \frac{x-3}{4-3} & ; \ 3 \le x \le 4 \\ \frac{7-x}{7-4} & ; \ 4 \le x \le 7 \\ 0 & ; \ x > 7 \end{cases}$$

By definition of Union:

$$\mu_{\tilde{A} \cup \tilde{B}}(x) = \mu_{\tilde{C}}(x) = \max \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x) \quad ; \forall \ x \in X$$



$$\mu_{\tilde{A} \cup \tilde{B}}(x) = \begin{cases} 0 & ; \ x < 2 \\ \frac{x-2}{5-2} & ; \ 2 \le x \le \frac{7}{2} \\ \frac{x-3}{4-3} & ; \ \frac{7}{2} \le x \le 4 \\ \frac{7-x}{7-4} & ; \ 4 \le x \le \frac{9}{2} \\ \frac{x-2}{5-2} & ; \ \frac{9}{2} \le x \le 5 \\ \frac{6-x}{6-5} & ; \ 5 \le x \le \frac{11}{2} \\ \frac{7-x}{7-4} & ; \ \frac{11}{2} \le x \le 7 \\ 0 & ; \ 7 < x \end{cases}$$

P → $\frac{x-2}{5-2} = \frac{x-3}{4-3}$

$\frac{x-2}{3} = \frac{x-3}{1}$

$x - 2 = 3x - 9$

$9 - 2 = 3x - x$

$\frac{7}{2} = x$

Q → $\frac{x-2}{5-2} = \frac{7-x}{7-4}$

$3(x-2) = 3(7-x)$

$x + x = 7 + 2$

$2x = 9$

$x = 9/2$

R → $\frac{6-x}{6-5} = \frac{7-x}{7-4}$

$3(6-x) = 7 - x$

$18 - 3x = 7 - x$

$18 - 7 = 3x - x$

$x = 11/2$

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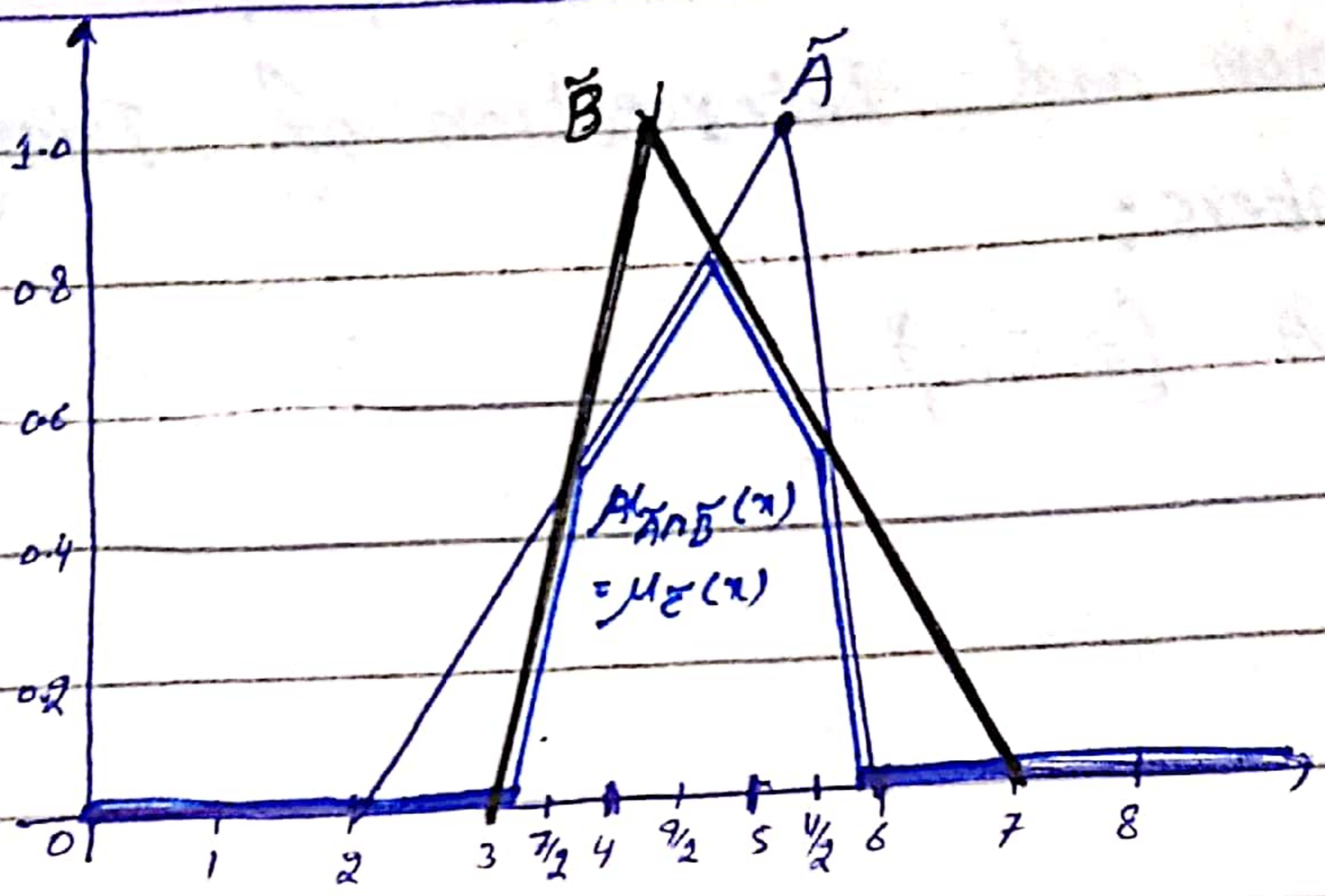

$$\mu_{\tilde{A}\cap\tilde{B}}(x)=\mu_{\tilde{C}}(x)=\begin{cases}0 & ; \quad x<3\\ \frac{x-3}{4-3} & ; \quad 3\le x\le 7/2\\ \frac{x-2}{5-2} & ; \quad 7/2\le x\le 9/2\\ \frac{7-x}{7-4} & ; \quad 9/2\le x\le 11/2\\ \frac{6-x}{6-5} & ; \quad 11/2\le x\le 6\\ 0 & ; \quad 6<x\end{cases}$$

Addition of Triangular fuzzy numbers:

Let $\tilde{A}=(a_1,b_1,c_1)$ and $\tilde{B}=(a_2,b_2,c_2)$ be two triangular fuzzy numbers, then their addition is defined as

$$\tilde{A}+\tilde{B}=(a_1,b_1,c_1)+(a_2,b_2,c_2)=$$

$$=(a_1+a_2,\ b_1+b_2,\ c_1+c_2)$$

Example: $\tilde{A}=(2,3,5)$, $\tilde{B}=(4,7,8)$

$$\tilde{A}+\tilde{B}=(2+3,5)+(4,7,8)$$

$$=(2+4,3+7+5+8)$$

$$\tilde{A}+\tilde{B}=(6,10,11)$$

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Negative of Triangular Fuzzy number:

Let $\tilde{A} = (a, b, c)$ be a triangular fuzzy number, then $-\tilde{A}$ is defined as

$$-\tilde{A} = (-c, -b, -a)$$

Example: $\tilde{A} = (3, 5, 9)$; $a \leq b \leq c$

$-\tilde{A} = (-9, -5, -3)$; $-9 \leq -5 \leq -3$

$-\tilde{A} = (-3, -5, -9)$; $-3 \not\leq -5 \not\leq -9$ Does not Hold

Lecture # 10

Subtraction of Triangular fuzzy numbers:

Let $\tilde{A} = (a_1, b_1, c_1)$ and $\tilde{B} = (a_2, b_2, c_2)$ be two triangular fuzzy numbers, then their subtraction is defined as: $\tilde{A} - \tilde{B} = (a_1, b_1, c_1) - (a_2, b_2, c_2)$

$$= (a_1 - c_2, b_1 - b_2, c_1 - a_2)$$

Examples: $\tilde{A} = (4, 7, 8)$, $\tilde{B} = (2, 3, 5)$

$\tilde{A} - \tilde{B} = (4, 7, 8) - (2, 3, 5)$

$= (4, 7, 8) + (-5, -3, -2)$

$= (4-5, 7-3, 8-2)$

$= (-1, 4, 6)$

Multiplication of Triangular fuzzy numbers:

Let $\tilde{A} = (a_1, b_1, c_1)$ and $\tilde{B} = (a_2, b_2, c_2)$ be two triangular fuzzy numbers, then their multiplication is defined as: $\tilde{A} \times \tilde{B} = (a_1, b_1, c_1) \times (a_2, b_2, c_2)$

$$= (a_1 a_2, b_1 b_2, c_1 c_2)$$

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Example: $\tilde{A} = (2,3,5)$, $\tilde{B} = (3,4,7)$

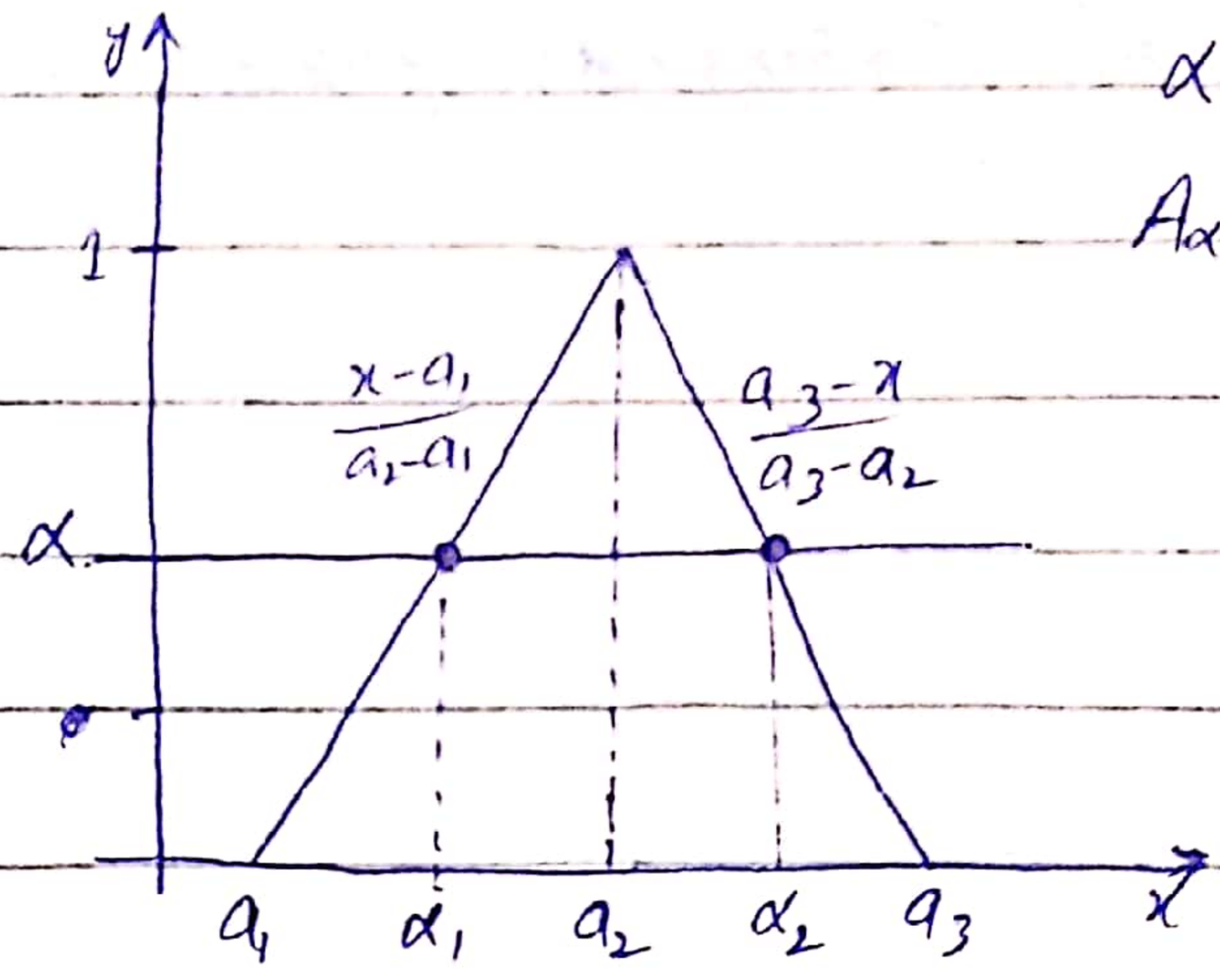
$\tilde{A} \times \tilde{B} = (2,3,5) \times (3,4,7)$

$= (2)(3), (3)(4), (5)(7)$

$= (6, 12, 35)$

α cut of triangular fuzzy numbers:

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; x < a_1, \\ \dfrac{x-a_1}{a_2-a_1} & ; a_1 \le x \le a_2 \\ \dfrac{a_3-x}{a_3-a_2} & ; a_2 \le x \le 3 \\ 0 & ; x \ge a_3 \end{cases}$$



$\alpha \in [0,1]$

$A_\alpha \in [\alpha_1, a_2]$

$$\alpha = \frac{x-a_1}{a_2-a_1}$$

$$\alpha(a_2-a_1) = x-a_1$$

$$\boxed{a_1 + \alpha(a_2-a_1) = x = \alpha_1}$$

$$\alpha = \frac{a_3-x}{a_3-a_2}$$

$$\alpha(a_3-a_2) = a_3-x$$

$$\boxed{\alpha_2 = x = a_3 - \alpha(a_3-a_2)}$$

$$A_\alpha = [\alpha_1, \alpha_2]$$

$$A_\alpha = [a_1 + \alpha(a_2-a_1),\ a_3 - \alpha(a_3-a_2)]$$

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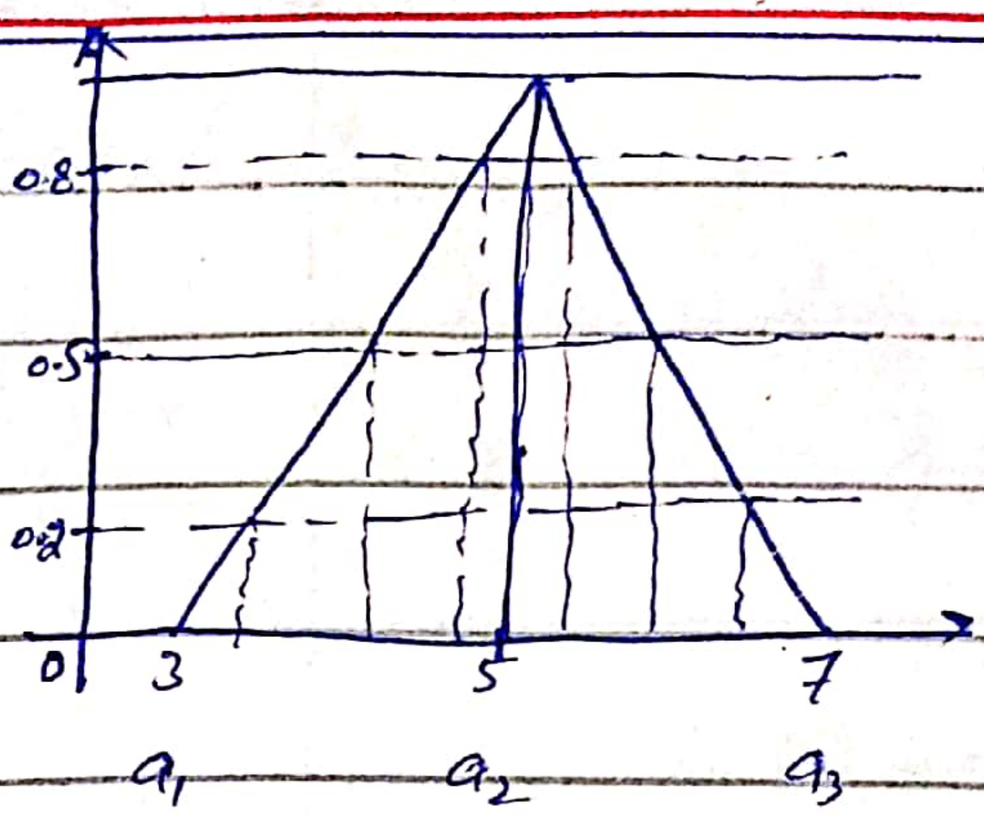
Example:

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; x < 3 \\ \frac{x-3}{5-3} & ; 3 \le x \le 5 \\ \frac{7-x}{7-5} & ; 5 \le x \le 7 \\ 0 & ; 7 < x \end{cases}$$



$\tilde{A} = (3, 5, 7)$

$A_\alpha = [\alpha_1, \alpha_2] = [a_1 + \alpha(a_2 - a_1), a_3 - \alpha(a_3 - a_2)]$

$\alpha = 0.2, 0.5, 0.8$

$A_{0.2} = [3 + 0.2(5-3), 7 - 0.2(7-5)] = [3.4, 6.6]$

$A_{0.5} = [3 + 0.5(5-3), 7 - 0.5(7-5)] = [4, 6]$

$A_{0.8} = [3 + 0.8(5-3), 7 - 0.8(7-5)] = [4.6, 5.4]$

$A_{0.2} \supset A_{0.5} \supset A_{0.8}$

$A_\alpha, A_{\alpha'}$

$\alpha \le \alpha' \Rightarrow A_\alpha \supseteq A_{\alpha'}$

Trapezoidal Fuzzy Number:

A fuzzy number represented with four points is called trapezoidal fuzzy number. It is defined as follows:

$$\tilde{A} = (a_1, a_2, a_3, a_4)$$

This fuzzy number is interpreted as fuzzy numbers

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & ; x < a_1 \\ \frac{x-a_1}{a_2-a_1} & ; a_1 \le x \le a_2 \\ 1 & ; a_2 \le x \le a_3 \\ \frac{a_4-x}{a_4-a_3} & ; a_3 \le x \le a_4 \\ 0 & ; a_4 < x \end{cases}$$

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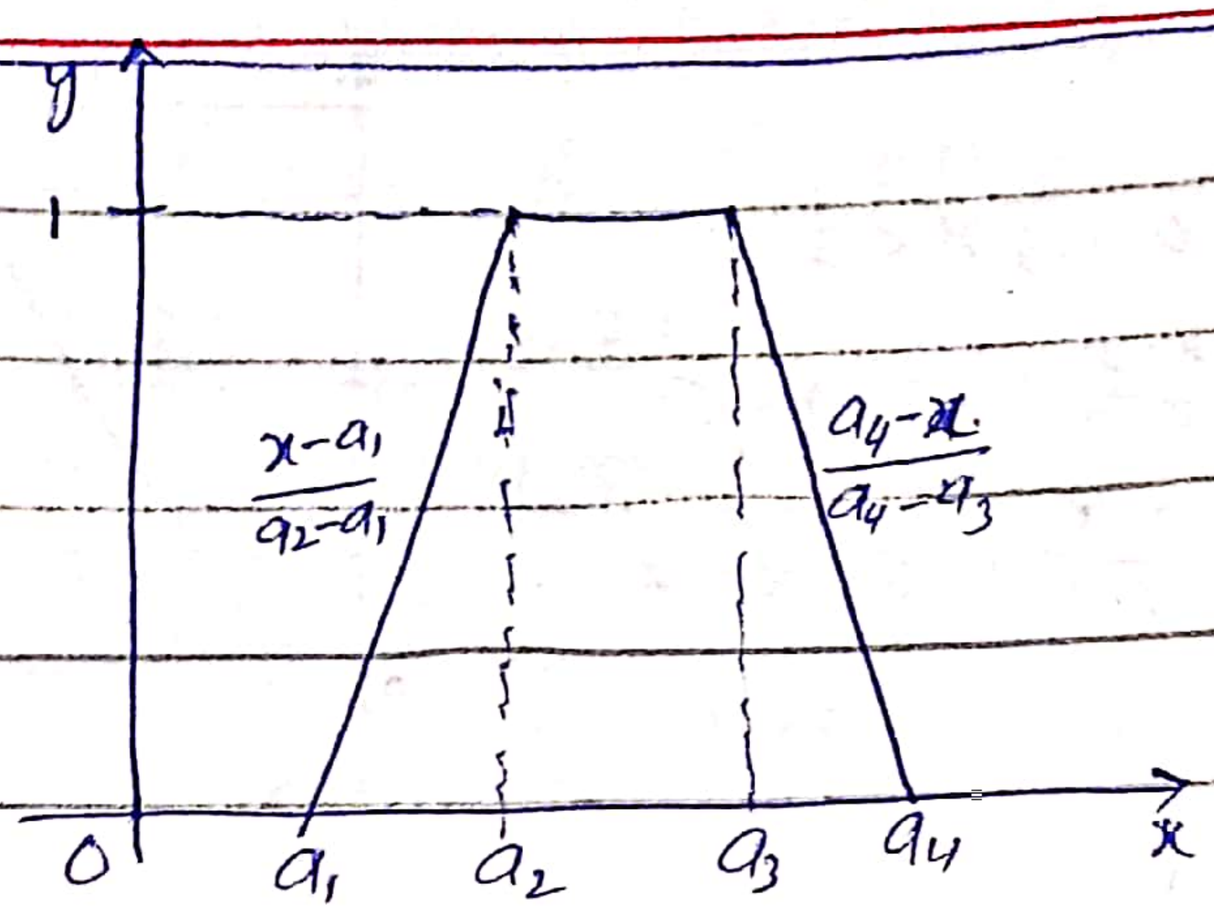

Note: If $a_2 = a_3$, Trapezoidal fuzzy number becomes a Triangular fuzzy number.

Addition of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their addition is defined as

$$\tilde{A} + \tilde{B} = (a_1, b_1, c_1, d_1) + (a_2, b_2, c_2, d_2)$$

$$= (a_1 + a_2, b_1 + b_2, c_1 + c_2, d_1 + d_2)$$

Example: $\tilde{A} = (2, 4, 5, 9)$; $\tilde{B} = (3, 5, 7, 8)$

$$\tilde{A} + \tilde{B} = (2+3, 4+5, 5+7, 9+8) = (5, 9, 12, 17)$$

$$\tilde{A}(x) = \begin{cases} 0 & ; \ x < 2 \\ \frac{x-2}{4-2} & ; \ 2 \le x \le 4 \\ 1 & ; \ 4 \le x \le 5 \\ \frac{9-x}{9-5} & ; \ 5 \le x \le 9 \\ 0 & ; \ 9 < x \end{cases}$$

$$\tilde{B}(x) = \begin{cases} 0 & ; \ x < 3 \\ \frac{x-3}{5-3} & ; \ 3 \le x \le 5 \\ 1 & ; \ 5 \le x \le 7 \\ \frac{8-x}{8-7} & ; \ 7 \le x \le 8 \\ 0 & ; \ 8 < x \end{cases}$$

$$\tilde{A} + \tilde{B} = \begin{cases} 0 & ; \ x < 5 \\ \frac{x-5}{9-5} & ; \ 5 \le x \le 9 \\ 1 & ; \ 9 \le x \le 12 \\ \frac{17-x}{17-12} & ; \ 12 \le x \le 17 \\ 0 & ; \ 17 < x \end{cases}$$

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Lecture # 11

Scaler Addition of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ be a trapezoidal fuzzy number, then its addition with a scaler k is defined by

$$\tilde{A} + k = (a_1 + k, b_1 + k, c_1 + k, d_1 + k), \quad k \in R$$

Example: $\tilde{A} = (2, 4, 5, 9)$, $k = 3$

$$\tilde{A} + k = (2+3, 4+3, 5+3, 9+3)$$

$$= (5, 7, 8, 12)$$

Negative of a Trapezoidal Fuzzy Number:

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then $-\tilde{A}$ is defined as

$$-\tilde{A} = (-d, -c, -b, -a)$$

Example: $\tilde{A} = (2, 3, 5, 8)$

$$-\tilde{A} = (-8, -5, -3, -2)$$

Subtraction of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their subtraction is defined as

$$\tilde{A} - \tilde{B} = (a_1, b_1, c_1, d_1) - (a_2, b_2, c_2, d_2)$$

$$= (a_1 - d_2, b_1 - c_2, c_1 - b_2, d_1 - a_2)$$

Example: $\tilde{A} = (2, 4, 5, 9)$, $\tilde{B} = (3, 5, 7, 8)$

$$\tilde{A} - \tilde{B} = (2, 4, 5, 9) - (3, 5, 7, 8)$$

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$\tilde{A}-\tilde{B} = (2,4,5,9) + (-8,-7,-5,-3)$

$= (2-8, 4-7, 5-5, 9-3)$

$= (-6, -3, 0, 6)$

Scaler Subtraction of Trapezoidal Fuzzy Number:

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then its subtraction with a scalar k is defined as

$$\tilde{A} - k = (a-k, b-k, c-k, d-k), \; k \in \mathbb{R}$$

Example: $\tilde{A} = (2,4,5,9)$, $k = 5$ $\quad k \in \mathbb{R}$

$\tilde{A} - k = \tilde{A} - 5$

$= (2,4,5,9) - (5)$

$= (2-5, 4-5, 5-5, 9-5)$

$\tilde{A} - k = (-3, -1, 0, 4)$

Multiplication of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their multiplication is difined as

$$\tilde{A} \times \tilde{B} = (a_1, b_1, c_1, d_1) \times (a_2, b_2, c_2, d_2)$$

$$= (a_1 a_2, b_1 b_2, c_1 c_2, d_1 d_2)$$

Example: $\tilde{A} = (1,3,4,5)$, $\tilde{B} = (2,3,5,7)$

$\tilde{A} \times \tilde{B} = (1,3,4,5) \times (2,3,5,7)$

$= (2, 9, 20, 35)$

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Scaler Multiplication of Trapezoidal fuzzy Numbers:

(i) Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number and $\alpha \in \mathbb{R}^+$, then its multiplication with α is defined as

$$\alpha\tilde{A} = (\alpha a, \alpha b, \alpha c, \alpha d)$$

Example: $\tilde{A} = (1, 3, 4, 5)$, $\alpha = 3 \in \mathbb{R}^+$

$$\alpha\tilde{A} = 3\tilde{A}$$

$$= 3(1, 3, 4, 5)$$

$$\alpha\tilde{A} = (3, 9, 12, 15)$$

(ii) Let $\tilde{A} = (a, b, c, d)$ be trapezoidal fuzzy number and $\alpha \in \mathbb{R}^-$, then its multiplication with α is defined as

$$\alpha\tilde{A} = (\alpha d, \alpha c, \alpha b, \alpha a)$$

Example: $\tilde{A} = (1, 3, 4, 5)$, $\alpha = -2 \in \mathbb{R}^-$

$$\alpha\tilde{A} = -2\tilde{A}$$

$$= -2(1, 3, 4, 5)$$

$$= (-10, -8, -6, -2)$$

Inverse of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then its inverse is defined as

$$\tilde{A}^{-1} = \left(\frac{1}{d}, \frac{1}{c}, \frac{1}{b}, \frac{1}{a}\right)$$

Example: $\tilde{A} = (1, 3, 4, 5)$

$$\tilde{A}^{-1} = \left(\frac{1}{5}, \frac{1}{4}, \frac{1}{3}, 1\right)$$

Division of Trapezoidal Fuzzy Numbers:

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their division is defined as

$$\tilde{A}/\tilde{B} = (a_1, b_1, c_1, d_1)/(a_2, b_2, c_2, d_2)$$

$$= \left(\frac{a_1}{d_2}, \frac{b_1}{c_2}, \frac{c_1}{b_2}, \frac{d_1}{a_2}\right)$$

Example: $\tilde{A} = (1, 3, 5, 6)$, $\tilde{B} = (1, 4, 6, 7)$

$$\tilde{A}/\tilde{B} = (1, 3, 5, 6)/(1, 4, 6, 7)$$

$$= \left(\frac{1}{7}, \frac{3}{6}, \frac{5}{4}, \frac{6}{1}\right)$$

$$\tilde{A}/\tilde{B} = \left(\frac{1}{7}, \frac{1}{2}, \frac{5}{4}, 6\right)$$

Lecture #12

Cartesian product or product set:

Let A and B be two non-empty sets, the product set or cartesian product $A \times B$ is defined as follows,

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$A_1, A_2, A_3, \ldots, A_n$

$$A_1 \times A_2 \times A_3 \times \cdots \times A_n = \{(a_1, a_2, \ldots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}$$

$$\prod_{i=1}^{n} A_i = A^n$$

$$A \times A = A^2, \quad R \times R \times R = R^3$$

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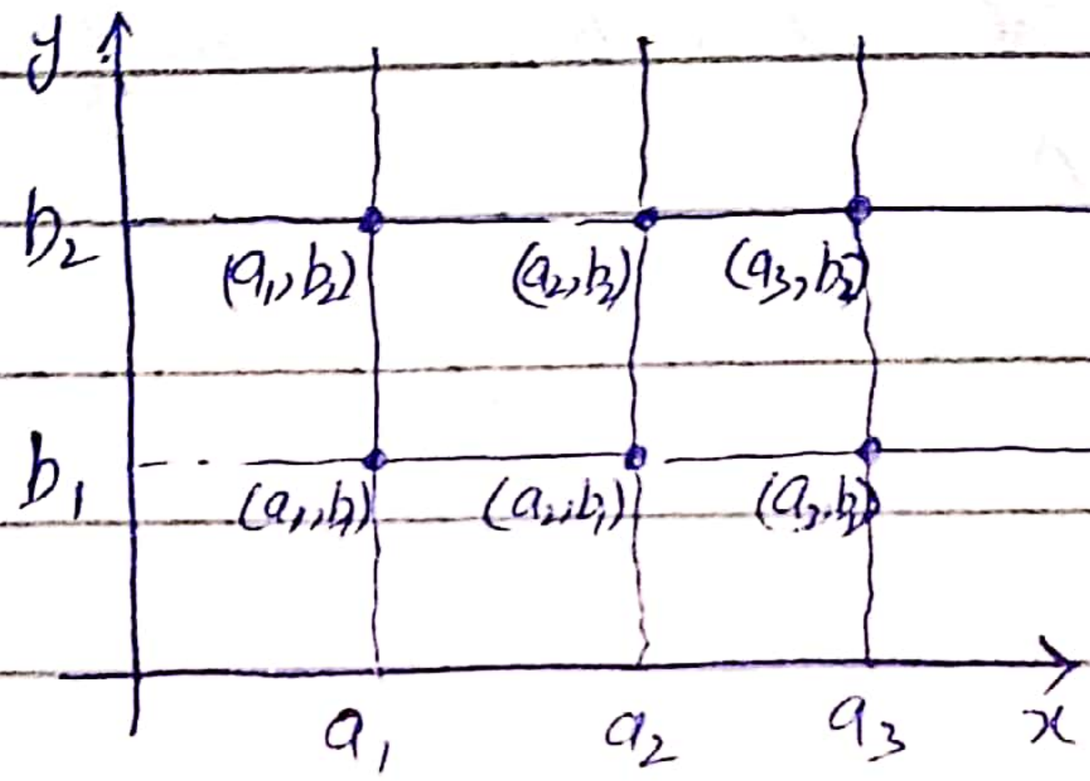
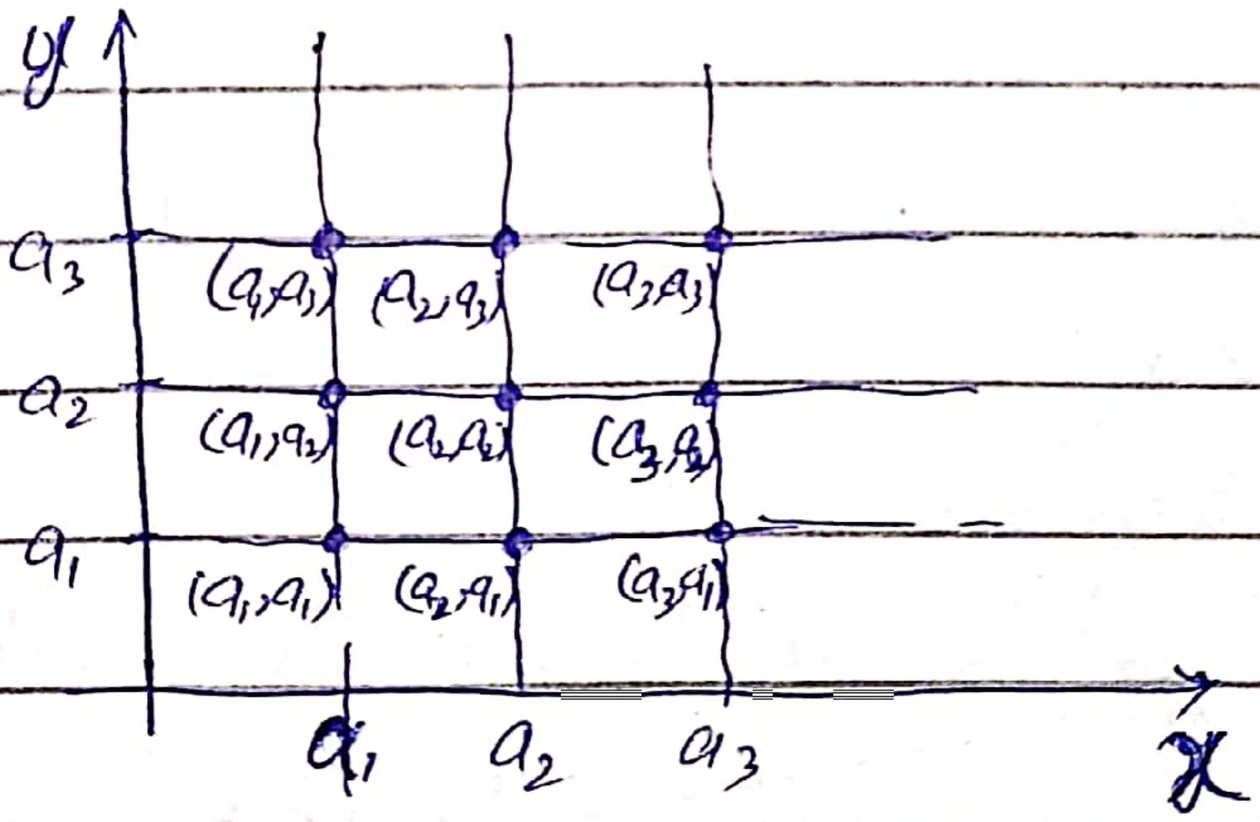
Example: When $A = \{a_1, a_2, a_3\}$, $B = \{b_1, b_2\}$, the cartesian product yields

$$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$$



Now $A \times A = \{(a_1, a_1), (a_1, a_2), (a_1, a_3), (a_1, a_1), (a_2, a_1), (a_2, a_3), (a_3, a_1), (a_3, a_2), (a_3, a_3)\}$



Binary Relation:

If A and B are two sets and there is a specific property between elements x of A and y of B, this property can be described using the ordered pair (x, y). A set of such (x, y) pairs, $x \in A$ and $y \in B$, is called a Relation R.

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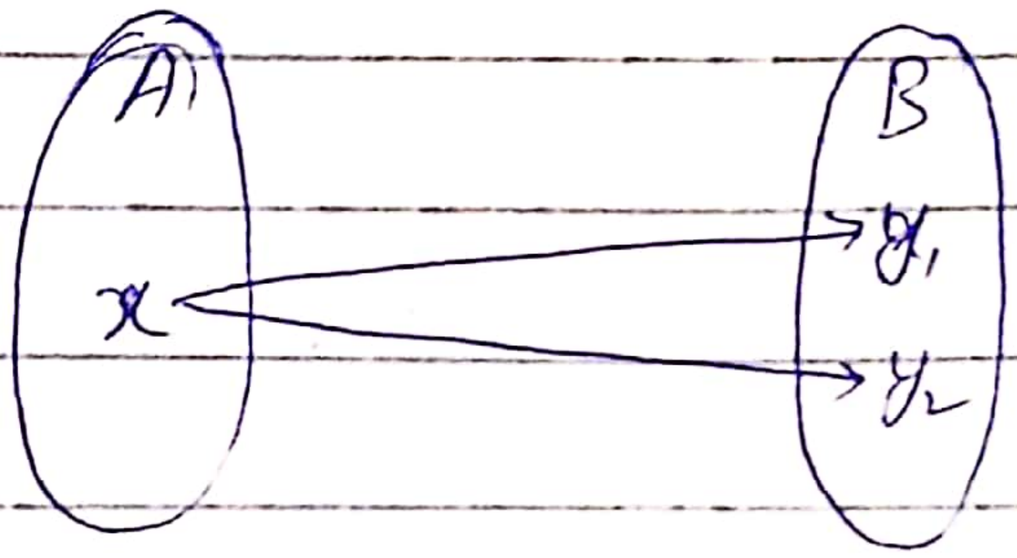
$$R = \{(x,y) \mid x \in A, y \in B\}$$

R is a binary relation and a subset of $A \times B$.

One to many:

R is said to be one-to-many if

$$\exists\, x \in A,\ y_1 \to y_2 \in B\ ;\ (x,y_1) \in R, (x,y_2) \in R$$



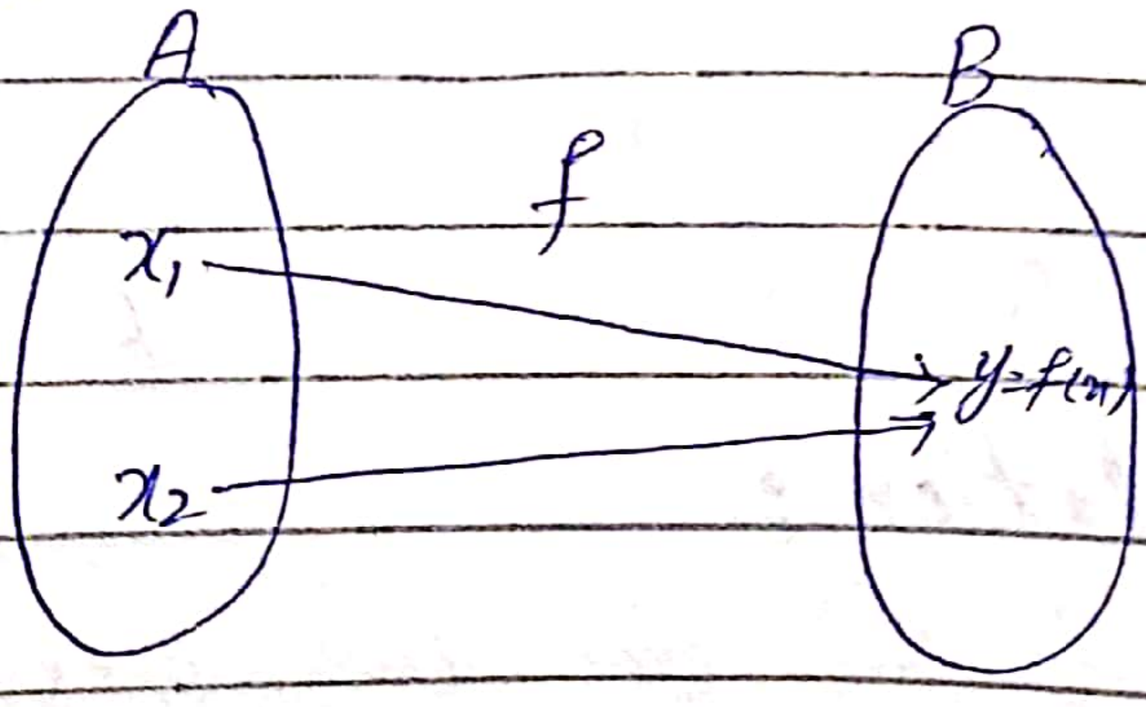
one-to-many relation (not a function)

Many to one (surjection):

R is said to be a surjection if

$$f(A) = B \quad \text{or} \quad \text{Rang}(R) = B$$

$$\forall\, y \in B,\ \exists\, x \in A,\ y = f(x)$$

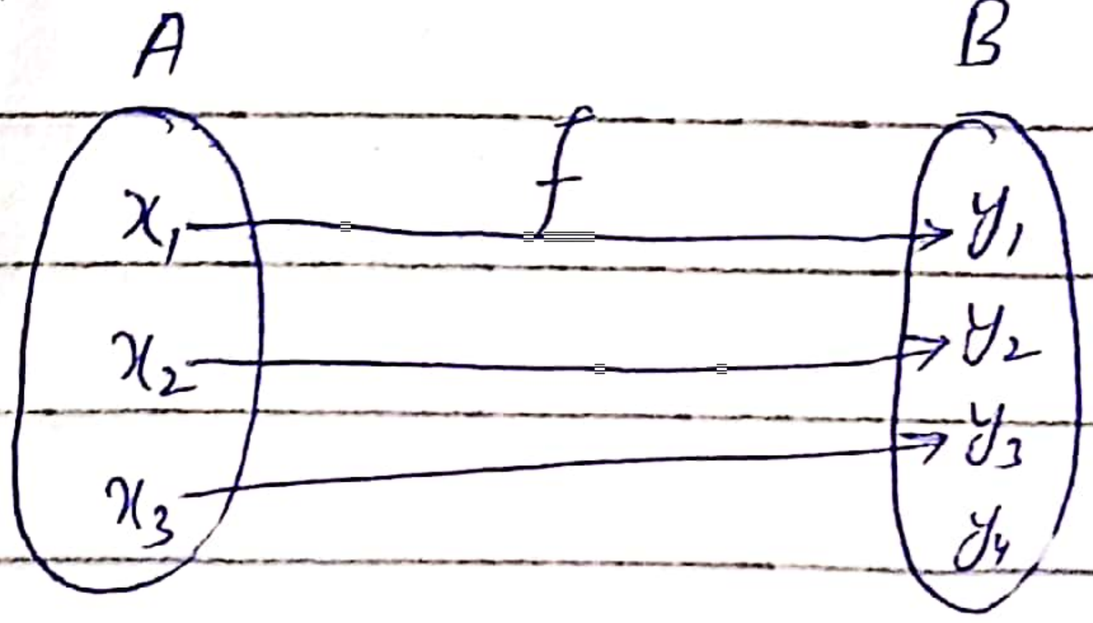


Surjection

Into, One to one (Injection):

R is said to be an injection if for all $x_1, x_2 \in A$, if $x_1 \neq x_2$, $f(x_1) \neq f(x_2)$

Therefore, if R is an injection, $(x_1, y) \in R$ $(x_2, y) \in R$ then $x_1 = x_2$

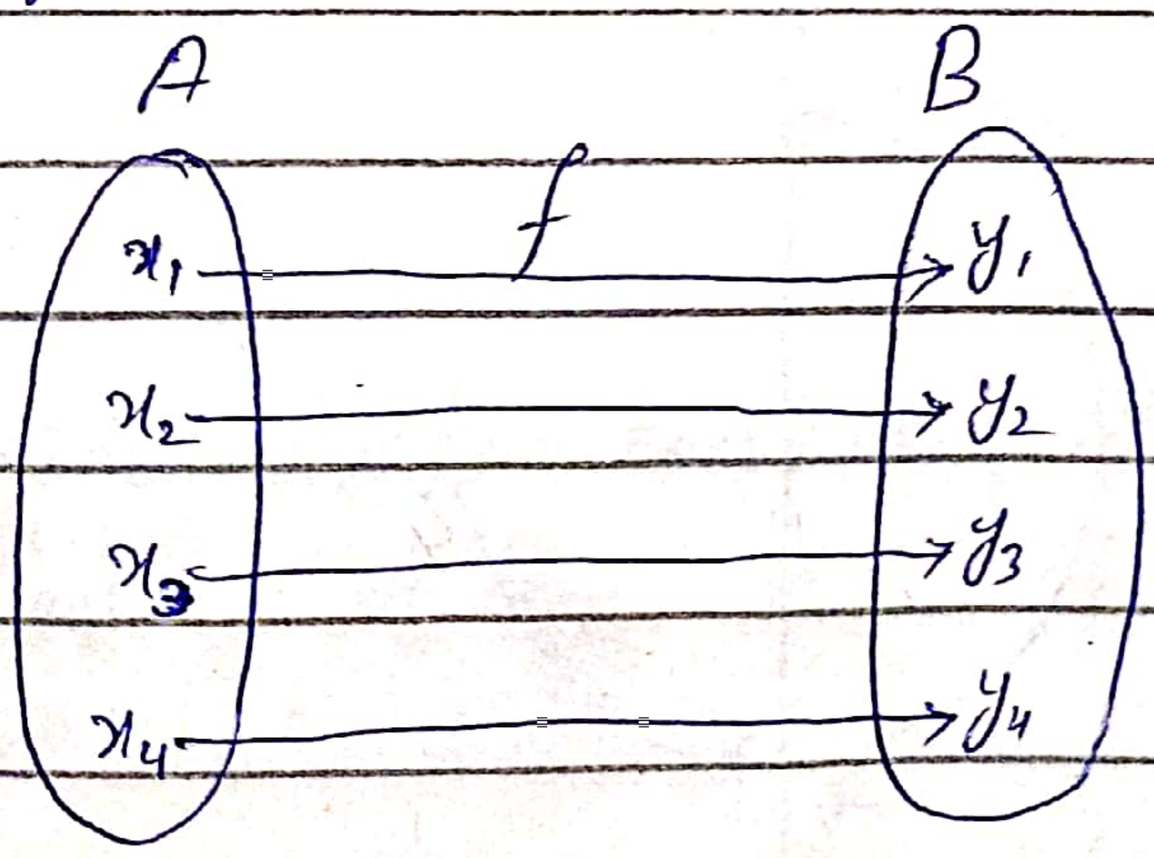


Injection:

One to one correspondence (Bijection):

R is said to be a bijection if it is both a surjection and an injection.

Assuming A and B are in bijection, this is an equivalence relation holds, i.e. the elements and the number of the elements correspond.



Bijection:

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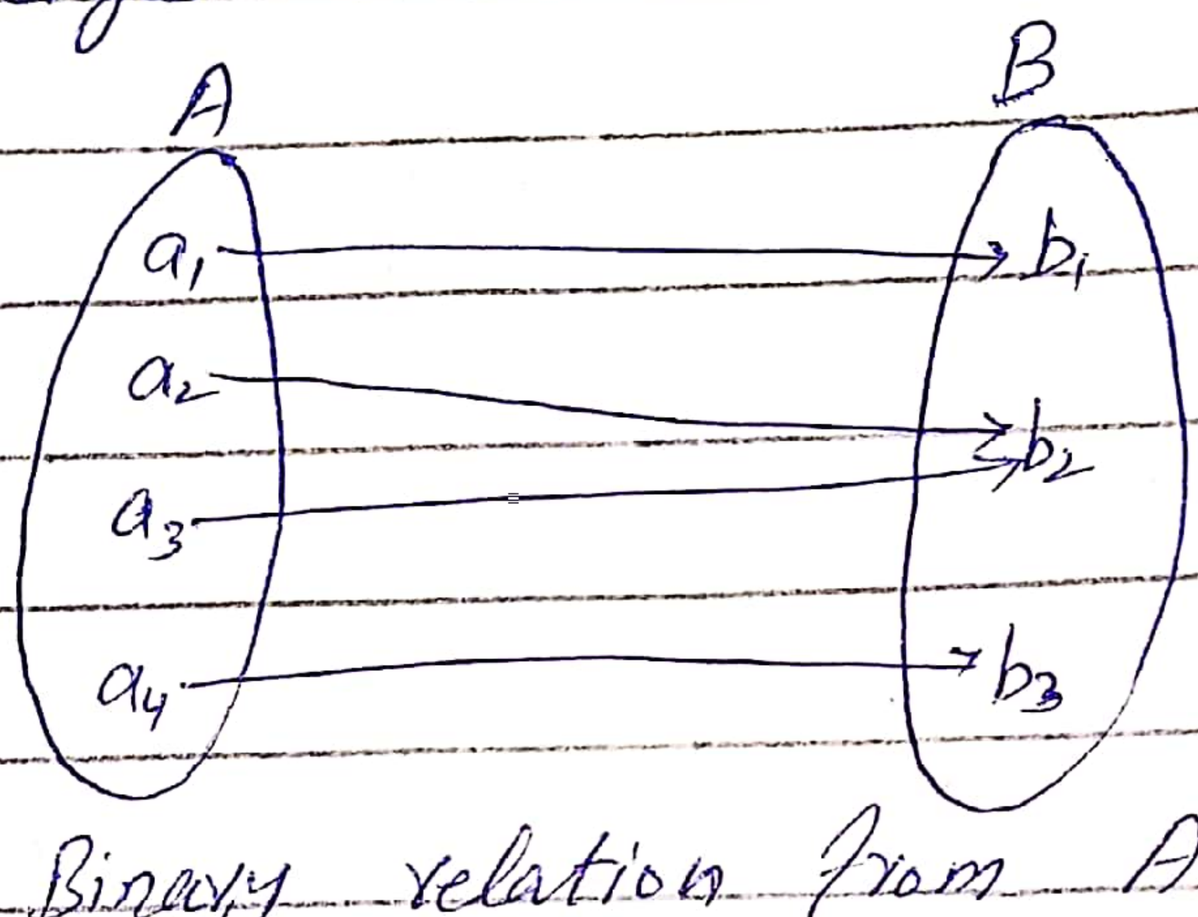
Representation Methods of Relations:

(i) Bipartite Graph:

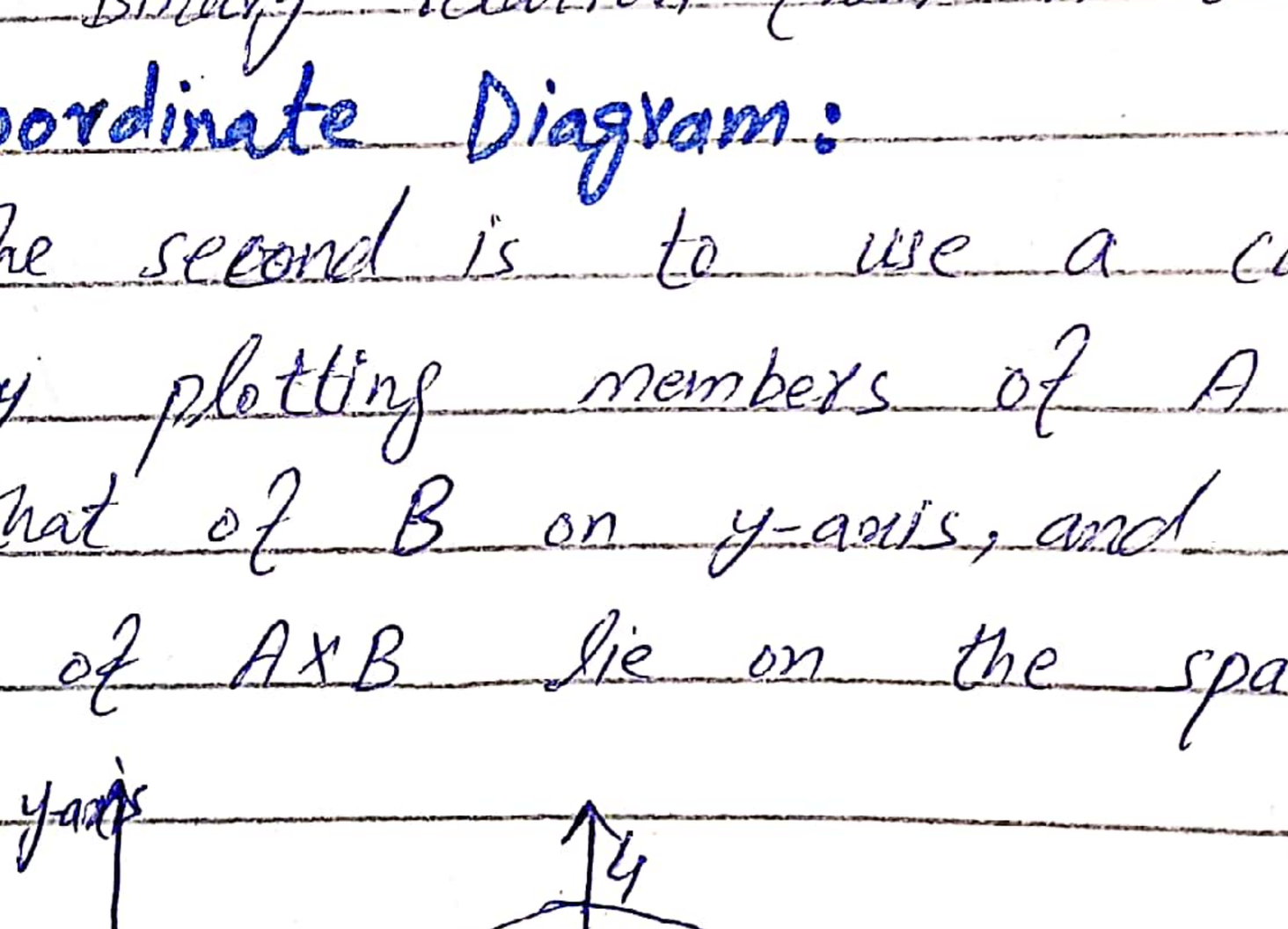
The first is by illustrating A and B in a figure and representing the relation by drawing arcs or edges



Binary relation from A to B.

(ii) Coordinate Diagram:

The second is to use a coordinate diagram by plotting members of A on x-axis and that of B on y-axis, and then the members of A×B lie on the space.



Relation of $x^2+y^2=4$

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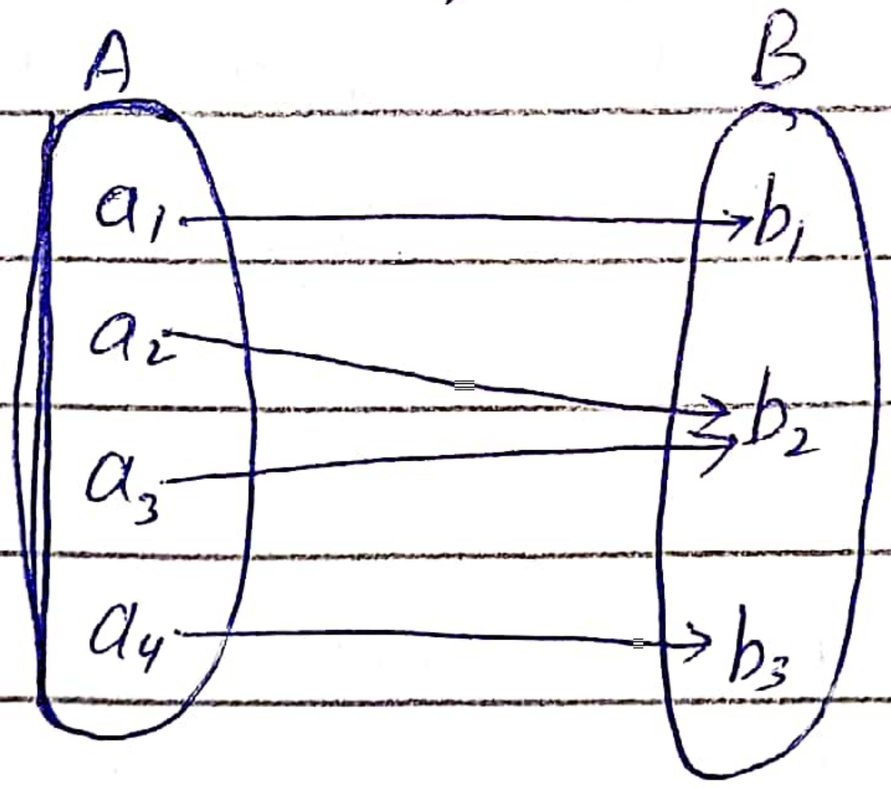
(iii) Matrix :

The third method is by manipulating relation matrix. let A and B be finite sets having m and n elements respectively. Assuming R is a relation between A and B, we may represent the relation by matrix $M_R = (m_{ij})$ which is defined as follows

$$M_R = (m_{ij})$$

$$m_{ij} = \begin{cases} 1, & (a_i, b_j) \in R \\ 0, & (a_i, b_j) \notin R \end{cases}$$

where $i = 1, 2, 3, \ldots, m$, $j = 1, 2, 3, \ldots, m$



R	b_1	b_2	b_3
a_1	1	0	0
a_2	0	1	0
a_3	0	1	0
a_4	0	0	1

$$R = \{(a_1, b_1), (a_2, b_2), (a_3, b_2), (a_4, b_3)\}$$

(iv) Digraph (directed graph) :

The forth method is the directed graph or digraph. Elements are represented as nodes, and relation between elements as directed edges.

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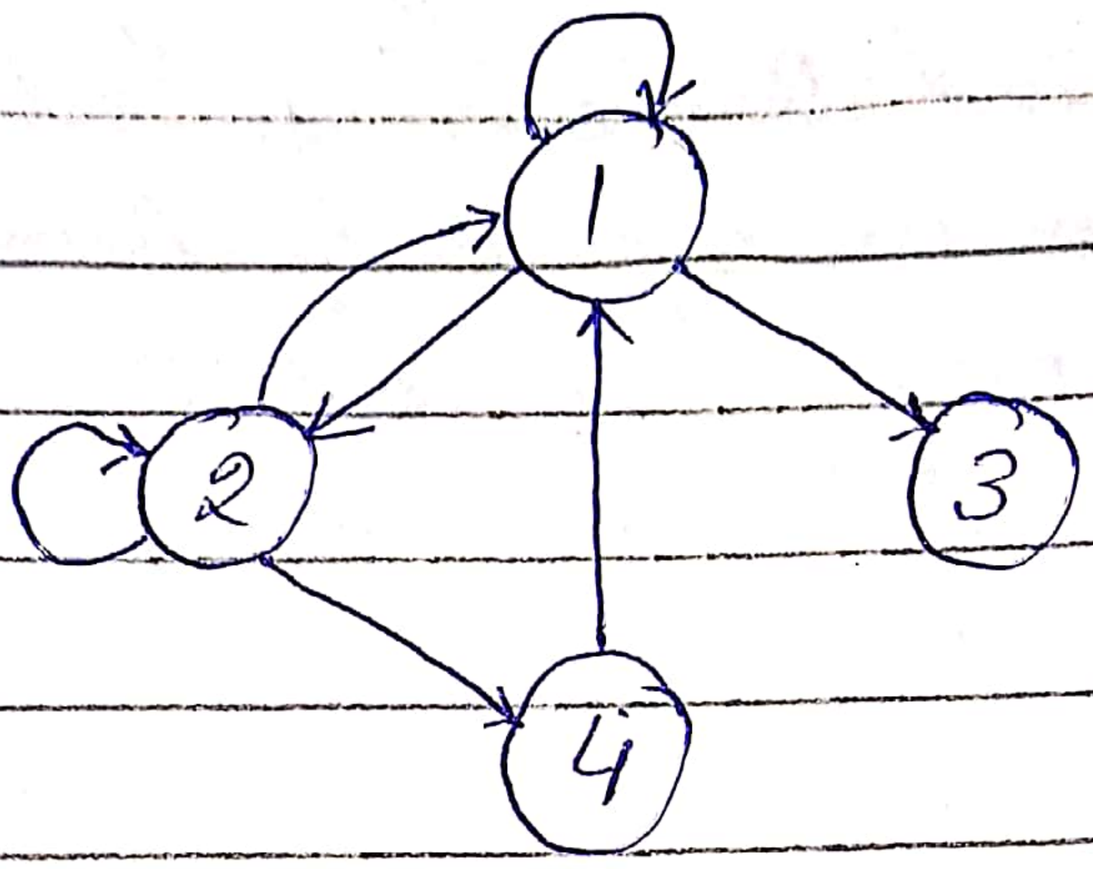
$A = \{1, 2, 3, 4\}$. and

$R = \{(1,1), (1,2), (2,1), (2,2), (1,3), (2,4), (4,1)\}$

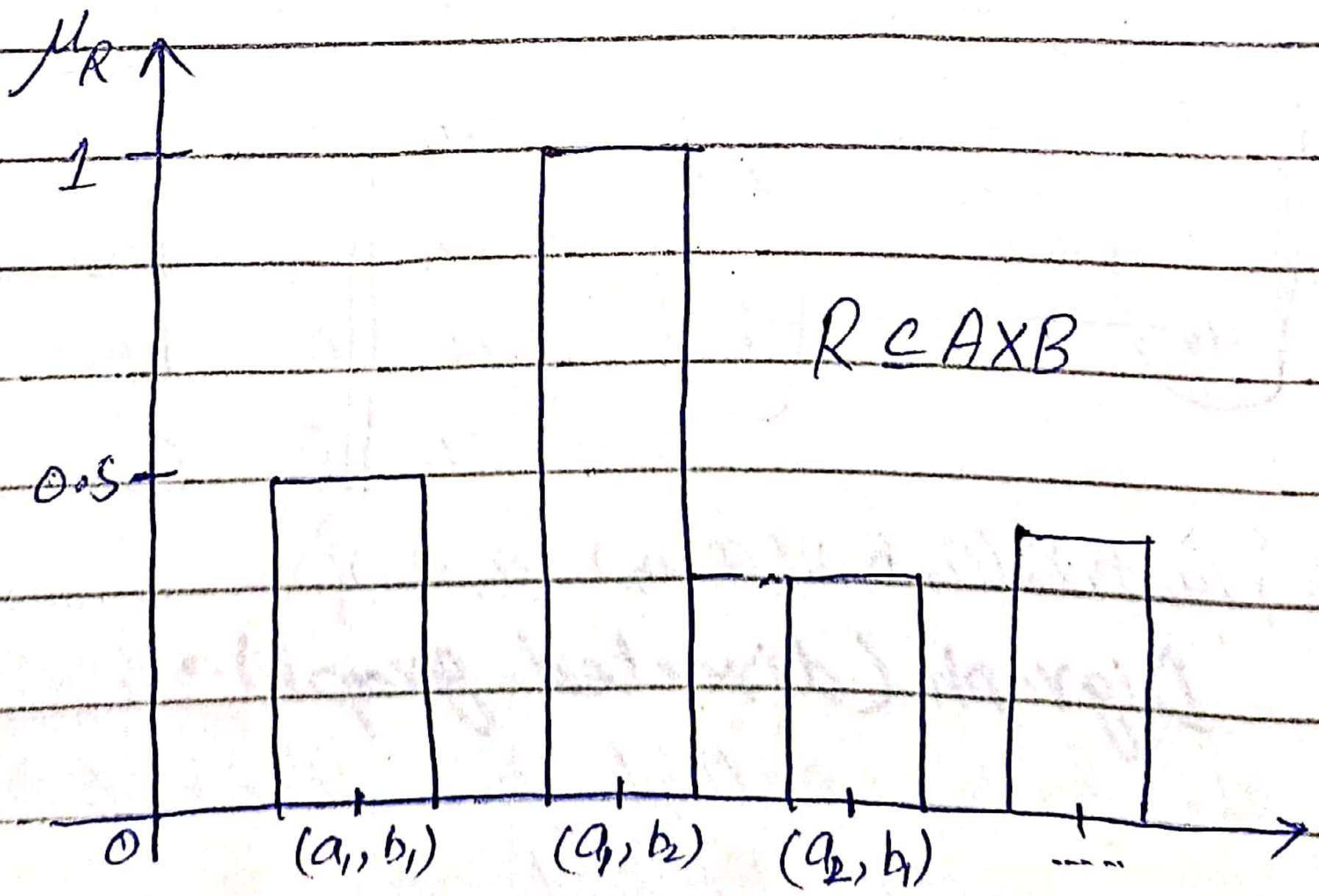


Fuzzy Relations:

Fuzzy relations has degree of membership whose values lies in $[0,1]$.

$$\mu_R : A \times B \rightarrow [0,1]$$

$$R = \{((x,y), \mu_R(x,y)) \mid \mu_R(x,y) \geq 0, x \in A, y \in B\}$$



Fuzzy relation as a fuzzy set $A \times B$

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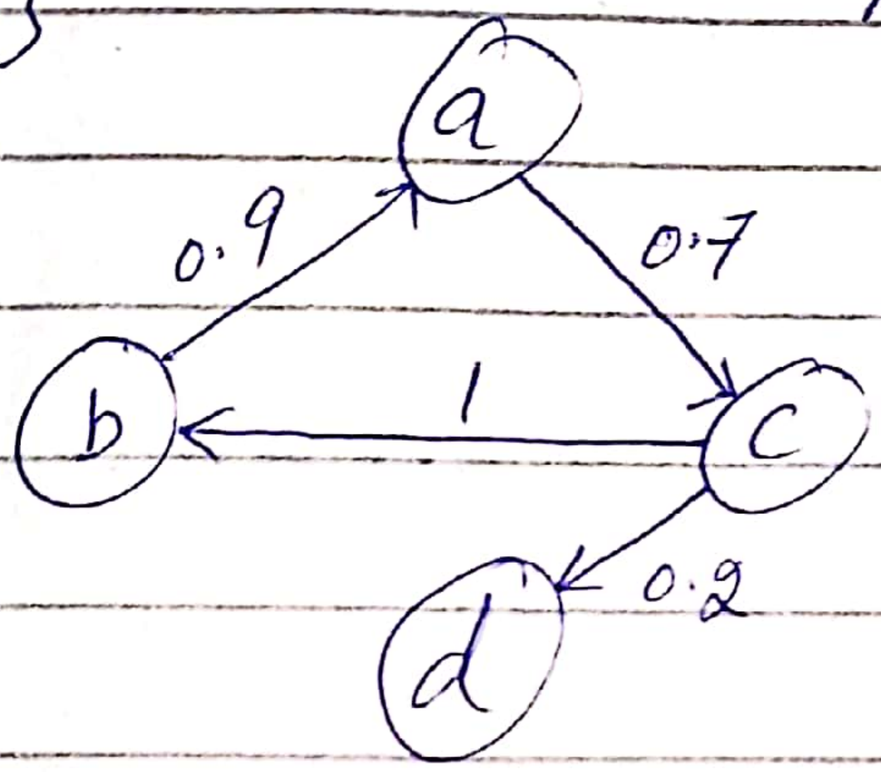
Lecture # 13

Example of Fuzzy Relations:

$R = \{(a,c), (b,a), (c,b), (c,d)\}$

$R = \{((a,c), 0.7), ((b,a), 0.9), ((c,b), 1), ((c,d), 0.2)\}$

$A = \{a, b, c, d\}$



$A \times A =$

A \ A	a	b	c	d
a	0.0	0.0	0.7	0.0
b	0.9	0.0	0.0	0.0
c	0.0	1.0	0.0	0.2
d	0.0	0.0	0.0	0.0

$$dom(R) = \{x \mid x \in A, y \in A, \mu_R(x,y) = 1\}$$

$$ran(R) = \{y \mid x \in A, y \in A, \mu_R(x,y) = 1\}$$

$R = \{(a,c), (b,a), (c,b), (c,d)\}$, $A = \{a, b, c, d\}$

$dom(R) = \{a, b, c\}$

$Ran(R) = \{c, a, b, d\}$

$$\mu_{dom(R)}(x) = \max_{y \in B} \mu_R(x,y)$$

$$\mu_{ran(R)}(x) = \max_{x \in A} \mu_R(x,y)$$

$X = \{x_1, x_2, x_3, x_4\}$

$$R = \begin{array}{c|cccc} & x_1 & x_2 & x_3 & x_4 \\ x_1 & 0.2 & 0 & 0.5 & 0 \\ x_2 & 0 & 0.3 & 0.7 & 0.8 \\ x_3 & 0.1 & 0 & 0.4 & 0 \\ x_4 & 0 & 0.6 & 0 & 0.1 \end{array}$$

$dom(R) = \{0.5, 0.8, 0.4, 0.6\}$

$ran(R) = \{0.2, 0.6, 0.7, 0.8\}$

$A \times B$, $dom(R) \subseteq A$

$ran(R) \subseteq B$

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Given a fuzzy matrix $A=(a_{ij})$ and $B=(b_{ij})$, we can perform following operations on these fuzzy matrices.

i) Sum $A+B = \max[a_{ij}, b_{ij}]$

ii) Max product $A \cdot B = AB = \max_k [\min[a_{ik}, b_{kj}]]$

iii) Scalex product λA where $0 \leq \lambda \leq 1$

$A =$

	a	b	c
a	0.2	0.5	0.0
b	0.4	1.0	0.1
c	0.0	1.0	0.0

$B =$

	a	b	c
a	1.0	0.1	0.0
b	0.0	0.0	0.5
c	0.0	1.0	0.1

$\max((a,a),(a,a))$

$A+B =$

+	a	b	c
a	1.0	0.5	0.0
b	0.4	1.0	0.5
c	0.0	1.0	0.1

$A \cdot B =$

·	a	b	c
a	0.2	0.1	0.5
b	0.4	0.1	0.5
c	0.0	0.6	0.5

$$\min\begin{Bmatrix} 0.2 & 0.5 & 0.0 \\ 1.0 & 0.0 & 0.0 \end{Bmatrix}$$

$\max[(0.2 \quad 0.0 \quad 0.0)]$

$= 0.2$

$a_{ij} \leq b_{ij} \iff A \leq B \iff SA \leq SB, AT \leq BT$

Fuzzy relation matrix:

If a fuzzy relation R is given in the form of fuzzy matrix, its elements represent the membership values of this relation. That is, if the matrix is denoted by M_R and membership values by $\mu_R(i,j)$ then $M_R = (\mu_R(i,j))$

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Lecture #14

Union of Fuzzy Relations:

Union of two relations R and S is defined as follows

$$\mu_{R \cup S}(x,y) = Max[\mu_R(x,y), \mu_S(x,y)], \forall (x,y) \in A \times B$$

$$= \mu_R(x,y) \vee \mu_S(x,y)$$

We generally use the sign $\vee$ for Max operation.

For n relatives, we extend it two the followings.

$$\mu_{R_1 \cup R_2 \cup R_3 \cup \dots \cup R_n}(x,y) = \bigvee_{R_i} \mu_{R_i}(x,y)$$

If expressing the fuzzy relation by fuzzy matrices, i.e. M_R and M_S, matrix $M_{R \cup S}$ concerning the union is obtained from the sum of two matrices $M_R + M_S$

$$M_{R \cup S} = M_R + M_S$$

Example: Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

Fuzzy relation matrices $M_{R \cup S}$ corresponding $R \cup S$

$M_{R \cup S}$	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.6	1.0	0.3

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$\vee$ Well $\wedge$ wedge
max min

$$RUS = \{((1,a), 0.3), ((1,b), 0.2), ((1,c), 1.0), ((2,a), 0.8), ((2,b), 1.0), ((2,c), 1.0), ((3,a), 0.6), ((3,b), 1.0), ((3,c), 0.3)\}$$

Intersection of Fuzzy Relations:

The intersection relation $R \cap S$ of set A and B is defined by the following membership function.

$$\mu_{R \cap S}(x,y) = \text{Min}[\mu_R(x,y), \mu_S(x,y)]$$
$$= \mu_R(x,y) \wedge \mu_S(x,y)$$

The symbol $\wedge$ is for the Min operation.
In the same manner, the intersection relation for n relations is defined by

$$\mu_{R_1 \cap R_2 \cap R_3 \cap \cdots \cap R_n}(x,y) = \bigwedge_{R_i} \mu_{R_i}(x,y)$$

Example: Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

Fuzzy relation matrices $M_{R \cap S}$ corresponding $R \cap S$.

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M_{RNS}	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.0	0.9	0.0

$RNS = \{((1,a),0.3), ((1,b),0.0), ((1,c),0.1),$
$((2,a),0.1), ((2,b),0.8), ((2,c),1.0),$
$((3,a),0.0), ((3,b),0.9), ((3,c),0.0)\}$

Complement of Fuzzy Relations:

Complement relation $\bar{R}(R')$ for fuzzy relation R shall be defined by the following membership function.

$$\mu_{\bar{R}}(x,y) = 1 - \mu_R(x,y), \forall (x,y) \in A \times B$$

Examples: Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

$M_{\bar{R}}$	a	b	c
1	0.7	0.8	0.0
2	0.2	0.0	0.0
3	1.0	0.0	1.0

$M_{\bar{S}}$	a	b	c
1	0.7	1.0	0.9
2	0.9	0.2	0.0
3	0.4	0.1	0.7

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Inverse of Fuzzy relations:

When a fuzzy relation $R \subseteq A \times B$ is given, the inverse relation of R^{-1} is defined by the following membership function.

$$\mu_{R^{-1}}(y, x) = \mu_R(x, y), \quad \forall (x, y) \in A \times B$$

Examples: Two fuzzy relations matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

$\mu_{R^{-1}}(y, x) = \mu_R(x, y)$ $M_{S^{-1}}$

$M_{R^{-1}}$	1	2	3
a	0.3	0.2	1.0
b	0.8	1.0	1.0
c	0.0	1.0	0.0

$M_{S^{-1}}$	1	2	3
a	0.3	0.0	0.1
b	0.1	0.8	1.0
c	0.6	0.9	0.3

$$R = \{((1,a), 0.3), ((1,b), 0.2), ((1,c), 1.0), ((2,a), 0.8), ((2,b), 1.0), ((2,c), 1.0), ((3,a), 0.0), ((3,b), 1.0), ((3,c), 0.0)\}$$

$$R^{-1} = \{((a,1), 0.3), ((b,1), 0.2), ((c,1), 1.0), ((a,2), 0.8), ((b,2), 1.0), ((c,2), 1.0), ((a,3), 0.0), ((b,3), 1.0), ((c,3), 0.0)\}$$

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Lecture # 15

Composition of Fuzzy Relations:

Two fuzzy relations R and S are defined on set A, B and C. That is, $R \subseteq A \times B$, $S \subseteq B \times C$.

The composition $S \circ R = SR$ of two relations R and S is expressed by the relation from A to C, and this composition is defined by the following.

$$\forall (x,y) \in A \times B, (y,z) \in B \times C,$$

$$\mu_{S \circ R}(x,z) = \max_y [\min(\mu_R(x,y), \mu_S(y,z))]$$

$$= \bigvee_y [\mu_R(x,y) \wedge \mu_S(y,z)], \forall (x,z) \in A \times C$$

$S \circ R$ from this elaboration is a subset of $A \times C$.

That is, $S \circ R \subseteq A \times C$.

If the relation R and S are represented by matrices M_R and M_S, the matrix $M_{S \circ R}$ corresponding $S \circ R$ is obtained from the product of M_R and M_S.

$$M_{S \circ R} = M_R \circ M_S$$

Example: Consider fuzzy relations $R \subseteq A \times B$, $S \subseteq B \times C$. The set A, B and C shall be the set of events. By the relation R, we can see the possibility of occurrance of B after A, and by S, that of C after B. For example, by M_R, the possibility of $a \in B$ after $1 \in A$ is 0.1. By M_S, the of occurrence of α after a is 0.9.

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

Here, we cannot guess the possibility of C when A is occured. So our main job now will be the obtaining the composition $S \circ R \subseteq A \times C$.

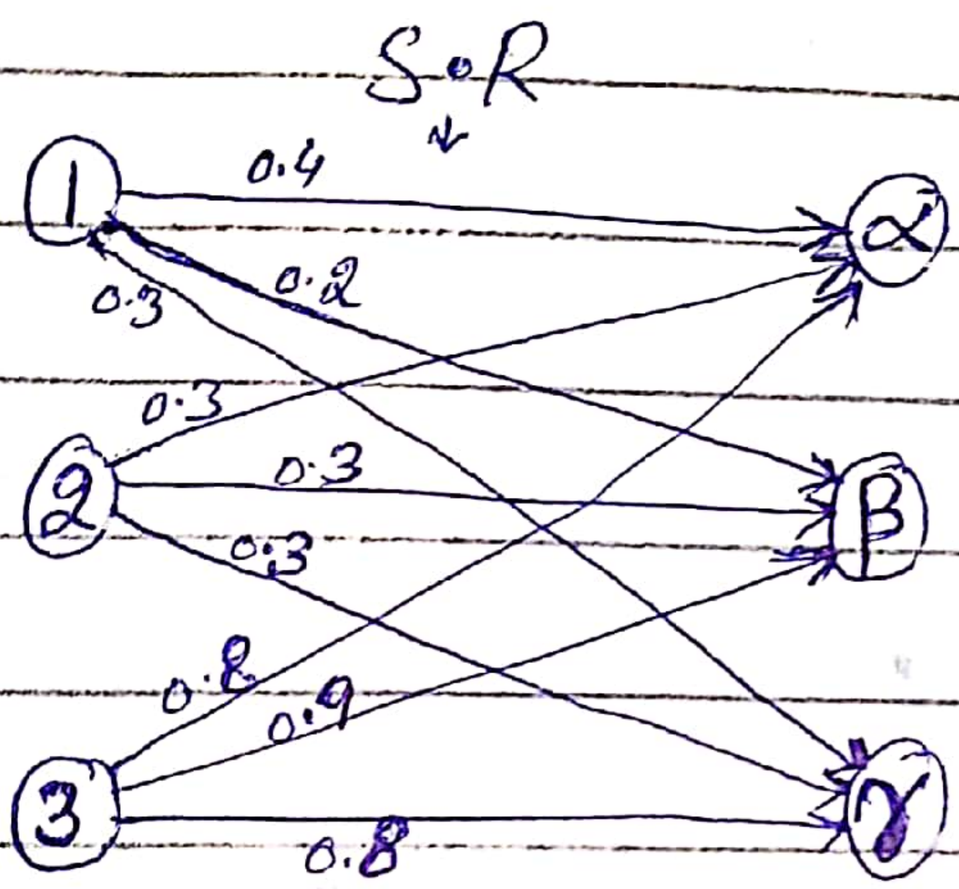
presuming that the relation R and S are the expressions of rules that guide the occurance of event or fact. that. Then the possibility of occurance of event B when event A is happened is guided by the rule R, and rule S indicates the possibility of C when B is existing. For further cases, the possibility of C when A has occured can be induced from the composition rule $S \circ R$. This manner is named as an "inference" which is a process producing new information.

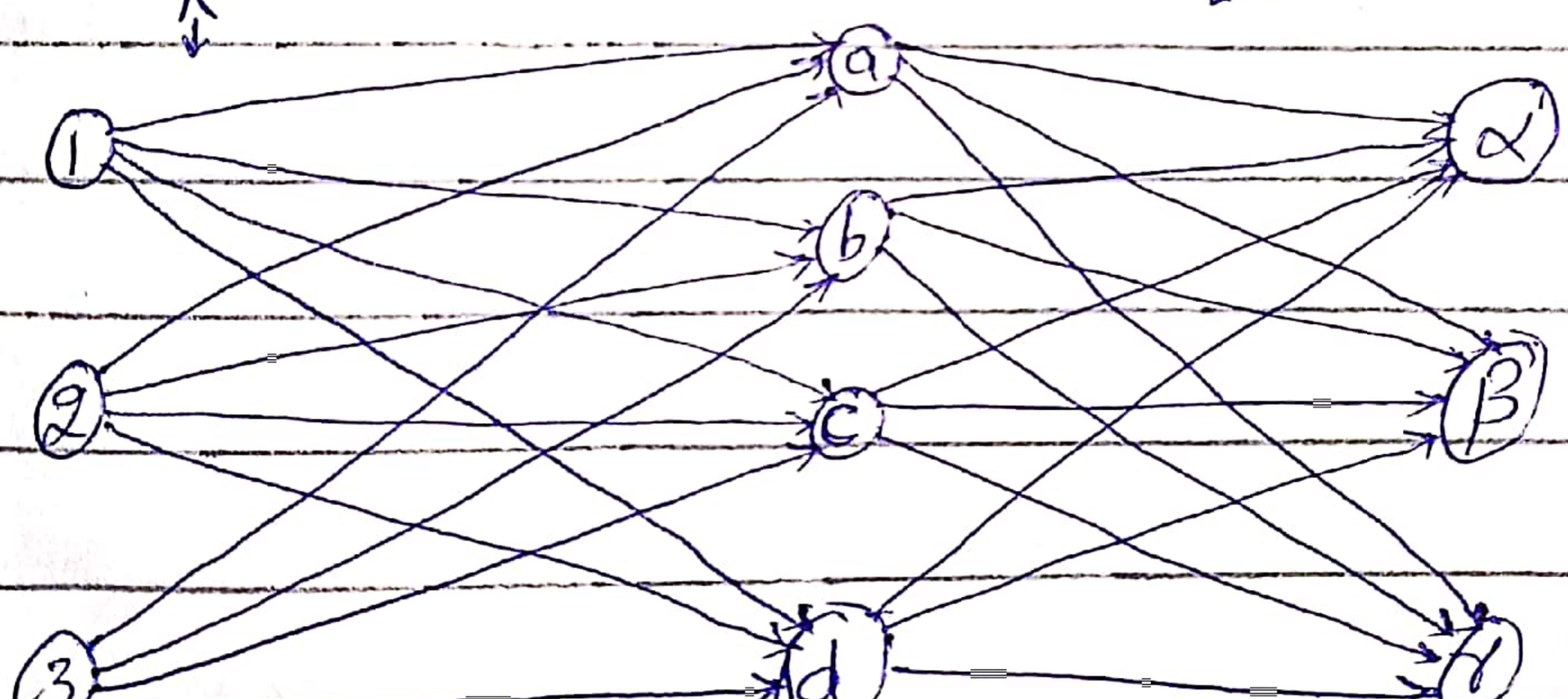
$$\mu_{S \circ R}(x, z) = \max_y [\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_y [\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ	Min
				(1, a), (a, α) 0.1
1	0.4	0.2	0.3	(1, b), (b, α) 0.2
2	0.3	0.3	0.3	(1, c), (c, α) 0.0
3	0.8	0.9	0.8	(1, d), (d, α) 0.4

Max = 0.4





Lecture # 16

Triangular Norms and Conorms:

Triangular norms and conorms generalize the basic connectives between fuzzy sets. They were first introduced in the theory of probabilistic metric spaces. Later these were found to be very suitable to be used with fuzzy sets.

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Triangular Norms:

A binary operation $\Delta : [0,1]^2 \to [0,1]$ is a triangular norm (t-norm) if it satisfies the following conditions:

i) $1 \Delta x = x$

ii) $x \Delta y = y \Delta x$

iii) $(x \Delta y) \Delta z = x \Delta (y \Delta z)$

iv) If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$.

Triangular Conorm:

A binary operation $\nabla : [0,1]^2 \to [0,1]$ is a triangular conorm (t-conorm) if it satisfies the following conditions:

i) $1 \nabla x = x$

ii) $x \nabla y = y \nabla x$

iii) $(x \nabla y) \nabla z = x \nabla (y \nabla z)$

iv) If $w \leq x$ and $y \leq z$, the $w \nabla y \leq x \nabla z$.

Theorem: If Δ is a t-norm then prove that $0 \Delta x = 0$

proof: We use 4th property of t-norm

If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$

or If $x \leq 1$ and $0 \leq 0$, then $x \Delta 0 \leq 1 \Delta 0$

$x \Delta 0 \leq 1 \Delta 0 = 0$ Using 1st property of t-norm

i.e. $1 \Delta x = x$

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So $x \Delta 0 \leq 0$ since 0 is at least number.

Hence $x \Delta 0 = 0$ which complete the proof.

Some examples of t-norms:

- $x \Delta_0 y = \begin{cases} x \wedge y ; \text{if } x \vee y = 1 \\ 0 ; \text{otherwise} \end{cases}$ (where $\wedge$ = min and $\vee$ = max)
- $x \Delta_1 y = 0 \vee (x + y - 1)$
- $x \Delta_2 y = \dfrac{xy}{2 - (x + y - xy)}$
- $x \Delta_3 y = xy$
- $x \Delta_4 y = \dfrac{xy}{x + y - xy}$
- $x \Delta_5 y = x \wedge y = \min(x, y)$

Lecture # 17

Example 1: Check min is t-norm or not?

Solution: This t-norm is also called Δ_5.

We check the four properties of the t-norm.

i) $1 \Delta x = \min(1, x) = x$

ii) $x \Delta y = \min(x, y) = \min(y, x) = y \Delta x$

iii) $(x \Delta y) \Delta z = (\min(x, y)) \Delta z$

$= \min((x, y), z) = \min(x, y, z)$

$= \min(x, (y, z)) = x \Delta (\min(y, z))$

$(x \Delta y) \Delta z = x \Delta (y \Delta z)$

iv) If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$.

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Since $w \le x$ and $y \le z$

$\Rightarrow \min(w,y) \le \min(x,z)$

$\Rightarrow w \Delta y \le x \Delta z$

Since all the four properties hold so min is a t-norm.

Example 2: Check $x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$ is t-norm or not?

Solution: This (t-norm) is also called Δ_2.

We check the four properties of the t-norm.

i) $1 \Delta_2 x = \frac{(1)(x)}{2-(1+x-1 \cdot x)} = \frac{x}{2-1} = x$

ii) $x \Delta_2 y = \frac{xy}{2-(x+y-xy)} = \frac{yx}{2-(y+x-yx)} = y \Delta_2 x$

iii) $x \Delta_2 (y \Delta_2 z) = x \Delta_2 \left(\frac{yz}{2-(y+z-yz)}\right)$

$$= \frac{(x)\left(\frac{yz}{2-(y+z-yz)}\right)}{2-\left(x+\frac{yz}{2-(y+z-yz)}\right)-(x)\left(\frac{yz}{2-(y+z-yz)}\right)}$$

$$= \frac{xyz}{2(2-(y+z-yz))-[x(2-(y+z-yz))+yz-xyz]}$$

$$= \frac{xyz}{4-2y-2z+2yz-2x+xy+xz-xyz+yz+xyz}$$

$$= \frac{xyz}{4-2x-2y-2z+yz+xy+xz}$$

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$$x \Delta_2 (y \Delta_2 z) = \frac{xyz}{4 - 2(x+y+z) + xy + yz + zx}$$

$$x \Delta_2 (y \Delta_2 z) = (x \Delta_2 y) \Delta z$$

iv) If $x \le y$ and $z \le w$, then $x \Delta_2 z \le y \Delta_2 w$.

So we show that

$$\frac{xz}{2-(x+z-xz)} \le \frac{yw}{2-(y+w-yw)}$$

or $\Rightarrow$ $$\frac{xz}{2-x-z+xz} \le \frac{yw}{2-y-w+yw}$$

Since $x \le y$ and $z \le w$

$\Rightarrow -x \ge -y$ and $-z \ge -w$

$\Rightarrow 1-x \ge 1-y$ and $1-z \ge 1-w$

$\Rightarrow (1-x)(1-z) \ge (1-y)(1-w)$

$\Rightarrow 1-x-z+xz \ge 1-y-w+yw$

$\Rightarrow 2-x-z+xz \ge 2-y-w+yw$

$\Rightarrow \frac{1}{2-x-z+xz} \le \frac{1}{2-y-w+yw} \longrightarrow (1)$

Also $xz \le yw \longrightarrow (2)$

Multiplying eq. (1) and (2)

$$\frac{xz}{2-x-z+xz} \le \frac{yw}{2-y-w+yw}$$

$$\Rightarrow \frac{xz}{2-(x+z-xz)} \le \frac{yw}{2-(y+w-yw)} \Rightarrow x \Delta_2 z \le y \Delta_2 w$$

Since all the four properties hold so Δ_2 is a t-norm.

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Lecture #18

Example 3: Check $x \Delta_3 y = xy$ is t-norm or not?

Solution: We check the four properties of the t-norm.

i) $1 \Delta_3 x = (1)(x) = x$

ii) $x \Delta_3 y = (x)(y) = (y)(x) = y \Delta_3 x$

iii) $x \Delta_3 (y \Delta_3 z) = x \Delta_3 (y)(z)$

$= (x)(y)(z) = (x)(y) \Delta_3 z$

$x \Delta_3 (y \Delta_3 z) = (x \Delta_3 y) \Delta_3 z$

iv) If $x \leq y$ and $z \leq w$, then $x \Delta_3 z \leq y \Delta_3 w$.

So we show that $xz \leq yw$

Since $x \leq y$ and $z \leq w$

$\Rightarrow xz \leq yw$

$\Rightarrow x \Delta_3 z \leq y \Delta_3 w$

Since all the four properties hold so Δ_3 is a t-norm.

Example 4: Check $x \Delta_4 y = \frac{xy}{x+y-xy}$ is t-norm or not?

Solution: We check the four properties of the t-norm.

i) $1 \Delta_4 x = \frac{(1)(x)}{1+x-(1)(x)} = \frac{x}{1} = x$

ii) $x \Delta_4 y = \frac{xy}{x+y-xy} = \frac{yx}{y+x-yx} = y \Delta_4 x$

iii) $L.H.S = x \Delta_4 (y \Delta_4 z) = x \Delta_4 \left(\frac{yz}{y+z-yz}\right)$

$= \frac{(x)\left(\frac{yz}{y+z-yz}\right)}{x + \frac{yz}{y+z-yz} - x\left(\frac{yz}{y+z-yz}\right)}$

$L.H.S = \frac{xyz}{x(y+z-yz)+yz-xyz} = \frac{xyz}{xy+yz+xz-2xyz}$

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Now $R.H.S = (x \Delta_4 y)\Delta_4 z$

$$= \left(\frac{xy}{x+y-xz}\right)\Delta_4 z$$

$$= \frac{\left(\frac{xy}{x+y-xz}\right)z}{\frac{xy}{x+y-xz}+z-\left(\frac{xy}{x+y-xy}\right)z}$$

$$= \frac{xyz}{xy+z(x+y-xz)-xyz}$$

$$R.H.S = \frac{xyz}{xy+yz+xz-2xyz} = L.H.S$$

So $x \Delta_4 (y \Delta_4 z) = (x \Delta_4 y)\Delta_4 z$

iv) If $x \le y$ and $z \le w$, then $x \Delta_4 z \le y \Delta_4 w$.

So we show that

$$\frac{xz}{x+y-xz} \le \frac{yw}{y+w-yw}$$

Since $x \le y$ and $z \le w$

$\Rightarrow xz \le yw \longrightarrow (1)$

Now $x-1 \le y-1 \le 0$ and $z-1 \le w-1 \le 0$

$\Rightarrow (x-1)(z-1) \ge (y-1)(w-1)$

$\Rightarrow xz-x-z+1 \ge yw-y-w+1$

$\Rightarrow 0 \ge xz-x-z \ge yw-y-w$

$$\Rightarrow \frac{1}{xz-x-z} \le \frac{1}{yw-y-w} \le 0 \longrightarrow (2)$$

Multiplying eq (1) and (2)

$$\Rightarrow \frac{xz}{xz-x-z} \ge \frac{yw}{yw-y-w}$$

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$$\Rightarrow \frac{xz}{-(x+z-xz)} \geq \frac{yw}{-(y+w-yw)}$$

$$\Rightarrow \frac{xz}{x+z-xz} \leq \frac{yw}{y+w-yw}$$

$$\Rightarrow x \Delta_y z \leq y \Delta_y w$$

Since all the four properties holds so Δ_y is a t-norm.

Lecture # 19

Idempotent ~~Norm~~ T-norm:

Δ is idempotent if $x \Delta x = x$, $\forall x \in [0,1]$.

Converse T-norm:

If $x \Delta y \leq c \leq x, \Delta y$,

So there exist r, s such that

$x \leq r \leq x$, and $y \leq s \leq y$,

such that $c = r \Delta s$

$\Rightarrow$ Δ is converse t-norm.

Theorem: If a t-norm Δ is convex and $a < b$, then their is $c \in [a,1]$ such that $a = b \Delta c$

proof:

We are given that Δ is t-norm:

If $a \leq a$ and $b \leq 1$

then $a \Delta b \leq a \Delta 1 = a$

$\Rightarrow$ $a \Delta b \leq a \leq b = 1 \Delta b$

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$\Rightarrow$ $a \Delta b \leq a \leq 1 \Delta b$

So by convexity there exist $a \leq b \leq 1$ and $b \leq c \leq b$.

$\Rightarrow$ $a = b \Delta c$ which is complete the proof.

Archimedean T-norm:

A t-norm Δ is archimedean if it is convex and for each $a, b \in (0,1)$, there exist a positive integer n such that

$[a]^n = a \Delta a \Delta a \cdots \Delta a < b$

Examples:

i) $[a]^3 = a \Delta a \Delta a < b$; for $a, b \in (0,1)$

ii) $\Delta_1, \Delta_2, \Delta_3$ and Δ_4 are all archimedean t-norms.

iii) min is not an archimedean t-norm.

Theorem: If Δ is archimedean t-norm then $\forall\ a, b \in (0,1)$, $a \Delta b < b$.

proof: Δ is archimedean t-norm

So $[a]^n < b$; $\forall\ a, b \in (0,1)$

Case I: If $a \leq b$

Take $a \leq b$ and $b \leq b$

then by t-norm $a \Delta b \leq b \Delta b < b$.

$\Rightarrow$ $a \Delta b < b$.

Case II: If $a > b$

Take $a < 1$ and $b \leq b$

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$\Rightarrow$ $a \Delta b < 1 \Delta b = b$

$\Rightarrow$ $a \Delta b < b$ which completes the proof.

Lecture #20

Exercise:

Question = $x \Delta y = 0 \vee \left(\frac{x+y-1+axy}{1+a} \right)$; for $a > -1$

Show that Δ is t-norm

Solution: We check the all four properties of t-norm.

i) $1 \Delta x = x$

$$L.H.S = 1 \Delta x = 0 \vee \left(\frac{1+x-1+a(1)(x)}{1+a} \right)$$

$$= 0 \vee \left(\frac{x+ax}{1+a} \right) = 0 \vee \left(\frac{x(1+a)}{1+a} \right)$$

$$1 \Delta x = 0 \vee x = x$$

$$L.H.S = R.H.S$$

ii) $x \Delta y = y \Delta x$

$$L.H.S = x \Delta y = 0 \vee \left(\frac{x+y-1+axy}{1+a} \right)$$

$$= 0 \vee \left(\frac{y+x-1+ayx}{1+a} \right)$$

$$= 0 \vee (y \Delta x)$$

$$\therefore x \Delta y = y \Delta x$$

$$L.H.S = R.H.S$$

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iii) $x \Delta (y \Delta z) = (x \Delta y) \Delta z$

$L.H.S = x \Delta (y \Delta z)$

$= x \Delta \left[0 \vee \left(\frac{y+z-1+ayz}{1+a} \right) \right]$

$= 0 \vee \left[\frac{x + \left(\frac{y+z-1+ayz}{1+a} \right) - 1 + a x \left(\frac{y+z-1+ayz}{1+a} \right)}{1+a} \right]$

$= 0 \vee \left[\frac{x(1+a) + (y+z-1+ayz) - 1(1+a) + a x (y+z-1+ayz)}{(1+a)^2} \right]$

$= 0 \vee \left[\frac{x+ax+y+z-1+ayz-1-a+axy+axz-ax+a^2xyz}{(1+a)^2} \right]$

$= 0 \vee \left[\frac{x+y+z+axy+ayz+axz-2+a^2xyz}{(1+a)^2} \right]$

$R.H.S = (x \Delta y) \Delta z$

$= \left[a \vee \left(\frac{x+y-1+axy}{1+a} \right) \right] \Delta z$

$= 0 \vee \left[\frac{\frac{x+y-1+axy}{1+a} + z - 1 + a \left(\frac{x+y-1+axy}{1+a} \right) z}{1+a} \right]$

$= 0 \vee \left[\frac{x+y-1+aaxy + z(1+a)x - 1 + a + axz + ayz - az + a^2xyz}{(1+a)^2} \right]$

$= 0 \vee \left[\frac{x+y+z+axy+ayz+axz-2+a^2xyz}{1+a^2} \right]$

$R.H.S = L.H.S$

iv) If $x \le y$ and $z \le w$, then $x \Delta z \le y \Delta w$.

we have to show that

$$\max\left(\frac{x+z-1+\alpha xz}{1+\alpha}\right) \le \max\left(\frac{y+w-1+\alpha yw}{1+\alpha}\right)$$

Since $x \le y$ and $z \le w$

$\Rightarrow x+z \le y+w$

$x+z-1 \le y+w-1 \longrightarrow (1)$

$x \le y$ and $z \le w$

$\Rightarrow xz \le yw$

$\Rightarrow \alpha xz \le \alpha yw \quad \because \alpha > -1$

$\Rightarrow \max 0, \alpha xz \le \max 0, \alpha yw \longrightarrow (2)$

Since $\vee$ (union) with 0 cannot be -ve.

$\max(0, x+z-1+\alpha xz) \le \max(0, y+w-1+\alpha yw)$

$\alpha > -1 \Rightarrow 1+\alpha > 0 \Rightarrow \frac{1}{1+\alpha} > 0$

So $\max\left(0, \frac{x+z-1+\alpha xz}{1+\alpha}\right) \le \max\left(0, \frac{y+w-1+\alpha yw}{1+\alpha}\right)$.

$\Rightarrow x \Delta z \le y \Delta w$

Since all the four properties hold, So Δ is t-norm.

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Lecture # 21

Theorem: Show that $x \Delta_0 y \leq x \Delta y \leq x \Delta_5 y$

proof: (i) $x \Delta_0 y \leq x \Delta y$

$$x \Delta_0 y = \begin{cases} \min(x,y) & \text{if } \max(x,y) = 1 \\ 0 & \text{if otherwise.} \end{cases}$$

Case I: If $\max(x,y) \neq 1$

$\Rightarrow$ $x \Delta_0 y = 0 \leq x \Delta y$

$\Rightarrow$ $x \Delta_0 y \leq x \Delta y \longrightarrow (1)$

Case II: If $\max(x,y) = 1$, let $\max(x,y) = y = 1$

$\Rightarrow$ $\min(x,y) = x$

$\Rightarrow$ $\min(x,y) = x \Delta_0 y = x \Delta_0 1 = x = x \Delta y$

$\Rightarrow$ $x \Delta_0 y \leq x \Delta y \longrightarrow (2)$

From equations (1) and (2)

$\Rightarrow$ $x \Delta_0 y \leq x \Delta y$

(ii) Now $x \Delta y \leq x \Delta_5 y$

$x \Delta_5 y = \min(x,y)$

Let $\min(x,y) = x$

Take $x \leq x$ and $y \leq 1$

$\Rightarrow$ $x \Delta y \leq x \Delta 1$ (by using 4th property of t-norm)

$\Rightarrow$ $x \Delta y \leq x = \min(x,y)$

$\Rightarrow$ $x \Delta y \leq x \Delta_5 y$

Hence $x \Delta_0 y \leq x \Delta y \leq x \Delta_5 y$

which is the complete the proof

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Theorem: Show that $\Delta_1 \leq \Delta_2 \leq \Delta_3$

proof: (i) $\Delta_1 \leq \Delta_2$

$$x \Delta_1 y = o v (x+y-1)$$

$$x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$$

$\because$ $0 \leq x \leq 1$ and $0 \leq y \leq 1$

$\Rightarrow$ $0 \leq x+y \leq 2$

$\Rightarrow$ $x+y-2 \leq 0 \longrightarrow (1)$

Also $1 \geq 1-x \geq 0$ and $1 \geq 1-y \geq 0$

$\Rightarrow$ $1 \geq (1-x)(1-y) \geq 0$

$\Rightarrow$ $0 \leq (1-x)(1-y) \leq 1$

$\Rightarrow$ $1 \leq 1+(1-x)(1-y) \leq 2$

$\Rightarrow$ $1 \leq 2-x-y+xy \leq 2$

$\Rightarrow$ $1 \geq \frac{1}{2-x-y+xy} \geq \frac{1}{2} \longrightarrow (2)$

From eq. (1) and (2)

$\Rightarrow$ $x+y-2 \leq \frac{x+y-2}{2-x-y+xy}$

$\Rightarrow$ $x+y-2+1 \leq \frac{x+y-2}{2-x-y+xy} + 1 = \frac{x+y-2+2-x-y+xy}{2-(x+y-2xy)}$

$\Rightarrow$ $o v \; x+y-1 \leq \frac{xy}{2-(x+y-2xy)}$

$\Rightarrow$ $x \Delta_1 y \leq x \Delta_2 y$

$\Rightarrow$ $\Delta_1 \leq \Delta_2 \longrightarrow (A)$

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(ii) Now $x \Delta_2 y \leq x \Delta_3 y$

$\because x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$ and $x \Delta_3 y = xy$

$\because 0 \leq x \leq 1$ and $0 \leq y \leq 1$

$\Rightarrow 0 \geq -x \geq -1$ and $0 \geq -y \geq -1$

$\Rightarrow 1 \geq 1-x \geq 0$ and $1 \geq 1-y \geq 0$

$\Rightarrow 1 \geq (1-x)(1-y) \geq 0$

$\Rightarrow 1 \geq 1-x-y+xy \geq 0$

$\Rightarrow 2 \geq 2-x-y+xy \geq 1$

$\Rightarrow \frac{1}{2} \leq \frac{1}{2-x-y+xy} \leq 1 \longrightarrow (3)$

$\because 0 \leq xy \leq 1 \longrightarrow (4)$

Multiplying eq. (3) by "xy"

$\frac{xy}{2} \leq \frac{xy}{2-(x+y-xy)} \leq xy$

$\Rightarrow \frac{xy}{2-(x+y-xy)} \leq xy$

$\Rightarrow x \Delta_2 y \leq x \Delta_3 y$

So $\Delta_2 \leq \Delta_3 \longrightarrow (B)$

Hence by eq. (A) and (B)

So $\Delta_1 \leq \Delta_2 \leq \Delta_3$

which is complete the proof.

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Lecture # 22

Triangular Norms and Conorm:

Given any t-norm T and t-conorms S we have $T(x,0)=0$ and $S(x,1)=1$, $\forall\ x \in [0,1]$.

proof: From T_1 we have $T(0,1)=0$, Then from T_4 it follows that

$$T(0,x) \leq T(0,1)=0, \quad \forall\ x \in [0,1]$$

i.e. $T(0,x)=0$, Then from T_2 we get

$T(x,0)=0$.

$T(x,0)=0$; $x \Delta 0 = 0$

$0 \leq 0$ and $x \leq 1$

T_4 : $0 \Delta x \leq 0 \Delta 1 = 0$

$\Rightarrow 0 \Delta x \leq 0$

$\Rightarrow 0 \Delta x = 0$

T_2 : $x \Delta 0 = 0$

Now $S(x,1)=1 \Rightarrow x \nabla 1 = 1$

$0 \leq x$ and $1 \leq 1$

S_4 : $0 \nabla 1 \leq x \nabla 1$

$\Rightarrow 1 \leq x \nabla 1$

$\Rightarrow x \nabla 1 = 1$

$\Rightarrow S(x,1)=1$

Similarly, we have $S(1,0)=1$ so,

$1 = S(1,0) \leq S(1,x)$

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Definition (Zadeh 154):

A triplet (S, T, N) is called a De Morgan triplet if T is a t-norm, S is a t-conorm, N is a strong negation, and if they fulfill De Morgan's law

$$S(x,y) = N(T(N(x), N(y)))$$

If (S, T, N) is a De Morgan triplet then we have $T(x,y) = N(S(N(x), N(y)))$.

Indeed,

$$S(N(x), N(y)) = N(T(N(N(x), N(y)))) = N(T(x,y)).$$

Then

$$N(S(N(x), N(y))) = N(N(T(x,y))) = T(x,y)$$

Generating T-norms:

Let $f: [0,1] \to [a,1]$; $0 \le a < 1$ is

i) One to one and onto mapping i.e. (isomorphism) and

ii) If $x \le y \Rightarrow f(x) \le f(y)$ also $(f(0)=a, f(1)=1)$.

Then we can make a new t-norm Δ_f such that

$$x \Delta_f y = f^{-1}([f(x) \Delta f(y)] \vee a)$$

Example: $\Delta_f = ?$ $\Delta = xy$, $f(x) = x$

Sol: $x \Delta_f y = f^{-1}([f(x) \Delta f(y)] \vee a)$

$= f^{-1}([f(x) f(y)] \vee a)$

$= f^{-1}([f(x) f(y)] \vee 0)$

$f(0) = 0$

$a = 0$

$$x \Delta_f y = f^{-1}(f(x) f(y))$$

$$= f^{-1}(xy) \qquad \because f(x) = x$$

$$\Rightarrow f^{-1}(x) = x$$

$$x \Delta_f y = xy$$

Example: $f(x) = x^\gamma$; $\gamma > 0$, $x \Delta y = xy$

Sol: Let $y = x^\gamma$

$$y^{1/\gamma} = x$$

$$\Rightarrow f^{-1}(x) = x^{1/\gamma}$$

Now $f(0) = 0^\gamma = 0$

$$\Rightarrow a = 0$$

$$x \Delta_f y = f^{-1}([f(x) \Delta f(y)] \vee a)$$

$$= f^{-1}([f(x) f(y)] \vee 0)$$

$$= f^{-1}(f(x) f(y))$$

$$= f^{-1}(x^\gamma y^\gamma)$$

$$x \Delta_f y = (x^\gamma y^\gamma)^{1/\gamma} = xy$$

MTH645

Fuzzy Logic and Applications

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